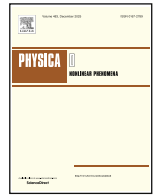


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Multi-front dynamics in spatially inhomogeneous Allen-Cahn equations [★]

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ABSTRACT

Recent studies of biological, chemical, and physical pattern-forming systems have started to go beyond the classic ‘near onset’ and ‘far from equilibrium’ theories for homogeneous systems to include the effects of spatial heterogeneities. In this article, we build a conceptual understanding of the impact of spatial heterogeneities on the pattern dynamics of reaction-diffusion models. We consider the simplest setting of an explicit, scalar, bi-stable Allen-Cahn equation driven by a general small-amplitude spatially-heterogeneous term $\varepsilon F(U, U_x, x)$. In the first part, we perform an analysis of the existence and stability of stationary one-, two-, and N -front patterns for general spatial heterogeneity $F(U, U_x, x)$. Then, for general dynamically-evolving N -front patterns, we explicitly determine the N -th order system of ODEs that governs to leading order the evolution of the front positions. In the second part, we focus on a particular class of spatial heterogeneities where $F(U, U_x, x) = H'(x)U_x + H''(x)U$, in which H is either spatially localized or spatially periodic. For localized heterogeneities, we determine all stationary N -front patterns, and show that these are unstable for $N > 1$. We find instead slowly evolving ‘trains’ of N -fronts that collectively travel to $\pm\infty$, either with slowly decreasing or increasing speeds. For spatially periodic heterogeneities, we show that the fronts of a multi-front pattern will get ‘pinned’ if the distances between successive fronts are sufficiently large, i.e., the multi-front pattern is attracted to a nearby stable stationary multi-front pattern.

1. Introduction

Traditionally, the mathematical analysis of the formation and dynamics of patterns, i.e., the dynamics of solutions of nonlinear systems of reaction-diffusion equations, has focused on systems of equations that do not explicitly vary in space (or time). For the theory of the onset of near-homogeneous patterns, see for instance [1] and the references therein. For the theory of localized, ‘far-from-equilibrium’ patterns, see [2–4] and the references therein.

In recent years, there has been a growing interest in understanding pattern dynamics in biological, chemical, and physical problems which are spatially heterogeneous. Patterns in biology are affected by the spatial variations of various rate constants, including for example in embryology where reaction or diffusion rates of various morphogen proteins may be modulated by preexisting spatial structure within the developing organism [5,6]. Patterns in ecological models are influenced by the topography of the terrain, the soil inclination, the local soil quality, and the availability of nutrients; and these have all been shown to affect the appearance, magnitude, movement, and growth (or decline)

of the patterns exhibited in models of ecosystems [7–11], including also that non-trivial spatial patterns may increase the resilience of an ecosystem [12]. Similarly, patterns in chemical reactions can depend on the spatially heterogeneity of reactors, spatially-dependent rate constants, or applied light sources [13]. Some other recent references to spatial heterogeneities in reaction-diffusion models include [14,15]. Therefore, from a broad scientific point of view, a general approach needs to be developed to understand – and preferably also to predict – the impact of spatial inhomogeneities on the dynamics of localized structures in multi-component reaction-diffusion systems defined on large domains.

1.1. The main model

In this article, we study the impact of spatial heterogeneities in a simple pattern generating model: the scalar, bistable Allen-Cahn model in one space dimension. In its standard form, this prototypical PDE is given by

$$U_t = U_{xx} + U - U^3. \quad (1)$$

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We analyze the impact of small-amplitude, spatially heterogeneous terms on the dynamics exhibited by the Allen-Cahn equation, i.e., we study the dynamics of patterns in

$$U_t = U_{xx} + U - U^3 + \varepsilon F(U, U_x, x). \quad (2)$$

A priori, we impose the natural assumptions that $F(U, V, x)$ is sufficiently smooth in all three arguments and that $\|x \mapsto F(U, V, x)\|_{C_b} \leq \mathcal{O}(1)$ for all $U, V \in \mathbb{R}$, though in Section 7 we will consider cases with unbounded F .

The analysis in the article will first focus largely on the case of general bounded heterogeneities F . In addition, especially in Secs. 5 and 6, we provide concrete examples of the main analytical formulas for two types of explicit forms of the heterogeneity F : a spatially localized F and a spatially-periodic F .

One particularly simple choice for $F(U, V, x)$ is

$$F(U, V, x) = f_1(x)U + f_2(x)V + f_3(x). \quad (3)$$

For example, in ecological models, $F(U, V, x)$ may represent the effect of a topography $H(x) \in C^2(\mathbb{R})$ with $f_2(x) = H'(x)$, $f_1(x) = f'_2(x)$, and $f_3(x) \equiv 0$, so that

$$F(U, V, x) = H'(x)V + H''(x)U \quad (4)$$

(see [8] for a derivation and note that $H'(x) \equiv \text{a constant}$ in the classical Klausmeier model [16]). This is the ‘topographical’ case that we will also consider as one of the main examples throughout the article.

Concretely, in this paper we will consider two prototypical heterogeneities. First, a localized heterogeneity, for which we will use the specific topographies in which $H(x)$ is a ‘unimodal hill’ (or ‘valley’) with either exponentially or algebraically decaying tails:

$$H_{\text{exp}}(x; \mu) = \frac{1}{\cosh^2(\frac{1}{2}\sqrt{2\mu}x)}, \quad H_{\text{alg}}(x; p) = \frac{1}{(1+x^2)^{\frac{1}{2}(p-1)}}. \quad (5)$$

Second, we consider spatially periodic heterogeneities with $F(U, V, x+X) = F(U, V, x)$ for some $X > 0$. To observe some of the typical behavior in an accessible and concrete manner, we consider examples of the form (3) with

$$f_1(x) = \alpha_1 \cos(kx), \quad f_2(x) = \alpha_2 \sin(kx), \quad f_3(x) = \alpha_3 \sin(kx), \quad (6)$$

where $\alpha_j, k \in \mathbb{R}$ ($j = 1, 2, 3$) and period $X = 2\pi/k$.

1.2. Simulations motivating the main questions

In this subsection, we show the outcomes of some representative multi-front simulations. These provide an initial, brief quantification of the effects of small heterogeneous terms F on the front dynamics in the Allen-Cahn Eq. (2). The results of these simulations motivate the main questions and quantitative analysis of this article.

In Figs. 1(a-c), the evolution in time of (2) from identical two-front initial data is shown for three kinds of inhomogeneities $F(U, U_x, x)$. Then, in Figs. 1(d-f), the same is done for three-front initial data (again identical in all three cases). Frames (a) and (d) illustrate ‘classical’ two-front and three-front dynamics in the homogeneous Allen-Cahn Eq. (1). Namely, two fronts collide, and the systems ‘coarsen’ to either the trivial state $U \equiv -1$ (Fig. 1(a)) or to a stationary one-front (Fig. 1(d)), as expected based on the classical literature applicable to Eq. (1) [17–19] (see also [20] for a recent account).

The significant effects of spatially localized heterogeneities are illustrated in Figs. 1(b) and (e). Instead of being attracted to each other (and eventually annihilating each other), the two fronts in Fig. 1(b) repel each other due to the localized heterogeneity. Indeed, the final pattern consists of a dynamically evolving two-front state, with the fronts continuing to move outward, though with decreasing speed, and with the solution being asymptotically close to $U = +1$ on the (growing) middle part between the fronts. Thus, the attracting $U \equiv -1$ state of the homogeneous case (Fig. 1(a)) is pushed further and further outward, to $-\infty$ and $+\infty$, and more and more of the domain is covered by the other

state. In the three-front case, see Fig. 1(e), the localized spatial inhomogeneity also pushes the two outer fronts to $\mp\infty$, but here the middle front is attracted to a specific position at which it settles. Hence, in both simulations, the final spatial patterns differ completely from those of the homogeneous cases, and interesting new multi-front dynamics are created by the localized heterogeneities.

Next, Figs. 1(c) and (f) show that spatially periodic inhomogeneities can also have significant effects on two-front and three-front solutions. Here, the fronts each get ‘pinned’ at specific locations. Hence, the spatially periodic heterogeneity also prevents the coarsening (seen in the homogeneous case) from happening, and it can generate new multi-front patterns with pinned fronts.

The phenomenon of pinning (of fronts or pulses) is perhaps the most studied impact of spatial inhomogeneities on the dynamics of ‘far from equilibrium’ localized structures – see for example [21–24] and the references therein, where we add that typically only solitary fronts or pulses are considered in the mathematical literature (however, see [25]). Moreover, in the case of spatially periodic inhomogeneities, the dynamics of N -front patterns is typically driven by an interplay between coarsening and pinning. Fronts that are ‘too close’ attract and annihilate each other, while fronts that are ‘sufficiently far’ apart get pinned. This is illustrated in the simulation in Fig. 2(a), where a 5-front pattern eventually ends up in a pinned (and thus stationary) three-front state. It is also illustrated by the variety of stationary multi-front solutions formed by pinning in Fig. 3.

The situation is different in the case of spatially localized inhomogeneities. For these, at most one front can get pinned, see again Fig. 1(e), and all other fronts necessarily start to travel. However, the speed of fronts that move in the same direction depends on their distance from (the center of) the spatial inhomogeneity. Fig. 2(b) shows the typical situation in which fronts catch up with those traveling directly ahead of them, which yields a stepwise process of successive annihilations, until there is at most one solitary traveling front. As we shall analyze, this is the expected behavior of multi-front patterns driven by localized inhomogeneities in the case in which the inhomogeneity is bounded as a function of x (i.e., if there is an $C = C(U, U_x) > 0$ such that $|F(U, U_x, x)| < C$ for all $x \in \mathbb{R}$). Finally, we observe that, although the analysis here requires that $F(U, U_x, x)$ is bounded, we can formally extrapolate beyond this and predict that it is possible that fronts may not catch up with their predecessors so that no mutual annihilation takes place, see Fig. 2(c).

The dynamics induced by various spatially heterogeneous terms, including those shown in Figs. 1, 2, and 3, motivate the following fundamental questions about multi-front patterns: *What kind of novel (stable) multiple-front patterns may be initiated by the heterogeneous effects? How do the different quantitative properties of the spatially heterogeneous terms affect the dynamics, pinning, and stability of the interacting fronts in multi-front patterns?* We address these analytically for the spatially heterogeneous Allen-Cahn model (2).

1.3. The main analytical results

The main results of this article are summarized in Table 1. First, for $0 < \varepsilon \ll 1$, we present a straightforward analysis showing the persistence in the spatially-inhomogeneous PDE (2) of two elementary (or basic) states of the homogeneous Allen-Cahn Eq. (1): the trivial background states $u_{\pm}(x) = \pm 1$, $u_0(x) = 0$ and the tanh-type one-front solutions that connect $u = \pm 1$. See Secs. 1.4 and 2. In particular, we obtain explicit results on the persistence of the background states and the stationary one-fronts under the general heterogeneities (2), by using a straightforward Melnikov analysis. We also determine the spectral stability of the background states and one-front solutions for the general nonlinearities in (2).

Next, we turn to two-front solutions of (2) in Section 3. These are the simplest multi-front patterns, and they are patterns that do not exist in the homogeneous Allen-Cahn equation. We study the existence and

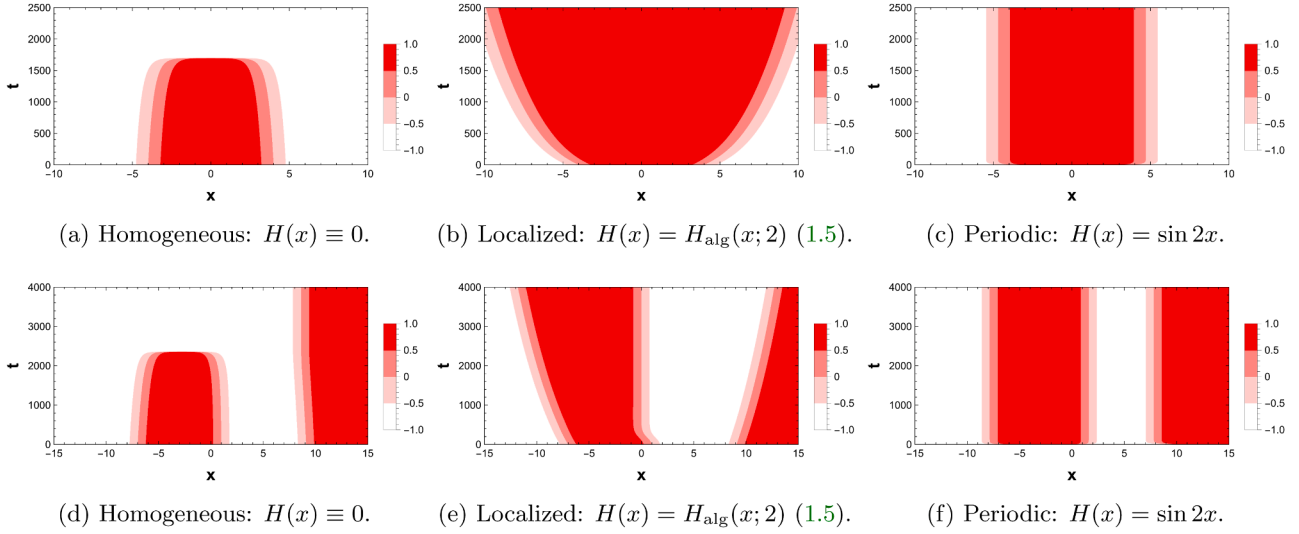


Fig. 1. Space-time plots of the multi-front dynamics of u obtained from numerical simulations of (2), subject to homogeneous Neumann boundary conditions, with topographical inhomogeneities $H(x)$ given by (4) and $\varepsilon = 0.1$. In (a), (b), and (c), the initial data are the two-front profiles $U(x, 0) = \tanh(x + 4) - \tanh(x - 4) - 1$, shown on $(-10, 10)$. In (d), (e), (f), the three-front initial data is $U(x, 0) = \tanh(x + 7) - \tanh(x - 1) + \tanh(x - 9.1)$ on $(-25, 25)$, plotted on $(-15, 15)$. The color indicates the magnitude of u .

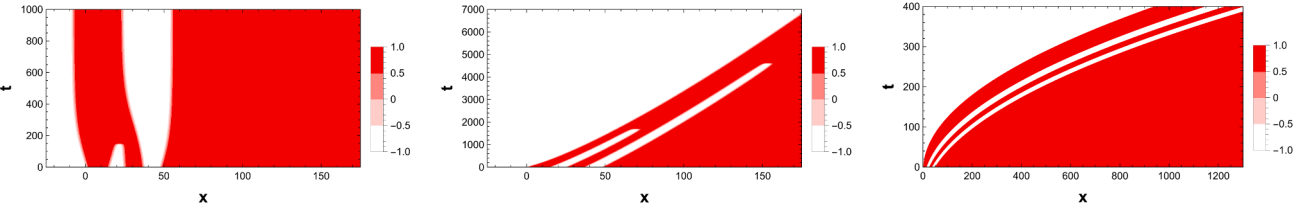


Fig. 2. Space-time plots of u obtained from direct numerical simulations of (2) with the five-front initial data $U(x, 0) = \tanh(x - 1) - \tanh(x - 15) + \tanh(x - 25) - \tanh(x - 37) + \tanh(x - 48)$ and $\varepsilon = 0.1$ on the interval $(-500, 1500)$ (plotted on relevant subintervals) with homogeneous Neumann boundary conditions. In (a), (2) is simulated with the periodic inhomogeneity (3) and $f_1(x) = \sin(\frac{1}{5}x)$, $f_2(x) \equiv f_3(x) \equiv 0$. In (b), the algebraically decaying spatially localized topographical inhomogeneity $H(x) = -H_{\text{alg}}(x; p)$ with $0 < p = .25$ is used, recall (4) and (5). In (c), the inhomogeneity $H(x) = -H_{\text{alg}}(x; p)$ with $p = -0.50 \in (-1, 0)$ is used.

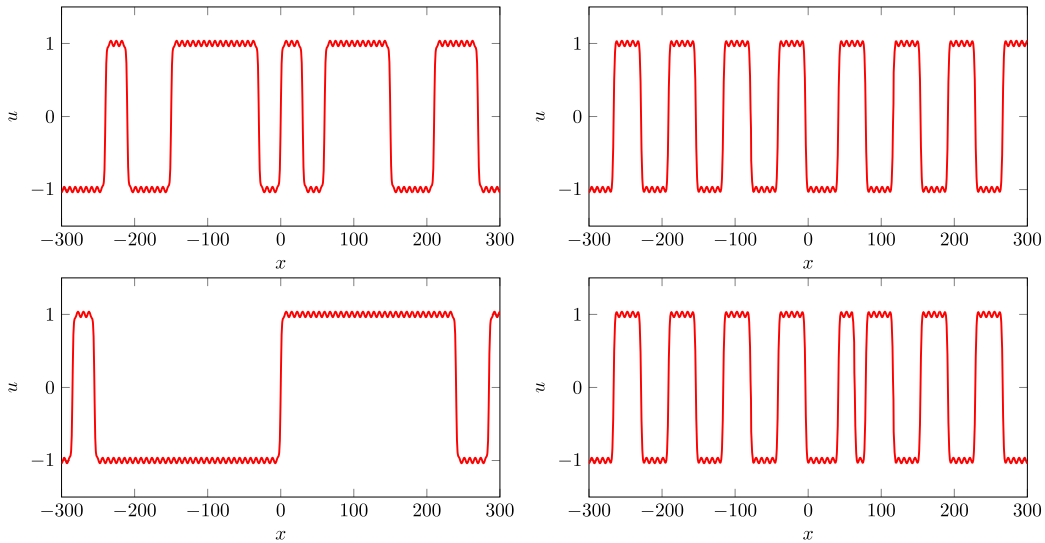


Fig. 3. Numerically stable, stationary multi-front patterns in the forced Allen-Cahn Eq. (2) showcasing the variety of stable stationary multi-front solutions of this equation. Each exhibits a different type of (irregular) front spacing, and all the fronts are formed by pinning. Here, the spatially periodic inhomogeneous term $F(U, V, x)$ is given by (3) with $f_1(x) = \cos(\frac{4}{15}\pi x)$, $f_2(x) \equiv 0$, $f_3(x) \equiv 0$, and $\varepsilon = 0.1$. (Simulations performed on $x \in (-300, 300)$ with homogeneous Neumann boundary conditions; patterns are (numerically) stable based on simulation up to $T = 10^{12}$).

Table 1

Overview of the primary results organized by section. The left column indicates the Theorem, Corollary, Claim, Equation, or Subsection in which the result can be found. Theorems and corollaries are proven. Other statements are based on formal analysis and/or numerical results. When applicable, illustrating figures are indicated in parenthesis. *An asterisk indicates that (spectral) stability is only proven if the (explicit) x -dependence of $F(U, U_x, x)$ is either localized or periodic, though we expect the results to hold in more generality.

Section 1. Introduction	
Section 1.4	Existence & stability* of 0-front solutions
Section 2. Stationary one-front solutions in the inhomogeneous Allen-Cahn equation	
Theorem 1	Existence & stability* of stationary 1-front solutions
Section 2.4.1	Application of Theorem 1 to localized topography F (5)
Section 2.4.2	Application of Theorem 1 to spatially periodic F (6)
Section 3. Stationary two-front solutions in the inhomogeneous Allen-Cahn equation	
Theorem 2	Existence & stability* of stationary 2-front solutions
Section 3.3.1	Application of Theorem 2 to localized topography F (5)
Section 3.3.2	Application of Theorem 2 to spatially periodic F (6)
Section 4. The (slow) dynamics of interacting fronts	
Eqn (78)	Formally derived ODE for the evolution of N -front solutions
Section 5. Front dynamics driven by localized topographies (localized topography F per (5))	
Theorem 3	Existence & stability of stationary N -front solutions (case of exponentially decaying tails)
Corollary 1	Existence & stability of stationary N -front solutions (case of algebraically decaying tails)
Section 5.1	Classification of dynamically-evolving 2-front solutions: • One front travels to $-\infty$ and the other to $+\infty$ (Figs. 1(b), 11(b)) • One front gets pinned and the other travels to $\pm\infty$ (Fig. 20(d)) • The fronts attract and annihilate each other (Fig. 11(d)) • The fronts travel as a train to either $+\infty$ or $-\infty$
Section 5.2	Dynamics of 3-front and 4-front solutions (Fig. 11)
Section 5.3	Generalizations of the 2-front dynamics for N -front dynamics (Fig. 12)
Section 6. The spatially periodically forced Allen-Cahn Equation (spatially periodic F per (6))	
Theorem 4	Existence & stability N -front solutions
Claim 1	Existence of nearby attractors for N -fronts sufficiently far apart (Fig. 19)
Section 6.1	Dynamics of 2-front solutions (Fig. 13, 14, 15)

spectral stability of all stationary two-front solutions in Section 3, and the main results are presented in Theorem 2.

Then, for N -front solutions with general N and general spatially heterogeneous terms F , we derive the N -dimensional dynamical systems that govern the interactions of the dynamically-evolving fronts of (2) to leading order. This system of N ODEs is given by (78). (Here, we follow the approach also used in [26,27].)

The system (78) of ODEs has insightful and explicitly tractable forms for (at least) two large general classes of spatial heterogeneities F : one class in which F has the ‘topographical’ character of (4) and another class in which F is spatially periodic. Hence, in Secs. 5 and 6, we analyze these two classes, respectively. In Section 5, we present the main analysis of the multi-front solutions for topographical F , where the system of ODEs (78) reduces to (84). For exponentially decaying topographies, we establish the existence of large families of stationary N -front patterns. These patterns correspond to equilibria of the system (78) of ODEs, and we prove that the maximal number is $(K + 1)N - 1$, where K is the number of zeros of an associated Melnikov function. Also, the stability of these stationary N -front patterns is determined by the stability type of the corresponding equilibria of the system of ODEs. See Theorem 3. Then, we also establish the existence of stationary N -front solutions in the presence of algebraically decaying topographies. See Corollary 1. However, we show that, for $N > 1$, these stationary N -front solutions are unstable, in both the cases of exponential decay and algebraic decay. This general result builds on a detailed analysis of the special cases $N = 2, 3, 4$.

The analysis of these exponentially and algebraically decaying topographies in Secs. 5.3 also provides insights into the fate of

dynamically-evolving multi-front patterns that start from initial data consisting of N -front configurations. Namely, after an initial process of coarsening during which one front may get pinned at a special location (recall for example Fig. 1(e)), ‘trains’ of M -front patterns (with $M \leq N$) are formed that collectively travel to $+$ or $-\infty$. We show that, as long as $H'(x)$ decays, fronts within such a train will eventually catch up with their predecessors, thereby driving a slow coarsening process – recall also Fig. 2(b).

Next, in Section 6, we turn to investigate the system of N ODEs (78) in the case of spatially periodic heterogeneities. The dynamics of N -front patterns for these spatially-periodic problems are summarized in Theorems 4 and Claim 1. The former establishes the existence of stationary N -front patterns for patterns in which neighboring fronts are sufficiently far apart. In fact, near any initial N -front configuration with sufficiently separated successive fronts, there are K^N (distinct) stationary N -fronts of which $(\frac{1}{2}K)^N$ are stable, where $K \geq 0$ is the number of zeros per period of the associated periodic Melnikov function. This theorem also establishes the non-existence of stationary N -front patterns when the fronts are too close to each other. The process by which stationary fronts are created (as the initial distance between fronts increases) is studied in detail in the specific setting of two-fronts. It is shown how the underlying geometry of this process can be interpreted in terms of the well-established terminology of weakly driven nonlinear oscillators. Then, Claim 1 presents insights in dynamic N -fronts. Again, these are all new patterns, which cannot exist in the homogeneous Allen-Cahn equation.

We conclude the article with an extensive discussion in Section 7. First, the impact of the spatial heterogeneities on the process of coars-

ening is discussed, followed by a brief exploration into the realm of topographies beyond those considered in the earlier sections, namely, beyond the cases of spatially-periodic F and of spatially localized F with monotonically decaying tails. Then, the main topic of this section is the behavior of N -front patterns driven by topographies $H(x)$ for which $H'(x)$ ‘decays’ algebraically with rate $p < 0$ or even $p < -1$ (i.e., $H'(x) \sim 1/x^p$ as $x \rightarrow \pm\infty$ – see example $H_{\text{alg}}(x; p)$ in (5)). For such $H(x)$, the methods of [26,27] to settle the validity of the N -front interaction system cannot be applied directly, although there are indications that this should be possible for p down to -1 (i.e., for $p > -1$). It follows from the interaction equations – and is confirmed by numerical simulations (Fig. 2(c), Fig. 22) – that for $-1 < p < 0$, fronts within an ‘ M -train’ of jointly traveling fronts will travel slower than their predecessors. This implies that the M -trains will persist and slowly ‘spread out’. The situation becomes even more interesting as p passes through -1 . For that value, the (formal) front interaction equations predict fronts with increasing speeds that blow up in finite time, which is also confirmed by simulations, see Fig. 23. This last section is concluded by a brief discussion of the potential implications of our findings in the context of some scientific problems, especially ecosystem models which were a primary motivation.

Remark 1. Most of the numerical PDE simulations presented in this article were performed in Mathematica using the NDSolveValue and MethodOfLines routines. The exceptions are the results in Fig. 3, which are made in MATLAB using the pdepe routine.

Remark 2. The stationary problem associated to (2) is given by

$$0 = u_{xx} + u - u^3 + \varepsilon F(u, u_x; x) \quad \text{or} \quad \begin{cases} u_x &= v, \\ v_x &= u^3 - u - \varepsilon F(u, v; x). \end{cases} \quad (7)$$

It is of the form of a weakly driven (planar) nonlinear oscillator. This form will play a central role in the upcoming analysis. Also, since the classical theory of periodically driven oscillators predicts chaotic behavior (see [28], for example), one may thus expect (stationary) ‘chaotic multi-front patterns’ if $F(u, v; x)$ varies periodically in x . See Fig. 3 for stable patterns that appear to be of this nature.

Remark 3. There is a more extensive literature on inhomogeneous Allen-Cahn-type equations in which the inhomogeneity varies slowly in space (i.e., on long spatial scales) – see [29–31] and the references therein. In these references, (2) is typically written as a singularly perturbed equation, $U_t = \varepsilon^2 U_{XX} + G(U, X)$ which is equivalent to $U_t = U_{xx} + G(U, \varepsilon x)$ with $x = X/\varepsilon$. (In fact, in [31] $G(U, \varepsilon x) = (1 - U^2)(U - a(\varepsilon x))$ which corresponds to $\varepsilon F(U, U_x, x) = -a(\varepsilon x)(1 - U^2)$ in (2).) Here, we instead consider inhomogeneities that vary on the same spatial scale as $U(x, t)$. To set up the asymptotic analysis however, we need to assume that the amplitude of the inhomogeneities is small (which is not necessary when the inhomogeneities vary on long spatial scales). See also [32,33] and the references therein for the analysis of inhomogeneous systems of reaction-diffusion equations in which the inhomogeneity varies on a long spatial scale.

Remark 4. In order to consider a topography as being truly localized, one needs to impose that $\mu > 0$ for $H_{\text{exp}}(x; \mu)$ and $p > 1$ for $H_{\text{alg}}(x; p)$. However, by (4) we note that only $H'(x)$ enters into Eq. (2), so that it is natural to consider $p > 0$, or even $p \geq 0$ since also $F(U, V, x)$ remains bounded for $p = 0$, which corresponds for example to the classical Klausmeier case [16]. We find that our analysis may also be valid for $p \in (-1, 0)$ for which $H''_{\text{alg}}(x; p)$ remains bounded (but $H'_{\text{alg}}(x; p)$ not), see Section 7 and Fig. 2(c).

Remark 5. As we will see, the ‘driving force’ of the algebraically decaying tails of $H_{\text{alg}}(x; p)$ is much stronger than that of the exponentially decaying tails of $H_{\text{exp}}(x; \mu)$. Hence, most simulations will be performed with $H_{\text{alg}}(x; p)$. However, we will also see that $H_{\text{exp}}(x; \mu)$ is more accessible to explicit analysis. Therefore, it will appear more often in the examples presented at various places in the article.

1.4. Zero-front solutions: The persistence of the background states

In this brief subsection, we consider the existence and stability of the background states of (2) to which all multi-front patterns will converge in the limits $x \rightarrow \pm\infty$. In the homogeneous Allen-Cahn Eq. (1), there are three such background states: $u_{\pm}(x) \equiv \pm 1$ and $u_0(x) \equiv 0$. The following dispersion relations define the location of the (essential) spectra $\Sigma_{\pm,0}$ and $\Sigma_{0,0}$ associated to their stability:

$$\begin{aligned} u_{\pm} : \quad \lambda_{\pm}(k) &= -k^2 + (1 - 3u_{\pm}^2) = -k^2 - 2, \\ u_0 : \quad \lambda_0(k) &= -k^2 + (1 - 3u_0^2) = -k^2 + 1. \end{aligned} \quad (8)$$

Thus, the states $u_{\pm}$ are spectrally stable against small-amplitude perturbations, whereas the state u_0 is unstable.

For (2) with $\varepsilon > 0$, the classical solutions u_{-} , u_0 , and u_{+} are generally no longer stationary solutions. However, for sufficiently small ε , there are unique, bounded ‘zero-front’ solutions, $u_{\pm}^{\varepsilon}$, which lie $\mathcal{O}(\varepsilon)$ -close to the classical solutions $u_{\pm} = \pm 1$. These are given by

$$\begin{aligned} u_{\pm}^{\varepsilon}(x) &= \pm 1 + \varepsilon \tilde{u}_{\pm}^{\varepsilon}(x) := \pm 1 + \frac{\varepsilon}{2\sqrt{2}} \left[\int_x^{\infty} F(\pm 1, 0, z) e^{\sqrt{2}(x-z)} dz \right. \\ &\quad \left. + \int_{-\infty}^x F(\pm 1, 0, z) e^{-\sqrt{2}(x-z)} dz \right] + \mathcal{O}(\varepsilon^2). \end{aligned} \quad (9)$$

For future reference, we note that, in the case of topographic inhomogeneities (4),

$$u_{\pm}^{\varepsilon}(x) = \pm 1 \pm \frac{\varepsilon}{2\sqrt{2}} \left[\int_x^{\infty} H''(z) e^{\sqrt{2}(x-z)} dz + \int_{-\infty}^x H''(z) e^{-\sqrt{2}(x-z)} dz \right] + \mathcal{O}(\varepsilon^2), \quad (10)$$

i.e., $u_{\pm}^{\varepsilon}(x)$ only depends on $H''(x)$ (to leading order) and is bounded (for $x \in \mathbb{R}$) under the condition that $H''(x)$ is (recall (5): $H'(x)$ may thus be unbounded here).

In the extended (u, u_x, x) -phase space of the stationary problem (7), the sets $\{(u_{\pm}^{\varepsilon}(x), \frac{d}{dx} u_{\pm}^{\varepsilon}(x), x) | x \in \mathbb{R}\}$ representing the zero-front solutions are 1D saddle invariant manifolds, which we label $\mathcal{M}_{\pm}^{\varepsilon}$. These perturb for $0 < \varepsilon \ll 1$ from the lines $\{(u_{\pm}, 0, x) | x \in \mathbb{R}\}$ of saddle points of the $\varepsilon = 0$ system. For ε sufficiently small and suitably smooth functions F , their persistence is guaranteed by classical theory for normally hyperbolic invariant manifolds, see [34–36] for example. Moreover, by the same persistence theory for normally hyperbolic manifolds, the normally hyperbolic invariant manifolds $\mathcal{M}_{\pm}^{\varepsilon}$ have 2D local stable and unstable manifolds, $W^{s,u}(\mathcal{M}_{\pm}^{\varepsilon})$, for all ε sufficiently small. These manifolds are the cornerstones of the theory for the front solutions presented in this article.

Naturally, the stability of the zero-front solutions $u_{\pm}^{\varepsilon}(x)$ is determined by substituting $U(x, t) = u_{\pm}^{\varepsilon}(x) + \tilde{u} e^{it}$ into (2) and linearizing, which yields the spectral problem,

$$\left[\partial_x^2 + (1 - 3(u_{\pm}^{\varepsilon}(x))^2) + \varepsilon \left(F_u(u_{\pm}^{\varepsilon}(x), \partial_x u_{\pm}^{\varepsilon}(x), x) + F_{u_x}(u_{\pm}^{\varepsilon}(x), \partial_x u_{\pm}^{\varepsilon}(x), x) \partial_x \right) \right] \tilde{u} = \lambda \tilde{u}. \quad (11)$$

(as usual defined on $H^2(\mathbb{R})$ [37]). For $\varepsilon = 0$, this reduces to

$$[\partial_x^2 - 2] \tilde{u} = \lambda \tilde{u}, \quad (12)$$

the spectral problem associated to the trivial background states $u_{\pm} = \pm 1$ of the classical case of the Allen-Cahn Eq. (1), i.e.,

$$\Sigma_{\pm,0} = \{\lambda \in \mathbb{C} : \lambda \leq -2\} \quad (13)$$

(cf. (8)). More precisely, it follows from (9) that the spectral stability problem (11) can be rewritten as an $\mathcal{O}(\varepsilon)$ perturbation of the unperturbed spectral problem (13),

$$\left[(\partial_x^2 - 2) + \varepsilon \left(\mp 6\tilde{u}_{\pm}^{\varepsilon} + F_u(u_{\pm}^{\varepsilon}, \partial_x u_{\pm}^{\varepsilon}, x) + F_{u_x}(u_{\pm}^{\varepsilon}, \partial_x u_{\pm}^{\varepsilon}, x) \partial_x \right) + \varepsilon^2 (-3(\tilde{u}_{\pm}^{\varepsilon})^2) \right] \tilde{u} = \lambda \tilde{u}.$$

Under the additional assumption that $F(U, U_x, x)$ is either localized or periodic as a function of x (see the upcoming Theorems 1 and 1, as well as their proofs), it can now be shown by the theory of exponential dichotomies [38,39] that for ε sufficiently small, the spectrum associated

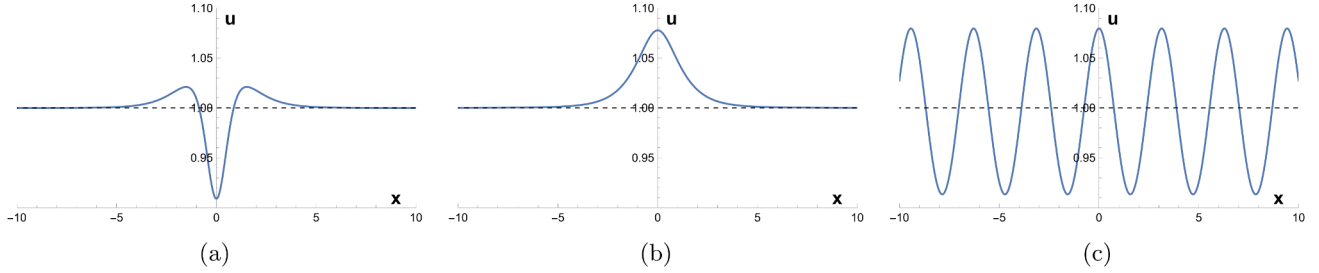


Fig. 4. The zero-front background states $u_{\pm}^{\varepsilon}(x)$ for inhomogeneous Eq. (2) with topographical inhomogeneities (4). (a) $H(x) = H_{\text{alg}}(x; 4.0)$, (b) $H(x) = H_{\text{alg}}(x; 0.0)$, (c) $H(x) = \sin 2x$ and $\varepsilon = 0.25$. Plots are obtained via numerical simulations; see also remark 1.

to the stability of $u_{\pm}^{\varepsilon}$ lies ε -close to $\Sigma_{\pm,0}$ (13) (see [7,25] for detailed analyses of more involved cases). We conclude that, if $F(U, U_x, x)$ is localized or periodic as a function of x , then the zero-front background states $u_{\pm}^{\varepsilon}$ are stable as solutions of (2), see Fig. 4.

2. Stationary one-front solutions in the inhomogeneous Allen-Cahn equation

In this section, we consider the existence and stability of stationary one-front patterns in the inhomogeneous Allen-Cahn Eq. (2), starting from a brief preparation on (stable) one-fronts in the homogeneous Eq. (1). See Fig. 5(b) for a homogeneous one-front pattern and Fig. 5(c) and (d) for examples of stable stationary one-front patterns in the Allen-Cahn equation with spatially localized and spatially periodic inhomogeneities, respectively.

2.1. One-front solutions in the homogeneous Allen-Cahn equation

The stationary problem associated to the homogeneous Allen-Cahn Eq. (1), i.e. (7) with $\varepsilon = 0$, is an integrable system with Hamiltonian

$$H(u, u_x) = \frac{1}{2}u_x^2 + \frac{1}{2}u^2 - \frac{1}{4}u^4. \quad (14)$$

The asymptotic states $u_{\pm} = \pm 1$ lie on the level set $H(u, u_x) = \frac{1}{4}$. Hence, there exist a heteroclinic connection $u_{\text{up}}(x)$ from u_- to u_+ and a heteroclinic connection $u_{\text{down}}(x)$ from u_+ to u_- , see Fig. 5(a). These are given explicitly by

$$u_{\text{up}}(x; \phi) = +\tanh\left(\frac{1}{2}\sqrt{2}[x - \phi]\right), \quad u_{\text{down}}(x; \phi) = -\tanh\left(\frac{1}{2}\sqrt{2}[x - \phi]\right) \quad (15)$$

(see Fig. 5(b)), where ϕ is an arbitrary phase shift. As a preparation for the upcoming analysis of the non-autonomously perturbed system (7), it is useful to view these heteroclinic orbits in the 3D (u, u_x, x) extended phase space. For $\varepsilon = 0$, branches of the manifolds $W^u(\mathcal{M}_+^0)$ and $W^s(\mathcal{M}_+^0)$ coincide, as do branches of $W^u(\mathcal{M}_-^0)$ and $W^s(\mathcal{M}_-^0)$. These form 2D surfaces connecting $\mathcal{M}_+^0$ and $\mathcal{M}_-^0$, and they are foliated by the orbits (15), taken over all x and ϕ .

The spectral stability of a one-front pattern $u_h(x)$ – where $u_h(x)$ is either $u_{\text{up}}(x)$ or $u_{\text{down}}(x)$ – is determined by the Sturm-Liouville problem,

$$(\mathcal{L}_0(x; \phi) - \lambda)\tilde{u} = 0 \quad (16)$$

on $H^2(\mathbb{R})$ with (Sturm-Liouville) operator $\mathcal{L}_0 : H^2(\mathbb{R}) \rightarrow L^2(\mathbb{R})$ given by

$$\mathcal{L}_0(x; \phi) = \partial_x^2 + 1 - 3u_h^2(x; \phi) = \partial_x^2 + 1 - 3\tanh^2\left(\frac{1}{2}\sqrt{2}[x - \phi]\right). \quad (17)$$

We observe that the one-front patterns $u_{\text{up/down}}(x)$ have the same spectrum. For notational clarity, we take $\phi = 0$ in the rest of this section. However, in the forced system, the phase shift is no longer arbitrary, and only specific phase shifts will be allowed. Since $u_{\text{up/down}}(x)$ is asymptotic to $u_{\pm}$, the essential spectrum associated to the stability of these one-front patterns is given by

$$\Sigma_{\text{ess},0} = \Sigma_{\pm,0} = \{\lambda \in \mathbb{C} : \lambda \leq -2\}, \quad (18)$$

recall (13) and [37]).

The point spectrum Σ_{pt} can be determined by explicitly solving the eigenvalue problem (16). We find that for any $\lambda \in \mathbb{C}$ the eigenvalue problem (16) has two (Jost) solutions,

$$\begin{aligned} \tilde{u}_+(x; \lambda) &= e^{-\sqrt{2+2\lambda}x} \left(1 + \frac{2}{3}\lambda + \tanh^2\left(\frac{1}{2}\sqrt{2}x\right) + 2\sqrt{1 + \frac{1}{2}\lambda \tanh\left(\frac{1}{2}\sqrt{2}x\right)} \right), \\ \tilde{u}_-(x; \lambda) &= e^{\sqrt{2+2\lambda}x} \left(1 + \frac{2}{3}\lambda + \tanh^2\left(\frac{1}{2}\sqrt{2}x\right) - 2\sqrt{1 + \frac{1}{2}\lambda \tanh\left(\frac{1}{2}\sqrt{2}x\right)} \right). \end{aligned} \quad (19)$$

For all $\lambda \in \mathbb{C} \setminus \Sigma_{\text{ess},0}$, one of these solutions approaches 0 as $x \rightarrow \infty$ and the other as $x \rightarrow -\infty$. The eigenvalues may be found as the roots of an Evans function (see e.g. [3,40] and references therein), which is given by

$$D_0(\lambda) := \begin{vmatrix} \tilde{u}_+(0; \lambda) & \tilde{u}_-(0; \lambda) \\ \tilde{u}'_+(0; \lambda) & \tilde{u}'_-(0; \lambda) \end{vmatrix} = \frac{4}{9}\lambda(3+2\lambda)\sqrt{2+\lambda}. \quad (20)$$

Therefore, the eigenvalues are simple. They are $\lambda_0 = 0$, $\lambda_1 = -3/2$, and $\lambda_2 = -2$, the latter of which lies on the edge of $\Sigma_{\text{ess},0}$, recall (13). The eigenvalue $\lambda_0 = 0$ corresponds to the translation invariance of (1), and the other eigenvalues have negative real parts. Hence, the one-front patterns $u_{\text{up/down}}(x)$ have a spectrum that is bounded away from the imaginary axis, except for the simple translation eigenvalue at the origin. This implies that the fronts are (nonlinearly) stable, i.e., stable against all (small) perturbations up to phase shifts/translations of the front solution (recall [37]).

2.2. The existence of stationary one-fronts

For $0 < \varepsilon \ll 1$, (2) is no longer translation invariant. Thus, the phase shifts ϕ in (15) can no longer be chosen arbitrarily like in the case of $\varepsilon = 0$. Instead, the heterogeneous terms force specific phases, or perhaps more intuitively, specific locations for the heteroclinic connections. To determine which phases can be selected (if any), we need to determine which heteroclinic solutions persist for $0 < \varepsilon \ll 1$.

Let u_h^{ε} be a stationary solution for $\varepsilon \neq 0$ that lies close to a heteroclinic solution u_h of the $\varepsilon = 0$ problem (i.e., u_h is either u_{up} or u_{down} as in (15)). That is, let

$$u_h^{\varepsilon}(x) = u_h(x; \phi) + \varepsilon u_1(x) + \mathcal{O}(\varepsilon^2). \quad (21)$$

Substitution in (2) yields

$$\mathcal{L}_0 u_1 = -F(u_h(x; \phi), \partial_x u_h(x; \phi), x) + \mathcal{O}(\varepsilon), \quad (22)$$

with $\mathcal{L}_0$ as defined in (17). The homogeneous equation $\mathcal{L}_0 \Psi = 0$ has two linearly independent solutions, Ψ_b and Ψ_u . Since $\lambda = 0$ is an eigenvalue of (16), the solutions $\Psi_{b/u}$ are not spanned by $\tilde{u}^{\pm}(x; 0)$ (19): $\tilde{u}^+(x; 0) = \tilde{u}^-(x; 0) = \text{sech}^2(\frac{1}{2}\sqrt{2}x) = \pm\sqrt{2}\partial_x u_h(x; 0)$ (cf. (A.2)). Thus, the non-empty kernel of $\mathcal{L}_0$ is spanned by $\Psi_b(x; \phi) = \partial_x u_h(x; \phi)$, while $\Psi_u(x; \phi)$ is unbounded, with neither the $x \rightarrow \infty$ nor the $x \rightarrow -\infty$ limit existing (see App. A for explicit expressions of $\Psi_{b/u}(x; \phi)$).

As a consequence, inhomogeneous Eq. (22) only has a bounded solution if a solvability condition is satisfied. Let

$$\mathcal{R}(\phi) := \int_{-\infty}^{\infty} F(u_h(x; \phi), \partial_x u_h(x; \phi), x) \partial_x u_h(x; \phi) dx. \quad (23)$$

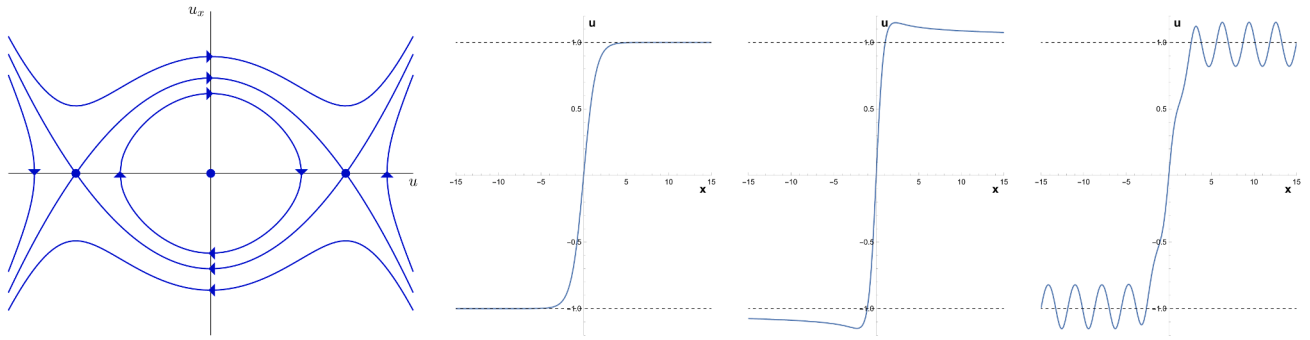


Fig. 5. (a) The phase portrait for the unperturbed ($\varepsilon = 0$) planar system (7) (determined by the level sets of $\mathcal{H}(u, u_x)$ (14)), (b) the unperturbed one-front pattern $u_{\text{up}}(x; 0)$ (15) as stable solution of (2) with $\varepsilon = 0$ and (c-d) (stable) stationary patterns in (2) with $\varepsilon = 0.5$ with (c) a 'localized' topography and (d) a spatially periodic topography (4) ((c) $H(x) = H_{\text{alg}}(x; -0.6)$, (d) $H(x) = \sin 2x$). (Note that we have chosen $p < 0$ in (c) to better illustrate the impacts of the heterogeneity, whereas with $p < 0$ the topography is not really localized (see also Section 7)).

The solvability condition is that $\mathcal{R}(\phi)$ has a simple zero. It can be verified that solutions satisfying (22) which stay bounded for $x \rightarrow -\infty$ have the following form (see App. A):

$$u_1(x; \phi) = A(x; \phi)\Psi_b(x; \phi) + B(x; \phi)\Psi_u(x; \phi), \quad (24)$$

where

$$\begin{aligned} A(x; \phi) &= \int_{\phi}^x F(u_h(z; \phi), u'_h(z; \phi), z) \Psi_u(z; \phi) dz, \\ B(x; \phi) &= -\int_{-\infty}^x F(u_h(z; \phi), u'_h(z; \phi), z) \Psi_b(z; \phi) dz. \end{aligned} \quad (25)$$

Hence, u_1 stays bounded for $x \rightarrow \infty$ if and only if the solvability condition that $\mathcal{R}(\phi)$ has a simple zero is satisfied, where $\mathcal{R}$ is given by (23) (see App. A for a comprehensive study of the inhomogeneous problem). Thus, for suitable ϕ , i.e., for those that satisfy $\mathcal{R}(\phi) = 0$, a heteroclinic connection exists that connects the asymptotic state u_+^e to u_-^e , or u_-^e to u_+^e . Examples are given in Fig. 5 and Section 2.4.

Finally, we observe that the solvability condition is possibly different for the persistence of front solutions connecting u_-^e to u_+^e and the persistence of front solutions connecting u_+^e to u_-^e . In particular, the former persist if

$$\mathcal{R}_{\text{up}}(\phi) := \int_{-\infty}^{\infty} F(u_{\text{up}}(x; \phi), \partial_x u_{\text{up}}(x; \phi), x) \partial_x u_{\text{up}}(x; \phi) dx \quad (26)$$

has a simple zero, and the latter if

$$\mathcal{R}_{\text{down}}(\phi) := \int_{-\infty}^{\infty} F(u_{\text{down}}(x; \phi), \partial_x u_{\text{down}}(x; \phi), x) \partial_x u_{\text{down}}(x; \phi) dx \quad (27)$$

does. These are the key conditions that determine the parameter values for which stationary one-front solutions persist in (2) with $0 < \varepsilon \ll 1$.

Geometrically, these persistence conditions are equivalent to the condition that the Melnikov functions have simple zeros. In particular, one measures the splitting distance between $W^u(\mathcal{M}_{\pm}^e)$ and $W^s(\mathcal{M}_{\pm}^e)$ on fixed (u, u_x) cross-sections in the 3D extended phase space (using an outward pointing normal to the 2D surface foliated by the unforced heteroclinics). For ε sufficiently small, the perturbed local stable and unstable manifolds $W_{\text{loc}}^s(\mathcal{M}_{\pm}^e)$ and $W_{\text{loc}}^u(\mathcal{M}_{\pm}^e)$ lie $\mathcal{O}(\varepsilon)$ close to their unforced counterparts [41]. By expanding solutions on these manifolds on appropriate semi-infinite intervals, the splitting distance can be measured, and the coefficient on the leading order ($\mathcal{O}(\varepsilon)$) term is the Melnikov function $\mathcal{R}(\phi)$, up to a normalization factor. For values of ϕ at which $\mathcal{R}$ is not equal to zero, there is a non-zero distance between the perturbed manifolds as measured on that cross-section, and the unperturbed heteroclinics for those values of ϕ do not persist. In contrast, for values of ϕ that correspond to simple zeros of $\mathcal{R}$, the perturbed stable and unstable manifolds intersect transversely for ε sufficiently small (as measured along the normal on that cross-section), and each transverse intersection point corresponds to a perturbed heteroclinic connection in the full 3D extended phase space. For more detailed information about Melnikov theory, we refer the reader to [28, Chapter 4] in the case of periodic perturbations and to [42] for general $\mathcal{O}(\varepsilon)$ perturbations. Also, in Secs. 2.4.1 and 2.4.2,

we illustrate the Melnikov function and the use of these conditions to find one-front solutions in some examples.

2.3. Spectral stability of stationary one-front solutions

For $0 < \varepsilon \ll 1$, the spectral stability of the one-front patterns can be determined by the properties of the linear stability problem associated to the fronts in the $\varepsilon = 0$ system via a combination of the theory of exponential dichotomies and the Evans function approach (similar to [7]). It follows that the spectrum lies ε -close to the spectrum of the fronts in the $\varepsilon = 0$ system (see Appendix B). In particular, for ε small enough, the essential spectrum is again to the left of the imaginary axis and bounded away from it. Then, the eigenvalues $\lambda_1 = -3/2$ and $\lambda_2 = -2$ are perturbed slightly, by amounts of $\mathcal{O}(\varepsilon)$, but remain well in the left half of the complex plane for $0 < \varepsilon \ll 1$. Only the $\lambda_0 = 0$ eigenvalue can potentially move into the right half of the complex plane and cause destabilization (see Fig. 6).

Let $u_*(x)$ denote an exact one-front solution $u_h^e(x)$ of the $\varepsilon \neq 0$ system. To determine what happens to the $\lambda_0 = 0$ eigenvalue when $\varepsilon \neq 0$, we employ regular perturbation theory. The linearized problem associated to the spectral stability of $u_*(x)$ is

$$(\mathcal{L}_* - \lambda)\tilde{u} = 0, \quad (28)$$

with $\mathcal{L}_* : H^2(\mathbb{R}) \rightarrow L^2(\mathbb{R})$ given by

$$\mathcal{L}_* = \partial_x^2 + (1 - 3(u_*(x))^2) + \varepsilon \left[F_u(u_*(x), \partial_x u_*(x), x) + F_{u_x}(u_*(x), \partial_x u_*(x), x) \partial_x \right]. \quad (29)$$

From the existence analysis, we know that

$$u_*(x) = u_h^e(x) = u_h(x; \phi_*) + \varepsilon u_1(x) + \mathcal{O}(\varepsilon^2) \quad \text{with} \quad \mathcal{R}(\phi_*) = 0, \quad (30)$$

recall (30). Thus, since $\lambda_0 = 0$ is an eigenvalue with eigenfunction $\partial_x u_h(x; \phi)$ for the $\varepsilon = 0$ system, it is expected that the eigenfunction for the system with $0 < \varepsilon \ll 1$ is close to the derivative $\partial_x u_h(x; \phi_*)$, and that the associated eigenvalue is of size $\mathcal{O}(\varepsilon)$. Hence, we set

$$\tilde{u}(x) = \partial_x u_h(x; \phi_*) + \varepsilon \tilde{u}_1(x) + \mathcal{O}(\varepsilon^2), \quad \lambda = \varepsilon \tilde{\lambda} + \mathcal{O}(\varepsilon^2). \quad (31)$$

By construction, substitution in (28) yields only terms of $\mathcal{O}(\varepsilon)$ and higher, since $\mathcal{L}_0 \partial_x u_h(x; \phi_*) = 0$ with $\mathcal{L}_0 = \mathcal{L}_0(x; \phi_*)$, recall (17). By dividing out a factor ε , we have

$$\tilde{\lambda} \tilde{u}_{h,x} = \mathcal{L}_0 \tilde{u}_1 + \left[F_u(u_h, u_{h,x}, x) + F_{u_x}(u_h, u_{h,x}, x) \partial_x - 6u_h u_1 \right] \tilde{u}_{h,x} + \mathcal{O}(\varepsilon), \quad (32)$$

where $u_{h,x} = \partial_x u_h(x; \phi_*)$. Since $u_h^e(x; \phi_*)$ is the exact solution of (2), we can formally take the derivative of (2) and again expand to find the relation

$$0 = \mathcal{L}_0 u_{1,x} + \left[F_u(u_h, u_{h,x}, x) + F_{u_x}(u_h, u_{h,x}, x) \partial_x + F_x(u_h, u_{h,x}, x) - 6u_h u_1 \right] u_{h,x} + \mathcal{O}(\varepsilon). \quad (33)$$

Then, using this relation in (32), we obtain

$$\mathcal{L}_0(\tilde{u}_1 - u_{1,x}) = \tilde{\lambda} \partial_x u_h(x; \phi_*) + F_x(u_h(x; \phi_*), \partial_x u_h(x; \phi_*), x) + \mathcal{O}(\varepsilon). \quad (34)$$

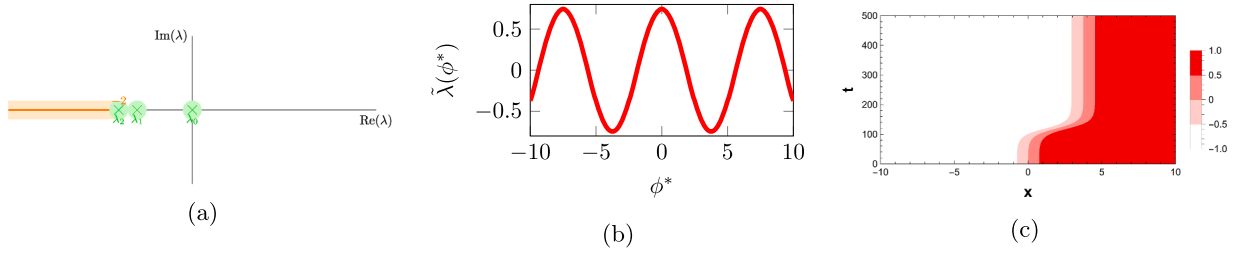


Fig. 6. (a) Sketch of (bounds on) the spectrum of heteroclinic connections in the $\varepsilon \neq 0$ system. (b) A plot of $\tilde{\lambda}(\phi_*)$ for $f_1(x) = \cos\left(\frac{4}{15}\pi x\right)$, $f_2(x) \equiv 0$, $f_3(x) \equiv 0$. Note $\tilde{\lambda}(0) \approx 0.75 > 0$, and $\tilde{\lambda}(15/4) \approx -0.75 < 0$. (c) Plot of $u(x, t)$ in the (x, t) plane, with the indicated color code for the amplitude of u , obtained using direct simulation of the full PDE. The initial condition is close to $u_h^0(x; 0)$, i.e., the unstable front.

A bounded solution $\bar{u}_1$ only exists if $\bar{u} - u_{1,x}$ is bounded. Hence, the terms in the right member must satisfy a (Fredholm) solvability condition, and this condition determines the eigenvalue $\tilde{\lambda}$. We find

$$\begin{aligned} \tilde{\lambda}(\phi_*) &= -\frac{\int_{-\infty}^{\infty} F_x(u_h(x; \phi_*), \partial_x u_h(x; \phi_*), x) \partial_x u_h(x; \phi_*) dx}{\|\partial_x u_h(x; \phi_*)\|_2^2} \\ &= -\frac{3\sqrt{2}}{4} \int_{-\infty}^{\infty} F_x(u_h(x; \phi_*), \partial_x u_h(x; \phi_*), x) \partial_x u_h(x; \phi_*) dx = -\frac{3\sqrt{2}}{4} \mathcal{R}'(\phi_*). \end{aligned} \quad (35)$$

(recall (23)). This asymptotic analysis can be summarized in the following theorem. (A similar theorem holds for the persistence of the stationary one-front solution u_{down} .)

Theorem 1. Consider the weakly heterogeneous Allen-Cahn Eq. (2) with bounded and sufficiently smooth inhomogeneity $F(U, U_x, x)$. Let the Melnikov function $\mathcal{R}(\phi)$ be given by (23) with $u_h = u_{\text{up}}$ and assume that $\mathcal{R}(\phi)$ has a non-degenerate zero at $\phi = \phi_*$ (i.e., $\mathcal{R}(\phi_*) = 0$ and $\mathcal{R}'(\phi_*) \neq 0$). Then, for $0 < \varepsilon \ll 1$ sufficiently small, (2) has a stationary one-front solution $U_{1-f}(x) = u_*(x)$ (30) that is heteroclinic to $u_{\pm}^{\varepsilon}(x) = \pm 1 + \mathcal{O}(\varepsilon)$ (9) (i.e., $\lim_{x \rightarrow \pm\infty} |U_{1-f}(x) - u_{\pm}^{\varepsilon}(x)| = 0$) and that is $\mathcal{O}(\varepsilon)$ close to $u_{\text{up}}(x; \phi_*)$ (15) (i.e., $\sup_{x \in \mathbb{R}} |U_{1-f}(x) - u_{\text{up}}(x; \phi_*)| = \mathcal{O}(\varepsilon)$).

If it is additionally assumed that the inhomogeneity $F(U, U_x, x)$ is either localized or periodic as a function of x , then the one-front pattern $U_{1-f}(x)$ is spectrally stable if $\mathcal{R}'(\phi_*) > 0$.

The proof of this theorem is given in App. B.

2.4. Stationary one-fronts in topographically driven systems and in spatially periodic systems

In this section, we apply the general theory to two main explicit classes of forcing functions $F(U, V, x)$ identified in the introduction: the ‘topographically driven’ case (4) in which $F(U, V, x)$ is determined by a topography $H(x)$ that is taken here to be spatially localized, and the spatially-periodically driven case in which $F(U, V, x)$ is given by (3), with $f_j(x)$ and $X > 0$ such that $f_j(x+X) \equiv f_j(x)$.

2.4.1. One-front solutions arising from weak topographical effects

With (4), the spatial inhomogeneity $F(U, V, x)$ may be interpreted as topographically-driven. The Melnikov function $\mathcal{R}(\phi)$, see (23), is calculated explicitly by translating the variable of integration to $y = x - \phi$,

$$\begin{aligned} \mathcal{R}(\phi) &= \int_{-\infty}^{\infty} (H'(y + \phi) \partial_y u_h(y; 0) + H''(y + \phi) u_h(y; 0)) \partial_y u_h(y; 0) dy \\ &= \int_{-\infty}^{\infty} (\partial_y [H'(y + \phi) u_h(y; 0) \partial_y u_h(y; 0)] - H'(y + \phi) u_h(y; 0) \partial_{yy} u_h(y; 0)) dy \\ &= \int_{-\infty}^{\infty} H'(y + \phi) W_h(y) dy \end{aligned} \quad (36)$$

with

$$W_h(y) := u_h^2(y; 0)(1 - u_h^2(y; 0)) \in [0, \frac{1}{4}]. \quad (37)$$

Since $u_{\text{up}}(x; \phi) = -u_{\text{down}}(x; \phi)$ by (15), it follows that

$$\begin{aligned} \mathcal{R}_{\text{up}}(\phi) &= \int_{-\infty}^{\infty} H'(y + \phi) u_{\text{up}}^2(y; 0)(1 - u_{\text{up}}^2(y; 0)) dy \\ &= \int_{-\infty}^{\infty} H'(y + \phi) u_{\text{down}}^2(y; 0)(1 - u_{\text{down}}^2(y; 0)) dy = \mathcal{R}_{\text{down}}(\phi). \end{aligned} \quad (38)$$

This is a useful explicit form of the Melnikov conditions for the topographical case.

Here, we highlight three direct consequences of these explicit formulas for $\mathcal{R}_{\text{up}}$ and $\mathcal{R}_{\text{down}}$. First, it follows directly from (36) (and (38)) that if the topography $H(x)$ is spatially periodic, i.e., if $H(x+X) \equiv H(x)$ for

some (minimal) $X > 0$, then also the Melnikov function $\mathcal{R}(\phi)$ is spatially periodic (with period X),

$$\mathcal{R}(\phi + X) = \int_{-\infty}^{\infty} H'(y + \phi + X) W_h(y) dy = \int_{-\infty}^{\infty} H'(y + \phi) W_h(y) dy = \mathcal{R}(\phi). \quad (39)$$

Second, for topographies $H(x)$ that are symmetric under $x \rightarrow -x$, i.e., for which $H(x) \equiv H(-x)$, one has that $H'(x) \equiv -H'(-x)$ and, hence, that

$$\mathcal{R}(0) = \left(\int_0^{\infty} + \int_{-\infty}^0 \right) H'(y) W_h(y) dy = \int_0^{\infty} H'(y) W_h(y) dy - \int_0^{\infty} H'(z) W_h(z) dz = 0. \quad (40)$$

Therefore, in the case of heterogeneities in which the topography $H(x)$ is symmetric around $x = 0$, there must be a stationary one-front of PDE (2) at $\phi_* = 0$. Also, these one-front patterns are spectrally stable for $\mathcal{R}'(0) > 0$, where

$$\mathcal{R}'(0) = 2 \int_0^{\infty} H''(y) W_h(y) dy. \quad (41)$$

Third, and of most interest, the equation $\mathcal{R}(\phi) = 0$ may have more solutions. As an explicit example, we consider the ‘unimodal hill’ with exponentially decaying tails $H_{\text{exp}}(x; \mu)$ (5), which gives for $\mu = 1$, with $\psi = \sqrt{2}\phi$ and $S_{\text{exp}}(\psi; \mu) = \mathcal{R}_{\text{exp}}(\phi; \mu)$,

$$S_{\text{exp}}(\psi; 1) = -\frac{16e^{\psi}[(3\psi - 13)e^{4\psi} + 2(27\psi - 47)e^{3\psi} + 126\psi e^{2\psi} + 2(27\psi + 47)e^{\psi} + (3\psi + 13)]}{3(e^{\psi} - 1)^6}. \quad (42)$$

As shown in Fig. 7, $\mathcal{R}_{\text{exp}}(\phi; 1)$ has three zeros. As the hill becomes less steep, i.e., as μ decreases, two of these merge into a pitchfork bifurcation (at $\mu_{\text{PF}} = 0.722133\dots$). The one-front at $\phi = 0$ is only stable for solitary hills that are sufficiently steep (i.e., for $\mu > \mu_{\text{PF}}$). Note on the other hand that if we change the hill into a valley, i.e., if we take $V_{\text{exp}}(x; \mu) = -H_{\text{exp}}(x; \mu)$ so that $\mathcal{R}_V(\phi; \mu) = -\mathcal{R}_{\text{exp}}(\phi; \mu)$, then the one-front at $\phi = 0$ is stable only if the slope of the ground is not too steep (i.e., for $0 < \mu < \mu_{\text{PF}}$). In this case, two stable states that are not centered at the deepest part of the valley appear as the steepness of the valley increases through μ_{PF} .

2.4.2. One-front solutions in systems with spatially-periodic heterogeneities

For general functions F (and/or f_j) that are X -periodic in x , i.e., $F(U, V, x+X) = F(U, V, x)$ for all $U, V, x \in \mathbb{R}$, the phase selection condition $\mathcal{R}(\phi) = 0$ is X -periodic, as well. Hence, generically, there will be an even number of $\phi \in [0, X]$ that satisfy the condition $\mathcal{R}(\phi) = 0$. Moreover, the function $\tilde{\lambda}(\phi_*)$ is X -periodic in ϕ_* . Therefore, generically, half of the phase shifts ϕ_* that satisfy $\mathcal{R}(\phi_*) = 0$ will have $\tilde{\lambda}(\phi_*) < 0$ (making these stable), and the other half have $\tilde{\lambda}(\phi_*) > 0$ (making these unstable). In the degenerate case that $\mathcal{R}(\phi) = 0$ has an odd number of solutions ϕ_* , there will be a ϕ_* for which $\tilde{\lambda}(\phi_*) = 0$ (i.e., in that case ϕ_* is an extreme point of $\mathcal{R}$), signaling a (saddle-node) bifurcation.

For explicit calculations, we focus on the case of (3) with

$$f_1(x) = \alpha_1 \cos(kx), \quad f_2(x) = \alpha_2 \sin(kx), \quad f_3(x) = \alpha_3 \sin(kx), \quad (43)$$

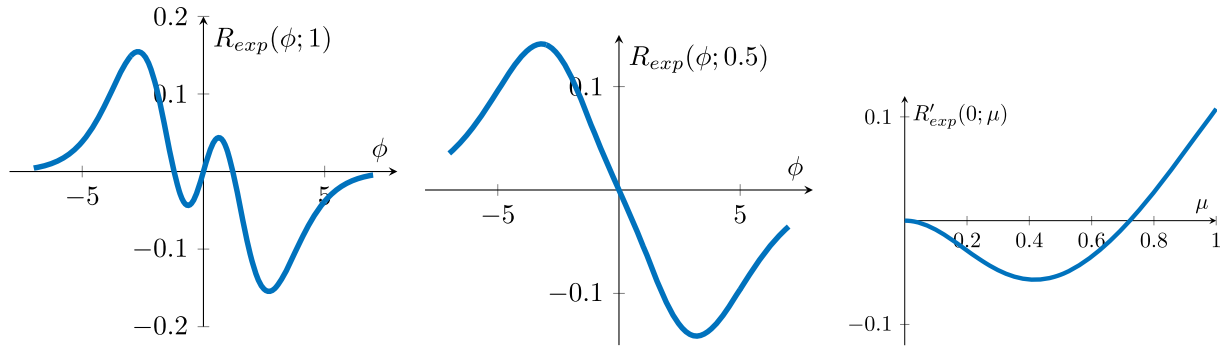


Fig. 7. (a) The Melnikov function $\mathcal{R}_{\text{exp}}(\phi; \mu)$ for a topography with a unimodal hill given by (5) with $\mu = 1$, see (42). There are three positions for which one-front solutions exist: the front at $\phi = 0$ is stable (since $\mathcal{R}'_{\text{exp}}(0; 1) > 0$), and the other two (with $\phi \neq 0$) are unstable. (b) The curve $\mathcal{R}_{\text{exp}}(\phi; \frac{1}{2})$. By decreasing μ from 1 to $\frac{1}{2}$, we see that the two unstable zeros have disappeared in a pitchfork bifurcation at the value $\mu_{\text{PF}} = 0.722133 \dots$, and the remaining one-front solution, which corresponds to the phase $\phi = 0$, has become unstable. (c) The graph of $\mathcal{R}'_{\text{exp}}(0; \mu)$ as a function of $\mu \in [0, 1]$ reveals the value of μ_{PF} .

with $\alpha_j, k \in \mathbb{R}$ ($j = 1, 2, 3$) and period $X = 2\pi/k$. This specific form enables one to observe some of the typical behavior in the most accessible and concrete way. In fact, the Melnikov functions $\mathcal{R}_{\text{up/down}}$ in (26) and (27) can be computed in closed form, as follows:

$$\begin{aligned} \mathcal{R}_{\text{up}}(\phi) &= \int_{-\infty}^{\infty} [\alpha_1 \cos(kx) u_{\text{up}}(x; \phi) \\ &\quad + \alpha_2 \sin(kx) u'_{\text{up}}(x; \phi) + \alpha_3 \sin(kx)] u'_{\text{up}}(x; \phi) dx \\ &= 16(A + B) \sin(k\phi); \end{aligned} \quad (44)$$

$$\begin{aligned} \mathcal{R}_{\text{down}}(\phi) &= \int_{-\infty}^{\infty} [\alpha_1 \cos(kx) u_{\text{up}}(x; \phi) \\ &\quad + \alpha_2 \sin(kx) u'_{\text{up}}(x; \phi) - \alpha_3 \sin(kx)] u'_{\text{up}}(x; \phi) dx \\ &= 16(A - B) \sin(k\phi), \end{aligned}$$

where A and B are defined by

$$16A := \frac{k\pi}{3 \sinh(k\pi/\sqrt{2})} [-3\alpha_1 k + \alpha_2(2 + k^2)], \quad 16B := \frac{k\pi}{3 \sinh(k\pi/\sqrt{2})} [3\sqrt{2}\alpha_3]. \quad (45)$$

Therefore, persistent one-front solutions exist at $k\phi = \ell\pi$ (to leading order) for all integers ℓ . Fig. 6 illustrates a particular case of this example, with $\alpha_1 = 1$, $\alpha_2 = \alpha_3 = 0$, and $k = 4\pi/15$. (Also, for completeness, we note that $B = 0$ in the topographic case, so that we recover $\mathcal{R}_{\text{up}}(\phi) \equiv \mathcal{R}_{\text{up}}(\phi)$, recall (38)).

3. Stationary two-front solutions in the inhomogeneous Allen-Cahn equation

As we have seen in the introduction, the classical homogeneous Allen-Cahn PDE (2) with $\varepsilon = 0$ (and $x \in \mathbb{R}$) has four primary stable solutions: the homogeneous background states $u_{\pm}$ and the one-front solutions (heteroclinics) u_{up} and u_{down} . These are illustrated in the phase portrait for $\varepsilon = 0$, recall Fig. 5. All other solutions are attracted to one of these four solutions. In particular, stationary multi-front solutions, with two or more fronts, are not present in (2) with $\varepsilon = 0$.

However, with $0 < \varepsilon \ll 1$, the symmetry is broken, and the system (2) may possess stationary two-front solutions. Geometrically, stationary two-front solutions will be found in the (transversal) intersections of the stable and unstable manifold of the same asymptotic state, i.e. intersections of $W^u(\mathcal{M}_{\pm}^{\varepsilon})$ and $W^s(\mathcal{M}_{\pm}^{\varepsilon})$, or of $W^u(\mathcal{M}_{+}^{\varepsilon})$ and $W^s(\mathcal{M}_{+}^{\varepsilon})$. The fronts in a stationary two-front solution will lie sufficiently far away from each other (i.e. the difference in their phases will be sufficiently large), since the invariant manifolds pass close by the other asymptotic state, that is, solutions on them pass asymptotically close to a saddle orbit of the system.

In this section, we construct stationary two-front solutions using regular expansions of solutions on invariant stable and unstable manifolds,

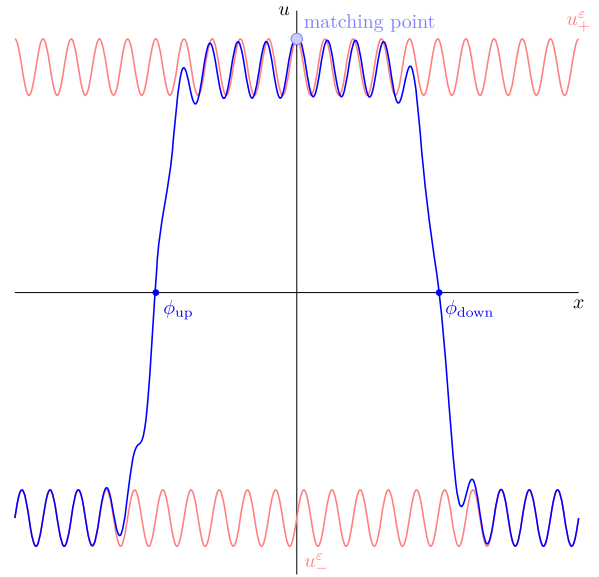


Fig. 8. Sketch of the construction of a two-front solution.

determine their spectral stability, and apply the general theory to the same two cases (topographical and spatially periodic heterogeneities) studied in the previous section. For clarity of notation, we focus on two-front solutions limiting on the asymptotic state u_{+}^{ε} ; the case of two-front solutions limiting on u_{-}^{ε} is similar.

Remark 6. Beyond the stationary two-fronts whose existence is established here, the spatially heterogeneous Allen-Cahn PDE (2) also has dynamically-evolving two-front solutions. We refer to the upcoming Secs. 5.1 and 6.1 – and various simulations shown throughout this article – for more on the dynamically-evolving two-front patterns and on the role played by the stationary patterns.

3.1. The existence of stationary two-front solutions

As the name suggests, two-front solutions consist, to leading order, of two fronts: one which goes (approximately) from the state u_{-}^{ε} to u_{+}^{ε} and the other which goes (approximately) from u_{+}^{ε} to u_{-}^{ε} . We denote the phase/location of the first front by ϕ_{up} and that of the second one by ϕ_{down} . We start with the following Ansatz for two-front solutions, which is based on an expansion around the former jump on the domain $x < 0$ and an expansion around the latter jump on the domain $x > 0$:

$$u_{2\text{-f}}(x; \phi_{\text{up}}, \phi_{\text{down}}) = \begin{cases} u_{\text{up}}(x; \phi_{\text{up}}) + \varepsilon u_1^-(x; \phi_{\text{up}}) + \mathcal{O}(\varepsilon^2) & \text{for } x < 0; \\ u_{\text{down}}(x; \phi_{\text{down}}) + \varepsilon u_1^+(x; \phi_{\text{down}}) + \mathcal{O}(\varepsilon^2) & \text{for } x > 0, \end{cases} \quad (46)$$

where $u_{\text{up/down}}$ are as defined in (15). See also Fig. 8 for a sketch of a stationary two-front solution and Section 3.3 for examples. A key element of the existence proof is to impose continuity of the two components of $u_{2-\varepsilon}$ at the point $x = 0$, as well as continuity of their first derivatives. One notes that, by themselves, the terms which are *prima facie* $\mathcal{O}(1)$ cannot be joined continuously at $x = 0$ for any $\mathcal{O}(1)$ choice of ϕ_{up} and ϕ_{down} , since otherwise two-front solutions would exist for the unperturbed Allen-Cahn equation. Therefore, in order to insure continuity of $u_{2-\varepsilon}$ and its first derivative at $x = 0$, one needs to examine simultaneously the terms which are *prima facie* $\mathcal{O}(1)$ and those which are *prima facie* $\mathcal{O}(\varepsilon)$.

Continuity of the components of (46) at $x = 0$ and continuity of their first derivatives at $x = 0$ hold when

$$\begin{aligned} u_{\text{up}}(0, \phi_{\text{up}}) + \varepsilon u_1^-(0, \phi_{\text{up}}) + \mathcal{O}(\varepsilon^2) &= u_{\text{down}}(0, \phi_{\text{down}}) + \varepsilon u_1^+(0, \phi_{\text{down}}) + \mathcal{O}(\varepsilon^2) \\ u'_{\text{up}}(0, \phi_{\text{up}}) + \varepsilon \partial_x u_1^-(0, \phi_{\text{up}}) + \mathcal{O}(\varepsilon^2) &= u'_{\text{down}}(0, \phi_{\text{down}}) + \varepsilon \partial_x u_1^+(0, \phi_{\text{down}}) + \mathcal{O}(\varepsilon^2). \end{aligned} \quad (47)$$

We set

$$\phi_{\text{up}} = -\frac{|\log \varepsilon|}{2\sqrt{2}} - \ell_{\text{up}}, \quad \phi_{\text{down}} = \frac{|\log \varepsilon|}{2\sqrt{2}} + \ell_{\text{down}}, \quad (48)$$

where ℓ_{up} and ℓ_{down} are $\mathcal{O}(1)$ constants to be determined (noting that only some choices will lead to two-front solutions). We examine each of the terms in (47). By (48), the terms which are *prima facie* $\mathcal{O}(1)$ in (47) are,

$$\begin{aligned} u_{\text{up/down}}(0; \phi_{\text{up/down}}) &= 1 - 2e^{\pm\sqrt{2}\phi_{\text{up/down}}} + \mathcal{O}(\varepsilon) = 1 - 2\sqrt{\varepsilon}e^{-\sqrt{2}\ell_{\text{up/down}}} + \mathcal{O}(\varepsilon) \\ u'_{\text{up/down}}(0; \phi_{\text{up/down}}) &= \pm 2\sqrt{2}\sqrt{\varepsilon}e^{-\sqrt{2}\ell_{\text{up/down}}} + \mathcal{O}(\varepsilon). \end{aligned} \quad (49)$$

Next, the terms which are *prima facie* $\mathcal{O}(\varepsilon)$ in (47) satisfy Eq. (22) with u_h being given by the corresponding one-front u_{up} or u_{down} . A comprehensive study of this inhomogeneous problem is given in App. A. In fact, by stipulating boundedness for u_1^- in the limit $x \rightarrow -\infty$ and for u_1^+ in the limit $x \rightarrow \infty$, we can determine the limit behavior for $u_1^\pm$ at $x = 0$. Using (A.6) and (48), we obtain

$$\begin{aligned} u_1^-(0; \phi_{\text{up}}) &= -\frac{\mathcal{R}_{\text{up}}(\phi_{\text{up}})}{8}e^{-\sqrt{2}\phi_{\text{up}}} + \mathcal{O}(1) = -\frac{1}{\sqrt{\varepsilon}}\frac{\mathcal{R}_{\text{up}}(\phi_{\text{up}})}{8}e^{\sqrt{2}\ell_{\text{up}}} + \mathcal{O}(1); \\ \partial_x u_1^-(0; \phi_{\text{up}}) &= -\frac{\sqrt{2}}{\sqrt{\varepsilon}}\frac{\mathcal{R}_{\text{up}}(\phi_{\text{up}})}{8}e^{\sqrt{2}\ell_{\text{up}}} + \mathcal{O}(1). \end{aligned} \quad (50)$$

Similarly, using (A.6) and (48), one finds

$$\begin{aligned} u_1^+(0; \phi_{\text{down}}) &= \frac{\mathcal{R}_{\text{down}}(\phi_{\text{down}})}{8}e^{\sqrt{2}\phi_{\text{down}}} + \mathcal{O}(1) \\ &= \frac{1}{\sqrt{\varepsilon}}\frac{\mathcal{R}_{\text{down}}(\phi_{\text{down}})}{8}e^{\sqrt{2}\ell_{\text{down}}} + \mathcal{O}(1), \end{aligned} \quad (51)$$

$$\partial_x u_1^+(0; \phi_{\text{down}}) = \frac{\sqrt{2}}{\sqrt{\varepsilon}}\frac{\mathcal{R}_{\text{down}}(\phi_{\text{down}})}{8}e^{\sqrt{2}\ell_{\text{down}}} + \mathcal{O}(1), \quad (52)$$

where $\mathcal{R}_{\text{up/down}}$ are as defined in (26) and (27). Hence, the continuity condition on u implies

$$\begin{aligned} 1 - 2\sqrt{\varepsilon}e^{-\sqrt{2}\ell_{\text{up}}} - \frac{\mathcal{R}_{\text{up}}(\phi_{\text{up}})}{8}e^{\sqrt{2}\ell_{\text{up}}} + \mathcal{O}(\varepsilon) \\ = 1 - 2\sqrt{\varepsilon}e^{-\sqrt{2}\ell_{\text{down}}} + \frac{\mathcal{R}_{\text{down}}(\phi_{\text{down}})}{8}e^{\sqrt{2}\ell_{\text{down}}} + \mathcal{O}(\varepsilon). \end{aligned}$$

Putting this together with a similar condition for the continuity of the derivatives at $x = 0$ (per (47)), one finds the following conditions at $\mathcal{O}(\sqrt{\varepsilon})$:

$$\begin{aligned} 2e^{-\sqrt{2}\ell_{\text{up}}} + \frac{\mathcal{R}_{\text{up}}(\phi_{\text{up}})}{8}e^{\sqrt{2}\ell_{\text{up}}} &= 2e^{-\sqrt{2}\ell_{\text{down}}} - \frac{\mathcal{R}_{\text{down}}(\phi_{\text{down}})}{8}e^{\sqrt{2}\ell_{\text{down}}} \\ 2e^{-\sqrt{2}\ell_{\text{up}}} - \frac{\mathcal{R}_{\text{up}}(\phi_{\text{up}})}{8}e^{\sqrt{2}\ell_{\text{up}}} &= -2e^{-\sqrt{2}\ell_{\text{down}}} - \frac{\mathcal{R}_{\text{down}}(\phi_{\text{down}})}{8}e^{\sqrt{2}\ell_{\text{down}}} \end{aligned} \quad (53)$$

Finally, addition and subtraction of these two conditions and recalling (48) allows us to rearrange them into the following condensed and insightful way:

$$\mathcal{R}_{\text{up}}(\phi_{\text{up}}) = -\mathcal{R}_{\text{down}}(\phi_{\text{down}}) = 16e^{-\sqrt{2}[\ell_{\text{up}} + \ell_{\text{down}}]} = \frac{16}{\varepsilon}e^{-\sqrt{2}[\phi_{\text{down}} - \phi_{\text{up}}]}. \quad (54)$$

Thus, we have shown that the PDE (2) possesses stationary two-front solutions for all parameter values that satisfy the conditions (54) on the Melnikov functions, and the fronts are located near $\phi_{\text{up/down}}$.

Remark 7. In the limit $\ell_{\text{up}} + \ell_{\text{down}} \rightarrow \infty$ (i.e., in the limit in which the two fronts are infinitely far apart from each other), condition (54) reduces to $\mathcal{R}_{\text{up}}(\phi_{\text{up}}) = 0$ and $\mathcal{R}_{\text{down}}(\phi_{\text{down}}) = 0$. This is the same as the existence condition for one-front solutions, recall (23). Thus, in this limit, interactions between fronts are only through their exponentially small tails, and their locations/phases are determined to leading order by the standard phase selection criterion.

3.2. Spectral stability of stationary two-front solutions

To determine the spectral stability of stationary two-front solutions, we observe that eigenvalues in the point spectrum lie close to those of the one-front solutions (see Appendix C). Hence, for sufficiently small ε , only the $\mathcal{O}(\varepsilon)$ eigenvalues (which are near $\lambda_0 = 0$) can possibly lie in the right-half plane, i.e., can possibly make the two-front solution unstable. In general, two such eigenvalues are expected here for two-front solutions. We determine the locations of these small eigenvalues using regular perturbation theory and a combination of techniques from the existence problem in Section 3.1 and the stability analysis of one-front solutions in Section 2.3. Specifically, we use regular perturbation theory with one expansion on $x < 0$ and another on $x > 0$, and we impose continuity of the eigenfunctions and their derivatives at $x = 0$. Furthermore, we observe that, for these small eigenvalues, the solutions to the eigenvalue problem should be multiples of the derivative of the (exact) two-front solutions to (2) to leading order in both regions, i.e., they should be multiples of $u'_{\text{up/down}}(x; \phi_{\text{up/down}})$. However, the multiples can be different in the two regions.

Specifically, we set

$$\begin{aligned} \bar{u}(x; \phi_{\text{up}}, \phi_{\text{down}}) &= \begin{cases} \gamma_- u'_{\text{up}}(x; \phi_{\text{up}}) + \varepsilon \bar{u}_1^-(x; \phi_{\text{up}}, \phi_{\text{down}}) + \mathcal{O}(\varepsilon^2) & \text{for } x < 0; \\ \gamma_+ u'_{\text{down}}(x; \phi_{\text{down}}) + \varepsilon \bar{u}_1^+(x; \phi_{\text{up}}, \phi_{\text{down}}) + \mathcal{O}(\varepsilon^2) & \text{for } x > 0; \end{cases} \\ \lambda &= \varepsilon \bar{\lambda} + \mathcal{O}(\varepsilon^2); \end{aligned} \quad (55)$$

(46) where $\gamma_{\pm}$ are constants. By the linearity of the eigenvalue problem, we are free to choose one of these constants, while the other needs to be determined via continuity. However, we should beware of the possibility that one of the constants is zero, and therefore we wait to specify the value of one of these constants.

Using a similar approach as in Section 2.3, we combine Eq. (2) and the x -derivative of (2) to find to leading order

$$\mathcal{L}_0 \bar{u}_1^\pm = \gamma_{\mp} \left[\pm \bar{\lambda} u'_{\text{up}} + F_x(\pm u_{\text{up}}, \pm u'_{\text{up}}, x) \right]. \quad (56)$$

Stipulating boundedness for $\bar{u}_1^-$ (respectively $\bar{u}_1^+$) in the limit $x \rightarrow -\infty$ (respectively in the limit $x \rightarrow +\infty$), we find $\bar{u}_1^\pm$ using formulae from App. A (after replacing the inhomogeneous terms). As in the existence problem, only the leading order behavior at the point $x = 0$ is important. Here, also by App. A, straightforward computations yield the leading order $\mathcal{O}(\sqrt{\varepsilon})$ terms at the point $x = 0$. In particular, imposing continuity of the two components and their derivatives at $x = 0$, we find the following conditions:

$$\begin{aligned} \sqrt{2} \gamma_- \left[2e^{-\sqrt{2}\ell_{\text{up}}} - \frac{\mathcal{R}_{\text{up}}(\phi_{\text{up}})}{8}e^{\sqrt{2}\ell_{\text{up}}} \right] \\ + \frac{\gamma_-}{8} \left[\bar{\lambda} \|u'_{\text{up}}\|_2^2 + \mathcal{R}'_{\text{up}}(\phi_{\text{up}}) \right] e^{\sqrt{2}\ell_{\text{up}}} \\ = \sqrt{2} \gamma_+ \left[-2e^{-\sqrt{2}\ell_{\text{down}}} - \frac{\mathcal{R}_{\text{down}}(\phi_{\text{down}})}{8}e^{\sqrt{2}\ell_{\text{down}}} \right] \\ - \frac{\gamma_+}{8} \left[\bar{\lambda} \|u'_{\text{up}}\|_2^2 + \mathcal{R}'_{\text{down}}(\phi_{\text{down}}) \right] e^{\sqrt{2}\ell_{\text{down}}}, \end{aligned}$$

and

$$\begin{aligned} -\sqrt{2} \gamma_- \left[2e^{-\sqrt{2}\ell_{\text{up}}} + \frac{\mathcal{R}_{\text{up}}(\phi_{\text{up}})}{8}e^{\sqrt{2}\ell_{\text{up}}} \right] \\ + \frac{\gamma_-}{8} \left[\bar{\lambda} \|u'_{\text{up}}\|_2^2 + \mathcal{R}'_{\text{up}}(\phi_{\text{up}}) \right] e^{\sqrt{2}\ell_{\text{up}}} \\ = -\sqrt{2} \gamma_+ \left[2e^{-\sqrt{2}\ell_{\text{down}}} - \frac{\mathcal{R}_{\text{down}}(\phi_{\text{down}})}{8}e^{\sqrt{2}\ell_{\text{down}}} \right] \\ + \frac{\gamma_+}{8} \left[\bar{\lambda} \|u'_{\text{up}}\|_2^2 + \mathcal{R}'_{\text{down}}(\phi_{\text{down}}) \right] e^{\sqrt{2}\ell_{\text{down}}}. \end{aligned}$$

Hence, using (54) which are the conditions for existence of two-front solutions, and adding and subtracting the above two equations, we find that these conditions reduce to

$$\begin{aligned} \gamma_- \left[\bar{\lambda} \|u'_{\text{up}}\|_2^2 + \mathcal{R}'_{\text{up}}(\phi_{\text{up}}) \right] &= -\gamma_+ \left[\bar{\lambda} \|u'_{\text{up}}\|_2^2 + \mathcal{R}'_{\text{down}}(\phi_{\text{down}}) \right] \\ &= 16\sqrt{2}(\gamma_- - \gamma_+)e^{-\sqrt{2}(\ell_{\text{up}} + \ell_{\text{down}})}. \end{aligned} \quad (57)$$

These are the conditions on the constants γ_- and γ_+ that must be satisfied for nontrivial eigenfunctions to exist.

Analysis of these conditions shows that there are no solutions of (57) with $\gamma_- = 0$ (except for in the limit $\ell_{\text{up}} + \ell_{\text{down}} \rightarrow \infty$; see also Remark 9). Hence, without loss of generality, we may scale the eigenfunctions by setting $\gamma_- = 1$. Then, the system of linear equations has two solutions given by

$$\begin{aligned} (\gamma_+)_{1,2} &= \frac{-\sqrt{2}}{64} e^{\sqrt{2}(\ell_{\text{up}} + \ell_{\text{down}})} \left[\mathcal{R}'_{\text{up}}(\phi_{\text{up}}) - \mathcal{R}'_{\text{down}}(\phi_{\text{down}}) \pm \sqrt{D} \right], \\ \bar{\lambda}_{1,2} &= \frac{1}{\|u'_{\text{up}}\|_2^2} \left[16\sqrt{2} e^{-\sqrt{2}(\ell_{\text{up}} + \ell_{\text{down}})} - \frac{\mathcal{R}'_{\text{up}}(\phi_{\text{up}}) + \mathcal{R}'_{\text{down}}(\phi_{\text{down}})}{2} \mp \frac{\sqrt{D}}{2} \right], \end{aligned} \quad (58)$$

where

$$D := \left(\mathcal{R}'_{\text{up}}(\phi_{\text{up}}) - \mathcal{R}'_{\text{down}}(\phi_{\text{down}}) \right)^2 + 2048 e^{-2\sqrt{2}(\ell_{\text{up}} + \ell_{\text{down}})}. \quad (59)$$

This completes the derivation of the eigenfunctions and eigenvalues.

We summarize the preceding analysis in a theorem – of which a proof is sketched in App. C – and illustrate our findings in two examples in the upcoming subsection.

Theorem 2. Consider the weakly heterogeneous Allen-Cahn Eq. (2) with bounded and sufficiently smooth inhomogeneity $F(U, U_x, x)$. Let the Melnikov functions $\mathcal{R}_{\text{up}}(\phi)$ and $\mathcal{R}_{\text{down}}(\phi)$ be given by (26) and (27). Assume that $(\phi_{\text{up}}, \phi_{\text{down}})$, with $\phi_{\text{up}} < \phi_{\text{down}}$, is a non-degenerate solution of (54) such that $|\mathcal{R}_{\text{up}}(\phi_{\text{up}})| = |\mathcal{R}_{\text{down}}(\phi_{\text{down}})| \gg \varepsilon$. Then, for $0 < \varepsilon \ll 1$ sufficiently small, (2) has a stationary two-front solution $U_{2-f}(x) = u_{2-f}(x; \phi_{\text{up}}, \phi_{\text{down}})$ (46) that is biasymptotic to $u_{\text{f}}^{\pm}(x) = -1 + \mathcal{O}(\varepsilon)$ (9) and that is $\mathcal{O}(\sqrt{\varepsilon})$ close to $u_{\text{up}}(x; \phi_{\text{up}}) + u_{\text{down}}(x; \phi_{\text{down}}) - 1$ (15) (i.e., $\sup_{x \in \mathbb{R}} |U_{2-f}(x) - (u_{\text{up}}(x; \phi_{\text{up}}) + u_{\text{down}}(x; \phi_{\text{down}}) - 1)| = \mathcal{O}(\sqrt{\varepsilon})$).

If it is additionally assumed that the inhomogeneity $F(U, U_x, x)$ is either localized or periodic as a function of x , then the one-front pattern $U_{1-f}(x)$ is spectrally stable if $\lambda_{1,2} = \varepsilon \bar{\lambda}_{1,2} < 0$ (58).

Remark 8. In the special case in which $\mathcal{R}'_{\text{up}}(\phi_{\text{up}}) = \mathcal{R}'_{\text{down}}(\phi_{\text{down}})$, one of the eigenfunctions is symmetric, and the other anti-symmetric. Specifically, in this case, we have

$$\begin{aligned} (\gamma_+)_{1,2} &= +1, & \bar{\lambda}_1 &= -\frac{\mathcal{R}'_{\text{up}}(\phi_{\text{up}})}{\|u'_{\text{up}}\|_2^2}, \\ (\gamma_+)_{2,2} &= -1, & \bar{\lambda}_2 &= -\frac{\mathcal{R}'_{\text{up}}(\phi_{\text{up}}) - 32\sqrt{2}e^{-\sqrt{2}(\ell_{\text{up}} + \ell_{\text{down}})}}{\|u'_{\text{up}}\|_2^2}. \end{aligned} \quad (60)$$

Remark 9. In the limit $\ell_{\text{up}} + \ell_{\text{down}} \rightarrow \infty$ (i.e., in the limit in which the two fronts are infinitely far apart from each other), conditions (57) reduce to $\gamma_- \left[\bar{\lambda} \|u'_{\text{up}}\|_2^2 + \mathcal{R}'_{\text{up}}(\phi_{\text{up}}) \right] = \gamma_+ \left[\bar{\lambda} \|u'_{\text{up}}\|_2^2 + \mathcal{R}'_{\text{down}}(\phi_{\text{down}}) \right] = 0$. Thus, in this limit, there are two eigenfunctions: one corresponds to the one-front eigenfunction of the left front (and to leading order nothing happens at the other front, i.e. $\gamma_+ = 0$), and the other to the one-front eigenvalue of the right front (and $\gamma_- = 0$). Moreover, the eigenvalues also reduce to those found in the stability analysis of the one-front solutions (35). Furthermore, (58) reduces to these values in the limit $\ell_{\text{up}} + \ell_{\text{down}} \rightarrow \infty$, as expected.

3.3. Stationary two-fronts in topographically driven systems and spatially periodic systems

3.3.1. Two-front solutions arising from weak topographical effects

In Section 2.4.1, we derived that $\mathcal{R}_{\text{up}}(\phi) = \mathcal{R}_{\text{down}}(\phi)$ in case (4) in which the driving term $F(U, V, x)$ models the impact of a topography $H(x)$. As a consequence, the existence condition (54) reduces to

$$\int_{-\infty}^{\infty} H'(y + \phi_{\text{up}}) W_h(y) dy = - \int_{-\infty}^{\infty} H'(y + \phi_{\text{down}}) W_h(y) dy = \frac{16}{\varepsilon} e^{-\sqrt{2}[\phi_{\text{down}} - \phi_{\text{up}}]}, \quad (61)$$

((36), (37), (48)). If we again also assume that $H(x) = H(-x)$ and additionally restrict our search for two-front patterns to symmetric patterns, i.e., assume that $\phi = \phi_{\text{down}} = -\phi_{\text{up}} > 0$, then we have

$$\mathcal{R}(\phi) = \int_{-\infty}^{\infty} H'(y + \phi) W_h(y) dy = - \int_{-\infty}^{\infty} H'(-y - \phi) W_h(y) dy$$

$$= - \int_{-\infty}^{\infty} H'(z - \phi) W_h(z) dz.$$

As a result, (61) simplifies to

$$\mathcal{R}(\phi) = -\frac{16}{\varepsilon} e^{-2\sqrt{2}\phi}. \quad (62)$$

In the case of an X -periodic topography $H(x)$, with also $\mathcal{R}(\phi + X) = \mathcal{R}(\phi)$ (39), it immediately follows that there is a critical $\bar{\phi}_{\text{min}} = \mathcal{O}\left(\frac{|\log \varepsilon|}{2\sqrt{2}}\right)$ such that (62) has countably many zeros $\phi_{n,j} \geq \bar{\phi}_{\text{min}}$ (see also Theorem 4 for a generalization to N -front patterns). For the unimodal hill $H_{\text{exp}}(x; \mu)$ (5), we obtain for $\psi = \sqrt{2}\phi \gg 1$ by the leading order behavior of (42) – see also Fig. 7(a) – that for $\mu = 1$ there is a symmetric two-front solution at $-\phi_{\text{up}} = \phi_{\text{down}} = \phi^*(1) \gg 1$ (the 1 referring to the μ -value) with $\phi^*(1)$ the (unique) solution of

$$\varepsilon \sqrt{2} \phi e^{\sqrt{2}\phi} = 1,$$

so that $\phi_* \in (\frac{1}{2} \nu \sqrt{2} |\log \varepsilon|, \frac{1}{2} \sqrt{2} |\log \varepsilon|)$ for any $\nu \in (0, 1)$. In fact, it will follow from the more general asymptotic analysis of Section 5.1 that $\phi^*(\mu) > 0$ exists for any $\mu > 0$ (see Fig. 7(b) for $\mu = \frac{1}{2}$). There is a symmetric two-front pattern for all unimodal hill topographies $H_{\text{exp}}(x; \mu)$. Moreover, it follows by the same arguments that there cannot be symmetric two-front patterns in the case of the unimodal valleys $V_{\text{exp}}(x; 1) = -H_{\text{exp}}(x; 1)$.

To determine the stability of the two-front patterns, we again first consider the more general setting of (4). Again, assume that $H(x)$ is symmetric and that the two-front is symmetric, i.e. $\phi = \phi_{\text{down}} = -\phi_{\text{up}} > 0$. We find

$$\begin{aligned} \mathcal{R}'_{\text{up}}(\phi_{\text{up}}) &= \int_{-\infty}^{\infty} H''(y - \phi) W_h(y) dy = \int_{-\infty}^{\infty} H''(-z - \phi) W_h(-z) dz \\ &= \int_{-\infty}^{\infty} H''(z + \phi) W_h(z) dz = \mathcal{R}'_{\text{down}}(\phi_{\text{down}}) \end{aligned}$$

Thus, this is the situation of the special case discussed in Remark 8. It follows by (54), (60), and (62) that the stability of symmetric two-front patterns at $\phi_{\text{down}} = -\phi_{\text{up}} = \phi^*$ is determined by

$$\bar{\lambda}_1 = -\frac{\mathcal{R}'(\phi^*)}{\|u'_{\text{up}}\|_2^2}, \quad \bar{\lambda}_2 = -\frac{\mathcal{R}'(\phi^*) + 2\sqrt{2}\mathcal{R}(\phi^*)}{\|u'_{\text{up}}\|_2^2}. \quad (63)$$

In the case that $H(x)$ is localized with monotonically decaying tails – as with the explicit hill and valley examples discussed above, it will be shown in Section 5 that both $\mathcal{R}(\phi)$ and $\mathcal{R}'(\phi) \rightarrow 0$ as $\phi \rightarrow \infty$. In fact, it follows in this case that if $\mathcal{R}(\phi) > 0$, respectively $\mathcal{R}(\phi) < 0$, for $\phi \gg 1$, then $\mathcal{R}(\phi)$ decreases to 0, i.e. $\mathcal{R}'(\phi) < 0$, resp. $\mathcal{R}(\phi)$ increases with $\mathcal{R}'(\phi) > 0$ (see Fig. 7). Therefore, since $\mathcal{R}(\phi^*) < 0$ by (62), it follows that $\bar{\lambda}_1 < 0$ and that the symmetric two-front pattern is stable if also $\bar{\lambda}_2 < 0$, that is by (63), if also

$$\mathcal{R}'(\phi^*) + 2\sqrt{2}\mathcal{R}(\phi^*) > 0. \quad (64)$$

Let us analyze this condition. First, we recall that $\mathcal{R}(\phi^*) < 0 < \mathcal{R}'(\phi^*)$. Hence, this condition may hold for certain localized, symmetric topographies $H(x)$. However, for the specific unimodal hill $H_{\text{exp}}(x; \mu)$ (5), we find explicitly from (42) for $\mu = 1$ that

$$\mathcal{R}(\phi^*) = -16\sqrt{2}\phi^* e^{-\sqrt{2}\phi^*} + \text{h.o.t.}, \quad \mathcal{R}'(\phi^*) = 32\phi^* e^{-\sqrt{2}\phi^*} + \text{h.o.t.}$$

This implies that $\bar{\lambda}_2 > 0$ and that the two-front pattern is unstable. Moreover, the case $\mu = 1$ is not exceptional. The analysis in Section 5.1 will show that the symmetric two-front pattern associated to the topography of $H_{\text{exp}}(x; \mu)$ is unstable for all $0 < \mu < 1$, as well (see Theorem 3).

3.3.2. Two-front solutions in systems with spatially-periodic heterogeneities

In this subsection, we consider the explicit case introduced in Section 2.4.2: (3) with $f_j(x)$ ($j = 1, 2, 3$) as in (43). Recalling the Melnikov functions $\mathcal{R}_{\text{up}}(\phi)$ and $\mathcal{R}_{\text{down}}(\phi)$ (44) with A and B given by (45), we observe that (54) provides an explicit condition on the existence of two-front patterns in this case,

$$(A + B) \sin(k\phi_{\text{up}}) = -(A - B) \sin(k\phi_{\text{down}}) = \frac{1}{\varepsilon} e^{-\sqrt{2}[\phi_{\text{down}} - \phi_{\text{up}}]} + \mathcal{O}(\varepsilon). \quad (65)$$

Here, the correction term represents the higher order terms so far neglected in this section (see the discussions in Secs. 5 and 7). Assuming the two fronts are logarithmically far apart, i.e., assuming that there is an $\rho > 0$ such that $0 < \sqrt{2}(\phi_{\text{down}} - \phi_{\text{up}})/|\log \epsilon| \rightarrow \rho$ as $\epsilon \rightarrow 0$ (see Theorem 4), it follows that

$$(A + B) \sin(k\phi_{\text{up}}) = -(A - B) \sin(k\phi_{\text{down}}) = \epsilon^{\rho-1} + \mathcal{O}(\epsilon),$$

which implies that there cannot be any stationary two-fronts for $0 < \rho < 1$. On the other hand, we also see that if $\rho > 1$ (and $A \pm B \neq 0$), there is a stationary two-front $\mathcal{O}(\epsilon^{\rho-1}, \epsilon)$ close to any pair $(n_{\text{up}}(\epsilon)\pi/k, n_{\text{down}}(\epsilon)\pi/k)$ with $(n_{\text{up}}(\epsilon), n_{\text{down}}(\epsilon)) \in \mathbb{Z}^2$ such that $0 < n_{\text{up}}(\epsilon) - n_{\text{down}}(\epsilon) \rightarrow k|\log \epsilon|/\pi\sqrt{2}$ as $\epsilon \rightarrow 0$. We refer to Theorem 4 for a similar statement on the more general case of N -front patterns in (2) with a periodic inhomogeneity (restricted to the topographic subcase (4)). The leading order expressions for the two eigenvalues (58) that determine the stability of these two-front patterns (i.e., with $\rho > 1$) also simplify considerably to

$$\lambda_{1,2} = -\frac{8k\epsilon}{\|u'_{\text{up}}\|_2^2} \left[(A+B) \cos(n_{\text{up}}\pi) + (A-B) \cos(n_{\text{down}}\pi) \pm \sqrt{((A+B) \cos(n_{\text{up}}\pi) - (A-B) \cos(n_{\text{down}}\pi))^2} \right]$$

(see (55), (59)). Hence,

$$\lambda_1 = -\frac{16k(A+B)\epsilon}{\|u'_{\text{up}}\|_2^2} \cos(n_{\text{up}}\pi), \quad \lambda_2 = -\frac{16k(A-B)\epsilon}{\|u'_{\text{up}}\|_2^2} \cos(n_{\text{down}}\pi) \quad (66)$$

to leading order in ϵ . We conclude that a two-front pattern with $\rho > 1$ is stable if both n_{up} and n_{down} are even, i.e., that one out of every quartet of these two-front patterns is stable (see again Theorem 4).

In Section 6.1 – and again in Section 6.2 – we consider the many bifurcations associated to increasing ρ from below $\rho < 1$, for which there are no stationary two-fronts, to $\rho > 1$, for which there are countably infinite. In Section 6.1, this will be done in the setting of the (reduced) two-dimensional ODE that governs the dynamics of the positions $(\phi_1(t), \phi_2(t))$ of an evolving two-front pattern solution of (2) (that will be derived in Section 4). It will also follow from this analysis that any two-front pattern $(\phi_1(t), \phi_2(t))$ will be attracted to a nearby stationary two-front pattern, i.e., that the limiting behavior of all possible $(\phi_1(t), \phi_2(t))$ -front patterns is determined by the family of stable stationary two-front patterns (see Claim 1 for the N -front case).

4. The (slow) dynamics of interacting fronts

In the homogeneous ($\epsilon = 0$) case, the dynamics of multi-front solutions are similar to that for the two-front solutions: fronts attract each other and consequently annihilate in pairs. The only possible stable end states are the trivial states u_- and u_+ or the (stationary) one-fronts u_{up} and u_{down} . By contrast, for $\epsilon \neq 0$, there are many stable and unstable stationary multi-front patterns possible. These stationary N -front solutions are given to leading order by a superposition of N fronts, where now each subsequent front should be of different type. Both the construction and the stability of these stationary multi-fronts can be established along the lines set out in Section 3 for two-front patterns. We refrain from following this approach here (however, see Section 5.3). The richness of the types of stable multi-front patterns exhibited by (2) is indicated for example by the attracting (and thus stable) stationary multi-front patterns presented in Figs. 3, as obtained by numerical simulations.

In the present section, we take a different approach. We have seen that (stationary) front solutions have small, $\mathcal{O}(\epsilon)$ eigenvalues in their associated point spectrum. These small eigenvalues originate from the translation invariance of 1-front solutions to (2) for $\epsilon = 0$ (which leads to an eigenvalue at $\lambda = 0$ with $u'_h(x)$ – the derivative of the unperturbed one-front pattern – as corresponding eigenfunction). When multiple fronts and/or heterogeneous terms (i.e. $\epsilon > 0$) are present, the small eigenvalues in the point spectrum have eigenfunctions that are to leading order determined by these derivatives, see Section 3.2. The dynamics induced by this so-called small spectrum can be captured by projecting the full state space onto a finite-dimensional space spanned by the associated

eigenspaces. Doing so for N -front solutions leads to an N -th order system of ODEs that describes the (slow) evolution of multi-front solutions to (2).

Indeed, the main result of this section is the derivation of the ODE system (78). The ODE system (78) governs the slow migratory movements of the multi-front solutions by tracking changes in the locations $\phi_i(t)$, $i = 1, \dots, N$, (or phase shifts) of the fronts to leading order in some small quantity – typically associated to the inverse of the distance between two successive fronts. The validity of this procedure, thus both of the low-dimensional approximation and of the ODE governing the low-dimensional dynamics of the weakly interacting fronts, has a rigorous basis and can be established by following the methods developed in the literature. For instance, see [26,27].

In this article, we refrain from going into the details of the validity analysis (see also Remark 10). Instead, we focus on the derivation of the governing ODEs describing the interactions of fronts to leading order and on their implications for the dynamics of multi-front patterns. More specifically, we derive the ODEs governing the interactions in N -front patterns in upcoming Section 4.1. Then, in the next sections of this article, we study various aspects of N -front interactions in the two explicit cases that were considered in the previous sections. Namely, in Section 5, we study the impacts of localized topographies, and in Section 6 we study the impacts of spatially periodic driving terms. Moreover, we will return to the above-mentioned stationary multi-front patterns, since they reappear as critical points of the interaction ODEs.

Remark 10. The assumption that the inhomogeneous term $F(U, V, x)$ in (2) is bounded for all x is essential to directly apply the methods of [26,27] to establish the finite dimensional character of front interactions and the governing ODEs. In Section 7, we will discuss observations indicating that the derived interaction equations may also be valid beyond this setting, see also Fig. 2(c) in which $F(U, V, x)$ is given by (4) with unbounded $H'(x)$. Thus, although we refrain from going into the details of the standard case (in which $F(U, V, x)$ is bounded), we motivate at several places in this work that setting up a detailed validity analysis that goes beyond that of the standard case would be an interesting and relevant mathematical enterprise.

4.1. Derivation of the governing ODEs

We consider an N -front pattern with front locations (or phase shifts) $\phi_1(t) < \phi_2(t) < \dots < \phi_N(t)$, see Fig. 9. We anchor the asymptotic analysis by introducing p_j ($j = 1, \dots, N$), with $p_j < p_{j+1}$, and $\ell_j(t)$ similar as in (48), using

$$\phi_j(t) = p_j \frac{|\log \epsilon|}{\sqrt{2}} + \ell_j(t). \quad (67)$$

The exact values of the p_j 's will not show up explicitly in the outcome of the analysis. However, the p_j 's determine the relative positions of the fronts and the driving terms, which will be important (among others) in the case of localized topographies (Section 5). By (67), the distance between two successive fronts $\Delta\phi_j$ is ‘anchored’ by

$$\Delta\phi_j := \phi_{j+1} - \phi_j = (p_{j+1} - p_j) \frac{|\log \epsilon|}{\sqrt{2}} + \ell_{j+1} - \ell_j, \quad (j = 1, \dots, N-1). \quad (68)$$

This is our chosen analytic expression for the requirement that the fronts are ‘sufficiently far apart’. As in (48), we assume in the upcoming analysis that $p_{j+1} - p_j = 1$ and $\ell_j(t) = \mathcal{O}(1)$, so that $\Delta\phi_j = \frac{|\log \epsilon|}{\sqrt{2}}$ to leading order in ϵ (see Fig. 9). We note however that we will go beyond that in our analysis of the ODE dynamics driven by localized topographies in Section 5.

By (68) the evolution of $\phi_j(t)$ will be slow and – anticipating the magnitude of $\frac{d\phi_j(t)}{dt}$ for $\ell_j(t) = \mathcal{O}(1)$ – we introduce $q_j(t)$ by

$$\frac{d\phi_j(t)}{dt} = \epsilon q_j. \quad (69)$$

To study the migratory movement of N -front patterns, i.e., to determine q_j to leading order in ϵ , we subdivide the domain $\mathbb{R}$ into N domains that

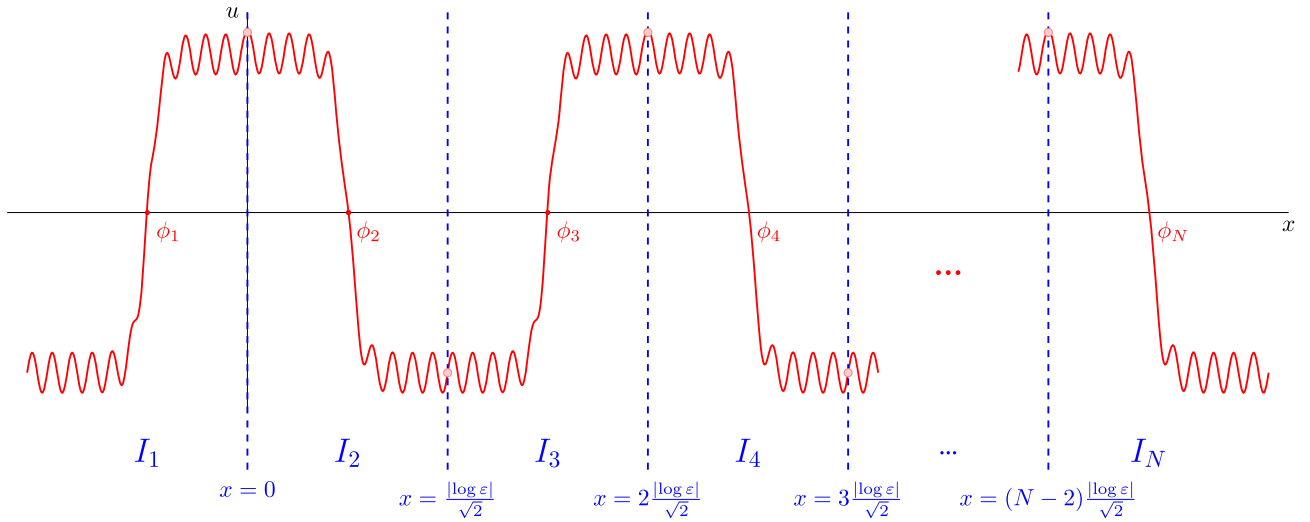


Fig. 9. Sketch of an N -front pattern anchored by (67) with $p_j = j - \frac{3}{2}$, a standard choice. Here, $\phi_1, \dots, \phi_N$ denote the locations – or phase shifts – of the fronts. The domain ($\mathbb{R}$) is divided into the regions $I_1, \dots, I_N$ as given in (70) in and solutions are matched at the boundaries of these regions.

each contains one front. (See Fig. 9.) Specifically, we define the regions

$$\begin{aligned} I_1 &:= \left(-\infty, \frac{1}{2}(p_1 + p_2) \frac{|\log \epsilon|}{\sqrt{2}}\right); \\ I_j &:= \left(\frac{1}{2}(p_{j-1} + p_j) \frac{|\log \epsilon|}{\sqrt{2}}, \frac{1}{2}(p_j + p_{j+1}) \frac{|\log \epsilon|}{\sqrt{2}}\right), \quad (j = 2, \dots, N-1); \\ I_N &:= \left(\frac{1}{2}(p_{N-1} + p_N) \frac{|\log \epsilon|}{\sqrt{2}}, \infty\right). \end{aligned} \quad (70)$$

Following the procedure used in Section 3.1, we now solve (2) separately in each region and match these expressions at the boundaries of the region. Doing so imposes conditions on q_j , which leads to a N -dimensional ODE describing the slow migratory movements of the front locations.

As before, we employ regular expansions to find approximate solutions in each of the regions. In regions with a front connecting (to leading order) u_-^ϵ to u_+^ϵ we set $u_j = u_{\text{up}} + u_{j,1}$ and in regions with a front connecting (to leading order) u_+^ϵ to u_-^ϵ we set $u_j = u_{\text{down}} + u_{j,1}$. For notational clarity, we assume that the odd regions are of the former type and the even regions of the latter type and we comment on other situation at the end of this section. So, we expand u as follows

$$u(x; \phi_1(t), \dots, \phi_N(t)) = \begin{cases} u_{\text{up}}(x; \phi_j(t)) + \epsilon u_{1,j}(x; \phi_j(t)) + h.o.t. & x \in I_j, j \text{ odd}; \\ u_{\text{down}}(x; \phi_j(t)) + \epsilon u_{1,j}(x; \phi_j(t)) + h.o.t. & x \in I_j, j \text{ even}. \end{cases} \quad (71)$$

Substitution in (2) then leads to the following equation at $\mathcal{O}(\epsilon)$:

$$\mathcal{L}_0 u_{1,j} = \begin{cases} -q_j u'_{\text{up}} - F(u_{\text{up}}, u'_{\text{up}}; x) & j \text{ odd}; \\ -q_j u'_{\text{down}} - F(u_{\text{down}}, u'_{\text{down}}; x) & j \text{ even}, \end{cases} \quad (72)$$

where $\mathcal{L}_0$, as defined in (17), is now given by

$$\mathcal{L}_0 = \partial_x^2 + 1 - 3 \tanh^2 \left(\frac{\sqrt{2}}{2} [x - \phi_j(t)] \right).$$

As this is essentially the same equation as before – only with slightly different inhomogeneous terms – the formulas from App. A can be used by replacing the inhomogeneous terms. Using the limit behavior of $u_{\text{up/down}}$, computed in (49) and of $u_{1,j}$, computed in (A.6), we can now match solutions at the boundaries $x = \frac{1}{2}(p_{j+1} + p_j) \frac{|\log \epsilon|}{\sqrt{2}}$ at order $\mathcal{O}(\sqrt{\epsilon})$ – similar to the procedure in Section 3.1.

Performing this procedure leads, for j odd, to the following set of conditions:

$$\begin{aligned} -2e^{\sqrt{2}\epsilon_j} + \frac{1}{8}(B_+) e^{-\sqrt{2}\epsilon_j} &= -2e^{-\sqrt{2}\epsilon_{j+1}} - \frac{1}{8}(B_-)_{j+1} e^{\sqrt{2}\epsilon_{j+1}}; \\ 2e^{\sqrt{2}\epsilon_j} + \frac{1}{8}(B_+) e^{-\sqrt{2}\epsilon_j} &= -2e^{-\sqrt{2}\epsilon_{j+1}} + \frac{1}{8}(B_-)_{j+1} e^{\sqrt{2}\epsilon_{j+1}}. \end{aligned}$$

Here, $(B_\pm)_j$ are defined consistently with (A.5); that is,

$$\begin{aligned} (B_\pm)_j &= \begin{cases} B_{0,j} - \int_{\phi_j}^{\pm\infty} [q_j u_{\text{up}}(z; \phi_j) + F(u_{\text{up}}(z; \phi_j), u'_{\text{up}}(z; \phi_j); z)] u'_{\text{up}}(z; \phi_j) dz & j \text{ odd}; \\ B_{0,j} - \int_{\phi_j}^{\pm\infty} [-q_j u_{\text{up}}(z; \phi_j) + F(-u_{\text{up}}(z; \phi_j), -u'_{\text{up}}(z; \phi_j); z)] u'_{\text{up}}(z; \phi_j) dz & j \text{ even}, \end{cases} \end{aligned} \quad (73)$$

where $B_{0,j}$ are (unknown) constants. By adding and subtracting these conditions, we arrive at the equivalent conditions – that hold for j odd:

$$(B_+)_j = -16e^{-\sqrt{2}(\epsilon_{j+1}-\epsilon_j)}, \quad (B_-)_{j+1} = 16e^{-\sqrt{2}(\epsilon_{j+1}-\epsilon_j)}.$$

Similarly, for j even, the same procedure results – for j even – in the conditions

$$(B_+)_j = 16e^{-\sqrt{2}(\epsilon_{j+1}-\epsilon_j)}, \quad (B_-)_{j+1} = -16e^{-\sqrt{2}(\epsilon_{j+1}-\epsilon_j)}.$$

Combining these sets of conditions, we thus have the following $2N$ conditions:

$$(B_+)_j = \begin{cases} -16e^{-\sqrt{2}(\epsilon_{j+1}-\epsilon_j)} & j \text{ odd} \\ +16e^{-\sqrt{2}(\epsilon_{j+1}-\epsilon_j)} & j \text{ even}, \end{cases} \quad (B_-)_j = \begin{cases} -16e^{-\sqrt{2}(\epsilon_j-\epsilon_{j-1})} & j \text{ odd} \\ +16e^{-\sqrt{2}(\epsilon_j-\epsilon_{j-1})} & j \text{ even}. \end{cases} \quad (74)$$

Note that we have $2N$ conditions here, since we also have $2N$ unknowns $(q_1, \dots, q_N)$ and $(B_{0,1}, \dots, B_{0,N})$. Since we are only interested in the values for q_j , we can subtract $(B_-)_j$ and $(B_+)_j$ to eliminate the other constants. That is, by (73), we have

$$\begin{aligned} (B_-)_j - (B_+)_j &= \begin{cases} \int_{-\infty}^{\infty} [q_j u_{\text{up}}(z; \phi_j) + F(u_{\text{up}}(z; \phi_j), u'_{\text{up}}(z; \phi_j); z)] u'_{\text{up}}(z; \phi_j) dz & j \text{ odd} \\ \int_{-\infty}^{\infty} [-q_j u_{\text{up}}(z; \phi_j) + F(-u_{\text{up}}(z; \phi_j), -u'_{\text{up}}(z; \phi_j); z)] u'_{\text{up}}(z; \phi_j) dz & j \text{ even}. \end{cases} \end{aligned} \quad (75)$$

Also, by (74), we have

$$(B_-)_j - (B_+)_j = (-1)^j 16 \left[e^{-\sqrt{2}(\epsilon_j-\epsilon_{j-1})} - e^{-\sqrt{2}(\epsilon_{j+1}-\epsilon_j)} \right]. \quad (76)$$

Hence, we can combine these two expressions to find N conditions that determine $q_1, \dots, q_N$. Moreover, the integrals can be worked out and related to the Melnikov integrals $\mathcal{R}_{\text{up/down}}$ from (26)–(27). This results in

$$q_j = \begin{cases} \frac{1}{\|u_{\text{up}}\|_2^2} \left[-\mathcal{R}_{\text{up}}(\phi_j) + 16 \left(e^{-\sqrt{2}(\epsilon_{j+1}-\epsilon_j)} - e^{-\sqrt{2}(\epsilon_j-\epsilon_{j-1})} \right) \right], & j \text{ odd} \\ \frac{1}{\|u_{\text{up}}\|_2^2} \left[-\mathcal{R}_{\text{down}}(\phi_j) + 16 \left(e^{-\sqrt{2}(\epsilon_{j+1}-\epsilon_j)} - e^{-\sqrt{2}(\epsilon_j-\epsilon_{j-1})} \right) \right], & j \text{ even}. \end{cases} \quad (77)$$

Recalling $\frac{d\phi(t)}{dt} = \varepsilon q_j$ and noting $\varepsilon e^{-\sqrt{2}(\varepsilon_j+1-\varepsilon_j)} = e^{-\sqrt{2}(\phi_{j+1}-\phi_j)} = e^{-\sqrt{2}\Delta\phi_j}$, we find that the slow dynamics of N -front patterns is determined to leading order in ε by the ODE,

$$\frac{d\phi_j}{dt} = \frac{1}{\|u'_{up}\|_2^2} \left[-\varepsilon R_j(\phi_j) + 16 \left(e^{-\sqrt{2}\Delta\phi_j} - e^{-\sqrt{2}\Delta\phi_{j-1}} \right) \right], \quad (78)$$

where $\Delta\phi_j := \phi_{j+1} - \phi_j$ ($j = 1, \dots, N-1$), $\Delta\phi_0 \rightarrow \infty$, $\Delta\phi_N \rightarrow \infty$, and

$$R_j = \begin{cases} \mathcal{R}_{up} & \text{region } I_j \text{ contains a front going up} \\ \mathcal{R}_{down} & \text{region } I_j \text{ contains a front going down.} \end{cases} \quad (79)$$

We now recall that the above derivation has been performed on N -front configurations in which the odd regions have fronts going up (from u_-^e to u_+^e). The above procedure can also be used on the other kind of N -front configurations – with odd regions that have fronts going down (from u_-^e to u_+^e). The end result is identical.

Hence, in general, the interactions of N -fronts anchored by (67) are governed to leading order by the N -dimensional system of ODEs (78). Also, system (78) confirms that the choice of p_j (in (67)) does not appear explicitly; however, it does appear implicitly through (73). Furthermore, the (neglected) correction terms in (78) are $\mathcal{O}(\varepsilon^2)$, as can be established by a straightforward procedure, under the assumption that the inhomogeneity $F(U, V, x)$ in (2) is bounded. This implies that (78) loses its validity for multi-fronts $(\phi_1, \phi_2, \dots, \phi_N)$ that are spread out so much that the magnitude of the right hand side of (78) has decreased to $\mathcal{O}(\varepsilon^2)$ – see also the extended discussions in Secs. 5 and 7.

Finally, since the system (78) of ODEs captures the dynamics of localized N -front patterns in the PDE (2) governed by the small spectrum, the small eigenvalues associated to specific N -front patterns can also be found via (78). More specifically, stationary N -front patterns in (2) correspond directly to fixed points of (78). By construction, the associated small eigenvalues of the N -front patterns correspond to the eigenvalues of these critical points (see Remark 12 for the case of one-front patterns). If the critical point is stable in (78), then the stability of the corresponding pattern as a solution of (2) immediately follows because all other elements of the spectrum necessarily lie to the left of the imaginary axis and are bounded away from it (see 2.3).

We refrain from summarizing the results of this section in the form of a general theorem that captures the validity of N -front interaction Eqs. (78) – like we did with Theorems 1 and 2 for the main results of Secs. 2 and 3. Indeed, it can be shown by the methods of [26,27] that there is an attracting, approximate N -dimensional manifold $\mathcal{M}_N^0$ for the (semi-)flow generated by (2) and that the flow on $\mathcal{M}_N^0$ is governed by (78) to leading order. However, the manifold $\mathcal{M}_N^0$ will have boundaries, and the nature of (parts of) these boundaries will vary. Indeed, the approximation scheme breaks down if the distance between (some) neighboring fronts becomes either too small or too large. Therefore, a precisely formulated validity result will ask for many technical details – see [43] for an example – and not add to the transparency of the present text, see also Section 7.

Remark 11. In the classical (unforced) case of the Allen-Cahn Eq. (2), $\mathcal{R}_{up/down}(\phi) \equiv 0$. Hence, the slow migration of fronts in a N -front configuration is given by the ODE

$$\frac{d\phi_j}{dt} = \frac{16}{\|u'_{up}\|_2^2} \left(e^{-\sqrt{2}\Delta\phi_j} - e^{-\sqrt{2}\Delta\phi_{j-1}} \right) \quad (80)$$

(see [17–19]). It is clear that this ODE cannot have any fixed points for $N > 1$. (Indeed, recalling that $\Delta\phi_0 = \infty$, we see that $\frac{d\phi_1}{dt}$ cannot vanish.) In fact,

$$\frac{d}{dt}(\phi_N - \phi_1) = -\frac{16}{\|u'_{up}\|_2^2} \left[e^{-\sqrt{2}(\phi_2 - \phi_1)} + e^{-\sqrt{2}(\phi_N - \phi_{N-1})} \right] < 0, \quad (81)$$

which implies that the extent of an N -front pattern – i.e., the distance $\phi_N - \phi_1$ between the two outermost fronts – always decreases. By contrast, in the case of the inhomogeneous Allen-Cahn Eq. (2), the forcing terms may strengthen or counteract the effects that fronts have on each other. Thus, system (78) can have critical points corresponding to stationary multi-front patterns in (2) – as we have already seen in the explicit one-front and two-front examples in the preceding sections. This observation forms the foundation for some of the main results in

the upcoming sections (see Theorems 3 and 4). Moreover, the inhomogeneity may ‘spread out’ an N -front pattern, i.e. may force $\frac{d}{dt}(\phi_N - \phi_1)$ to be positive (as in Fig. 2, see also Section 7).

Remark 12. The dynamics of a solitary front is described by

$$\frac{d\phi_1}{dt} = -\frac{\varepsilon}{\|u'_{up}\|_2^2} \mathcal{R}_{up}(\phi_1), \quad (82)$$

where we note that we considered the case of an ‘up-front’ connecting $u_-^e(x) = -1 + \mathcal{O}(\varepsilon)$ as $x \rightarrow -\infty$ to $u_+^e(x) = 1 + \mathcal{O}(\varepsilon)$ as $x \rightarrow +\infty$ (9), so that $\mathcal{R}_1(\phi_1) = \mathcal{R}_{up}(\phi_1)$ (79). We observe that this agrees fully with the analysis of the (stationary) one-fronts in Section 2. Namely, the existence of a stationary one-front is determined by the zeros ϕ_1^* of $\mathcal{R}_{up}(\phi_1)$, and their stability by $\lambda(\phi_1^*) = \varepsilon \tilde{\lambda}(\phi_1^*) = -\varepsilon \mathcal{R}'_{up}(\phi_1^*) / \|u'_{up}\|_2^2$ (see (35) which uses that $\|u'_{up}\|_2^2 = \frac{2}{3}\sqrt{2}$). By construction, Eq. (82) provides more information: it also shows that a one-front pattern will slowly move towards a zero ϕ_1^* of $\mathcal{R}_{up}(\phi_1)$ that has $\mathcal{R}'_{up}(\phi_1^*) > 0$ (if such a zero exists), i.e., it will move toward and ‘settle down’ (get pinned) at a nearby stable stationary one-front state (if such a state exists). (See also Claim 1 for a more precise statement in the setting of interacting N -front patterns driven by a spatially periodic topography (4).) Moreover, there can also be stable traveling one-front patterns that do not approach a stable stationary state, as shown in Fig. 2(b) (for $t > 4300$) and below in Fig. 21(a) (for $t > 3400$). The speed at which these fronts travel will in general not be constant; in fact, in the case of localized topographies the fronts typically slow down, see however Fig. 22 and the discussion in Section 7.

Remark 13. System (78) can be written as a gradient flow,

$$\frac{d\phi_j}{d\tau} = \frac{1}{\|u'_{up}\|_2^2} \frac{\partial}{\partial \phi_j} \left[-\varepsilon \sum_{i=1}^N \int_0^{\phi_i} \mathcal{R}_j(\varphi) d\varphi + 8\sqrt{2} \sum_{i=1}^{N-1} e^{-\sqrt{2}(\phi_{i+1}-\phi_i)} \right],$$

($j = 1, 2, \dots, N$).

This is common for the interaction equations governing the dynamics of localized structures (see [43]). Thus, for instance, we may conclude that the system of ODEs (78) cannot exhibit periodic dynamics.

5. Front dynamics driven by localized topographies

In this section, we consider the impact of a localized topography on the dynamics of (2), i.e. we consider (2) with $F(U, U_x, x)$ prescribed by a topography $H(x)$ (4) that is localized. As noted in Section 2.4.1, the fact that we assume that $F(U, U_x, x)$ is prescribed by a topography $H(x)$ implies that $\mathcal{R}_{up}(\phi) \equiv \mathcal{R}_{down}(\phi) \equiv \mathcal{R}(\phi)$ (see (38)) with $\mathcal{R}(\phi)$ given by (36). This is in fact the most important reason to restrict ourselves to this class of inhomogeneous perturbations $\varepsilon F(U, U_x, x)$ in (2): continuing to work with in principle distinct Melnikov functions $\mathcal{R}_{up}(\phi)$ and $\mathcal{R}_{down}(\phi)$ would make the ‘bookkeeping’ in the upcoming analysis more complicated. In other words, this choice is – also here – made to simplify the upcoming analysis as much as possible. The nature of our findings indicates that similar results will hold for the general case.

The simplification of the topographic setting can be made explicit by (re-)introducing

$$\psi_j(t) = \sqrt{2}\phi_j(t), \quad S(\psi_j) = \mathcal{R}(\phi_j), \quad \tau = \sqrt{2}t / \|u'_{up}\|_2^2. \quad (83)$$

(see (42)). By this rescaling, the system of N ODEs (78) can be written to leading order in ε as

$$\begin{cases} \frac{d\psi_1}{d\tau} = -\varepsilon S(\psi_1) + 16e^{-(\psi_2 - \psi_1)} \\ \frac{d\psi_2}{d\tau} = -\varepsilon S(\psi_2) + 16(e^{-(\psi_3 - \psi_2)} - e^{-(\psi_2 - \psi_1)}) \\ \vdots \\ \frac{d\psi_j}{d\tau} = -\varepsilon S(\psi_j) + 16(e^{-(\psi_{j+1} - \psi_j)} - e^{-(\psi_j - \psi_{j-1})}) \\ \vdots \\ \frac{d\psi_N}{d\tau} = -\varepsilon S(\psi_N) - 16e^{-(\psi_N - \psi_{N-1})}. \end{cases} \quad (84)$$

We recall that this system has been derived under the condition that $\psi_j < \psi_{j+1}$ with $\psi_{j+1} - \psi_j = \mathcal{O}(|\log \varepsilon|)$ ($j = 1, \dots, N-1$) and that the (neglected) leading order correction terms are of $\mathcal{O}(\varepsilon^2)$.

The main analytical result of this section is Theorem 3 on the existence and stability of stationary N -front patterns in the inhomogeneous Allen-Cahn Eq. (2) driven by a general localized topography $H(x)$ – see Section 5.3. The analysis in essence reduces to an investigation of the existence and stability of critical points of system (84) (with $S(\psi)$ given by (36) and (83)) – see Appendix D. The nature of the decay of topography $H(x)$ plays a crucial role in both the statement and the proof of

Theorem 3, not only directly but also through its interaction with the leading order corrections to (84). To build our understanding systematically and to make the analysis more transparent, we will first study the special cases $N = 2$ (Section 5.1) and $N = 3, 4$ (Section 5.2). In these sections, we will simultaneously consider the dynamics of the interacting fronts in two-, three-, and four-front patterns, i.e., the dynamics exhibited by system (84) (for $N = 2, 3, 4$) beyond just its critical points. This will also yield general insights and conjectures about the dynamics of N -front patterns, as will also be discussed in Section 5.3.

The reason why the decay properties of the topography $H(x)$ play such a crucial role in the upcoming analysis is that the limiting behavior of $H(x)$ – or more precisely of $H'(x)$ – as $x \rightarrow \pm\infty$ directly affects the limiting behavior of $S(\psi)$ as $\psi \rightarrow \pm\infty$. In fact, we first provide two technical lemmas in which the relation between the decay behavior of $H'(x)$ and that of $S(\psi)$ is precisely described. These are valid for general topographies $H(x)$ (that attain $\mathcal{O}(1)$ values for $x = \mathcal{O}(1)$), i.e. $H(x)$ need not be symmetric here, nor given by (5). However, we do assume that the derivative of the topography $H'(x)$ has uniformly decaying tails and that the rates of decay are the same for $x \rightarrow \pm\infty$ (see (85), (90) for more explicit statements). Both assumptions are made for simplicity, and we focus on two specific types of decay: exponential decay (Lemma 1) and algebraic decay (Lemma 2).

Lemma 1 (Exponential Decay). *Assume that there is a constant $\mu > 0$ and expressions $h_{\pm} = h_{\pm}(\mu) \neq 0$ such that*

$$\lim_{x \rightarrow \pm\infty} e^{\pm\mu\sqrt{2}x} H'(x) = h_{\pm}(\mu). \quad (85)$$

If $0 < \mu < 1$, then

$$\lim_{\psi \rightarrow \pm\infty} e^{\pm\mu\psi} S(\psi) = h_{\pm}(\mu) w_{\pm}(\mu), \quad (86)$$

with

$$w_{\pm}(\mu) := \int_{-\infty}^{\infty} e^{\mp\mu\sqrt{2}y} W_h(y) dy \quad (> 0); \quad (87)$$

recall (37). Finally, if $\mu > 1$, then

$$\lim_{\psi \rightarrow \pm\infty} e^{\pm\mu\psi} S(\psi) = 4\tilde{h}_{\pm}(\mu) = 4 \int_{-\infty}^{\infty} e^{\sqrt{2}z} H'(z) dz, \quad (88)$$

which implies that $S(\psi)$ cannot decay faster than $e^{\mp\mu\psi}$ as $\psi \rightarrow \pm\infty$, independent of the decay rate of $H'(x)$.

As the parameter μ tends to $\mu = 1$, the decay behavior as indicated by the lemma breaks down in the following way: the function $w_{\pm}(\mu) \rightarrow \infty$ as $\mu \uparrow 1$, since by (15) and (37)

$$W_h(y) = 4 \frac{(1 - e^{-\sqrt{2}y})^2}{(1 + e^{-\sqrt{2}y})^4} e^{-\sqrt{2}y} = 4 \frac{(1 - e^{\sqrt{2}y})^2}{(1 + e^{\sqrt{2}y})^4} e^{\sqrt{2}y}. \quad (89)$$

Moreover, $\tilde{h}_{\pm}(\mu) \uparrow \infty$ as $\mu \downarrow 1$, since the integral in (88) diverges as $\mu \downarrow 1$, by the assumed decay behavior of $H'(x)$. Furthermore, for the explicit example of the unimodal hill $H_{\text{exp}}(x; \mu)$ (5), the parameter μ in the definition of H_{exp} plays the same role as defined in (85) – for this example $h_{\pm}(\mu) = \mp 4\sqrt{2}\mu$. Moreover, as $\mu = 1$, the explicit expression (42) associated with $H_{\text{exp}}(x; 1)$ – with $\mathcal{R}_{\text{exp}}(\phi; \mu) = S_{\text{exp}}(\psi; \mu)$ (83) – indicates that

$$\lim_{\psi \rightarrow \infty} \frac{e^{\psi} S_{\text{exp}}(\psi; 1)}{\psi} = -16\sqrt{2} = h_+(1)w_+(1) = (-4\sqrt{2}) \times 4,$$

i.e., for $\mu = 1$, $S_{\text{exp}}(\phi)$ decays as $-16\sqrt{2}\psi e^{-\psi}$ for $\psi \rightarrow \infty$.

The case of algebraic decay is simpler.

Lemma 2 (Algebraic Decay). *Assume that there is a constant $p \in \mathbb{R}$ and $\tilde{h}_{\pm} = \tilde{h}_{\pm}(p) \neq 0$ such that*

$$\lim_{x \rightarrow \pm\infty} |x|^p H'(x) = \tilde{h}_{\pm}(p), \quad (90)$$

then

$$\lim_{\psi \rightarrow \pm\infty} |\psi|^p S(\psi) = \frac{1}{3} 2^{\frac{1}{2}(p+3)} \tilde{h}_{\pm}(p). \quad (91)$$

Note that we have thus allowed for topographies $H(x)$ that may also grow algebraically, see especially Section 7. Furthermore, for the unimodal topography with algebraically decaying/growing tails $H_{\text{alg}}(x; p)$ (5), we have $\tilde{h}_{\pm}(p) = \pm(1-p)$ – recall also (5).

Proofs of Lemmas 1 and 2. The derivations of (86), (88), and (91) follow from a number of straightforward observations, of which we only present the main lines. Moreover, we only consider the limits $x \rightarrow +\infty$, noting that the limits $x \rightarrow -\infty$ follow in a similar vein.

First, for $0 < \mu < 1$, we obtain, using (36), that

$$\begin{aligned} \lim_{\psi \rightarrow \infty} e^{\psi} S(\psi) &= \lim_{\psi \rightarrow \infty} \int_{-\infty}^{\infty} e^{\psi} H'(y + \frac{1}{2}\sqrt{2}\psi) W_h(y) dy \\ &= \int_{-\infty}^{\infty} \left(\lim_{\psi \rightarrow \infty} \left[e^{\mu\sqrt{2}(y + \frac{1}{2}\sqrt{2}\psi)} H'(y + \frac{1}{2}\sqrt{2}\psi) \right] e^{-\mu\sqrt{2}y} W_h(y) \right) dy \\ &= h_+(\mu) \int_{-\infty}^{\infty} e^{-\mu\sqrt{2}y} W_h(y) dy. \end{aligned}$$

Then, for $\mu > 1$, we obtain by (89),

$$\begin{aligned} \lim_{\psi \rightarrow \infty} e^{\psi} S(\phi) &= \int_{-\infty}^{\infty} \left(e^{\sqrt{2}z} H'(z) \lim_{\psi \rightarrow \infty} \left[e^{-\sqrt{2}(z - \frac{1}{2}\sqrt{2}\psi)} W_h(z - \frac{1}{2}\sqrt{2}\psi) \right] \right) dz \\ &= 4 \int_{-\infty}^{\infty} e^{\sqrt{2}z} H'(z) dz. \end{aligned}$$

Secondly, for the case of algebraic decay, we have

$$\begin{aligned} \lim_{\psi \rightarrow \infty} e^{\psi} S(\psi) &= \int_{-\infty}^{\infty} \left(\lim_{\phi \rightarrow \infty} \left[(y + \frac{1}{2}\sqrt{2}\psi)^p H'(y + \frac{1}{2}\sqrt{2}\psi) \frac{\psi^p}{(y + \frac{1}{2}\sqrt{2}\psi)^p} W_h(y) \right] \right) dy \\ &= \tilde{h}_+(p) \int_{-\infty}^{\infty} \left(\lim_{\psi \rightarrow \infty} \left[\frac{\sqrt{2}^p}{(1 + \frac{\sqrt{2}y}{\psi})^p} W_h(y) \right] \right) dy \\ &= 2^{\frac{1}{2}p} \tilde{h}_+(p) \int_{-\infty}^{\infty} W_h(y) dy = \frac{1}{3} 2^{\frac{1}{2}(p+3)} \tilde{h}_+(p). \end{aligned}$$

□

5.1. Interacting two-fronts

The leading order interaction ODE for two-front patterns in (2) with driving term (4) (that approach $u_{\pm}(x) = -1 + \mathcal{O}(\varepsilon)$ (9) as $x \rightarrow \pm\infty$) is given by

$$\begin{cases} \frac{d\psi_1}{dt} = -\varepsilon S(\psi_1) + 16e^{-(\psi_2 - \psi_1)} \\ \frac{d\psi_2}{dt} = -\varepsilon S(\psi_2) - 16e^{-(\psi_2 - \psi_1)}, \end{cases} \quad (92)$$

with the additional condition that $\psi_1(t) < \psi_2(t)$. A priori, one expects that the dynamics of two-dimensional system (92) are dominated by its critical points $(\tilde{\psi}_1, \tilde{\psi}_2)$. These satisfy

$$S(\tilde{\psi}_1) = -S(\tilde{\psi}_2) = \frac{16}{\varepsilon} e^{-(\tilde{\psi}_2 - \tilde{\psi}_1)}, \quad (93)$$

which coincides with (54). The $\mathcal{O}(1/\varepsilon)$ factor on the right hand side implies that $\tilde{\psi}_2 - \tilde{\psi}_1$ must be asymptotically large. By the assumptions on $H(x)$, $S(\psi)$ does not explicitly depend on ε and thus varies on the $\mathcal{O}(1)$ scale. This implies that either both $|\tilde{\psi}_1|, |\tilde{\psi}_2| \gg 1$ or only one of these. In other words, there are three kinds of (potential) critical points in (92):

- critical points of the first kind have $|\tilde{\psi}_j| \gg 1$ and, by (86), $|S(\tilde{\psi}_j)| \ll 1$ ($j = 1, 2$).
- critical points of the second kind have $\tilde{\psi}_1 = \mathcal{O}(1)$ and $\tilde{\psi}_2 \gg 1$.
- critical points of the third kind have $\tilde{\psi}_1 \ll -1$ and $\tilde{\psi}_2 = \mathcal{O}(1)$.

In Section 5.2, we will introduce a fourth kind, and in Section 5.3 we will establish that all possible stationary N -front patterns can be seen as extensions of these four fundamental kinds (see Theorem 3).

We start by searching for critical points of the first kind. For this, we introduce $v_{\pm} > 0$ and $\tilde{\varepsilon}_{\pm}$ and set

$$\tilde{\psi}_1 = -v_- |\log \varepsilon| + \tilde{\varepsilon}_- \ll -1, \quad \tilde{\psi}_2 = v_+ |\log \varepsilon| + \tilde{\varepsilon}_+ \gg 1 \quad (94)$$

so that (for topographies $H(x)$ that decay exponentially), we have to leading order (by (86)) that

$$\begin{aligned} 0 < \mu < 1 : S(\tilde{\psi}_1) &= \varepsilon^{\mu v_-} h_- w_- e^{\mu \tilde{\varepsilon}_-}, \quad S(\tilde{\psi}_2) = \varepsilon^{\mu v_+} h_+ w_+ e^{-\mu \tilde{\varepsilon}_+}, \\ e^{-(\tilde{\psi}_2 - \tilde{\psi}_1)} &= \varepsilon^{v_- + v_+} e^{-(\tilde{\varepsilon}_+ - \tilde{\varepsilon}_-)}, \end{aligned} \quad (95)$$

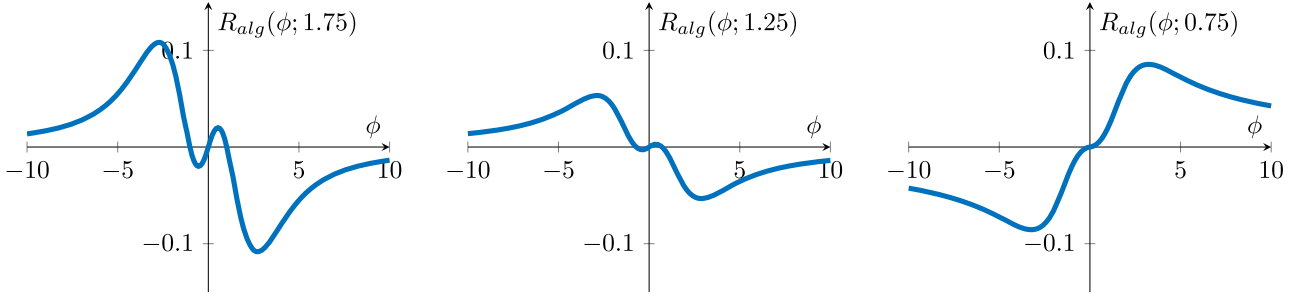


Fig. 10. The Melnikov function $R(\phi; p)$ for the topography $H_{\text{alg}}(x; p)$ of a symmetrical hill with algebraically decaying tails (5) (see (36)). By construction, p determines the leading order algebraic decay as in (90) (a) $p = 1.75$, (b) $p = 1.25$, (c) $p = .75$ (see Fig. 7).

$$\mu > 1 : S(\tilde{\psi}_1) = 4\epsilon^{\nu_-} \hat{h}_- e^{\tilde{\ell}_-}, S(\tilde{\psi}_2) = 4\epsilon^{\nu_+} \hat{h}_+ e^{-\tilde{\ell}_+}, \\ e^{-(\tilde{\psi}_2 - \tilde{\psi}_1)} = \epsilon^{\nu_- + \nu_+} e^{-(\tilde{\ell}_+ - \tilde{\ell}_-)}.$$

Here, we have suppressed the explicit dependence on μ in the functions $h_{\pm}$, $w_{\pm}$ and $\hat{h}_{\pm}$. Hence, by (93) this implies that for the critical points we need to take $\nu_- = \nu_+ := \nu$ and for $0 < \mu < 1$, $1 + \mu\nu = 2\nu$, i.e., $\nu = 1/(2 - \mu)$ while $\nu = 1$ for $\mu > 1$ (see below for a brief discussion of the case $\mu = 1$). It also follows that it is necessary for the existence of these kinds of critical points that $h_+, \hat{h}_+ < 0 < h_-, \hat{h}_-$ and that there exists a unique critical point if this condition holds. Note that the fact that $h_+, \hat{h}_+ < 0 < h_-, \hat{h}_-$ is necessary also implies that there cannot be stationary points $(\tilde{\phi}_1, \tilde{\phi}_2)$ with $(\tilde{\phi}_1, \tilde{\phi}_2)$ on the same tail of $S(\psi)$, i.e., with $\tilde{\phi}_1 \ll \tilde{\phi}_2 \ll -1$ or $1 \ll \tilde{\phi}_1 \ll \tilde{\phi}_2$. Note also that the symmetric critical point associated to the symmetrical topographies $H(x)$ considered in Section 3.3.1 coincides with the present critical point of the first kind.

At this point it also becomes clear that it may be natural to adapt the ‘anchor’ points (67) of the analysis by which system (92) was derived to the (interacting) patterns under consideration. For instance, for $0 < \mu < 1$, the choices $p_2 = \nu_+ = 1/(2 - \mu) = -p_1$ are self-evident (where we note that the factor $1/\sqrt{2}$ is incorporated in the definition of ψ).

More importantly, this is the first time we need to realize the potential impact of the neglected higher order effects in the derivation of the interaction ODEs (78). These higher order terms are represented by the correction terms $\epsilon^2 R_2^{j,2}(\psi_1, \psi_2; \epsilon)$, with $j = 1, 2$ corresponding to the correction to the $\frac{d\psi_j}{d\tau}$ equation. Moreover, it follows by standard techniques that, if the inhomogeneous term $F(U, V, x)$ in (2) is bounded (for $x \in \mathbb{R}$) and if $\psi_{1,2}$ remain bounded (in the appropriate norms), then there is a $C > 0$ such that $|R_2^{j,2}(\psi_1, \psi_2; \epsilon)| < C$ ($j = 1, 2$). In other words, the higher order terms are of $\mathcal{O}(\epsilon^2)$, which is the expected natural behavior of a regular perturbation expansion. However, since we did (and will) not evaluate (the leading order approximation of) $R_2^{j,2}(\psi_1, \psi_2; \epsilon)$, it is necessary to only consider critical points $(\tilde{\psi}_1, \tilde{\psi}_2)$ with $\epsilon |S(\tilde{\psi}_{1,2})| e^{-(\tilde{\psi}_2 - \tilde{\psi}_1)} \gg \epsilon^2$ (since the terms $R_2^{j,2}(\psi_1, \psi_2; \epsilon)$ may have a leading order effect if these conditions do not hold). Hence, we need to impose for $0 < \mu < 1$ that $1 + \mu\nu = 2\nu = \frac{2}{2-\mu} < 2$, i.e., that $\mu < 1$, which is consistent. However, for $\mu > 1$ we find $1 + \nu = 2\nu = 2 < 2$ and conclude that the $R_2^{j,2}(\psi_1, \psi_2; \epsilon)$ terms cannot be neglected. By working out the details of the case $\mu = 1$ in (85), we find that exactly the same holds for $\mu = 1$. Since we do not intend here to go deeper into the analysis of the impact of these higher order terms – see also Section 7 – we therefore impose the assumption that $0 < \mu < 1$ in our analysis of the critical points of the first kind.

The second kind of critical points have $\tilde{\psi}_1 = \mathcal{O}(1)$ and $\tilde{\psi}_2 \gg 1$, which implies that $|S(\tilde{\psi}_2)| \ll 1$. We note by (93) that $\tilde{\psi}_1$ must be asymptotically close to a zero of $S(\psi)$. Thus, we assume that there is a ψ_* such that $S(\psi_*) = 0$ and $S'(\psi_*) \neq 0$ (i.e., we assume that the zero ψ_* is non-degenerate); for example, see Fig. 7 for $S(\psi)$ associated to the unimodal hill (5), which has three such zeros for $\mu \in (\mu_{\text{PF}}, 1)$. Therefore, we change (94) into

$$\tilde{\psi}_1 = \psi_* + \epsilon^{\nu_1} \tilde{\psi}_1, \quad \tilde{\psi}_2 = \nu_2 |\log \epsilon| + \tilde{\ell}_2. \quad (96)$$

We deduce for $0 < \mu < 1$ (using similar reasoning as in the previous case) that the critical points have

$$\nu_1(\mu) = \mu/(1 - \mu), \quad \nu_2(\mu) = 1/(1 - \mu). \quad (97)$$

Also, we deduce that $\tilde{\psi}_1$ and $\tilde{\ell}_2$ can be solved uniquely (in terms of $S'(\psi_*) \neq 0$) if $h_+ < 0$ (note that for critical points of this second kind it is thus not necessary that $h_- > 0$ as is necessary for critical points of the first kind). For $\mu > 1$, we observe from (86) and the second identity of (93) that

$$\mathcal{O}(\epsilon^{1+\nu_2}) = |\epsilon S(\tilde{\psi}_2)| = 16e^{-(\tilde{\psi}_2 - \tilde{\psi}_1)} = \mathcal{O}(\epsilon^{\nu_2}),$$

which implies that there cannot be critical points of the second kind in case $\mu > 1$.

We again impose that the impact of the correction terms $\epsilon^2 R_2^{j,2}(\psi_1, \psi_2; \epsilon)$ ($j = 1, 2$) is indeed of higher order. Thus we find that this leading order outcome only is valid if $1 + \nu_1 = 1 + \mu\nu_2 < 2$, i.e., if $0 < \mu < \frac{1}{2}$. Note that this does not necessarily imply that there cannot be stationary two-front patterns of the second kind for $\mu \geq \frac{1}{2}$. In fact, it is not unlikely that these patterns exist for (some) $\mu \in [\frac{1}{2}, 1)$, since it is natural to expect that $|R_2^{j,2}(\psi_1, \psi_2; \epsilon)| \ll 1$ for $\tilde{\psi}_2 \gg 1$. However, to investigate this, one would need to develop a higher order analysis of (92), which we refrain from doing here.

Determination of the third kind of critical point follows completely along the same lines. We find that $0 < \mu < \frac{1}{2}$. However, in this case, (only) $h_- > 0$ appears as a necessary condition for existence.

We conclude that in a case like that of Fig. 7(a) for the unimodal hill $H_{\text{exp}}(x; \mu)$ (5) with $\mu \in (\mu_{\text{PF}}, 1)$ – that also has $h_+ < 0 < h_-$ – system (92) has in total $\mathcal{N}(2) = 7$ critical points: one of the first kind, three of the second, and three of the third kind. Moreover, we also conclude that there can be no critical points if neither $h_+ < 0$ nor $h_- > 0$ and that there are as many stationary two-fronts as there are stationary one-fronts, i.e., zeros of $S(\psi)$, if only $h_+ < 0$ – in which case they are all of the second kind – or only $h_- > 0$ (third kind).

However, all critical points are unstable, as we shall show in the rest of this subsection. We define

$$E_j := 16e^{-(\tilde{\psi}_j + 1 - \tilde{\psi}_j)}. \quad (98)$$

Hence, in the first case,

$$E_1 = 16\epsilon^{\frac{2}{2-\mu}} e^{-(\tilde{\ell}_+ - \tilde{\ell}_-)} = \epsilon^{\frac{2}{2-\mu}} |h_+| w_+ e^{-\mu\tilde{\ell}_+}$$

((93), (94), (95)). By the exponential nature of $S(\psi)$ (for $|\psi| \gg 1$) (86), we conclude that the spectral stability of the critical points $(\tilde{\psi}_1, \tilde{\psi}_2)$ is (to leading order) determined by the eigenvalues of the Jacobian matrix

$$\begin{pmatrix} -\epsilon S'(\tilde{\psi}_1) + E_1 & -E_1 \\ -E_1 & -\epsilon S'(\tilde{\psi}_2) + E_1 \end{pmatrix} = \epsilon^{\frac{2}{2-\mu}} |h_+| w_+ e^{-\mu\tilde{\ell}_+} \begin{pmatrix} 1 - \mu & -1 \\ -1 & 1 - \mu \end{pmatrix}. \quad (99)$$

Thus, $(\tilde{\psi}_1, \tilde{\psi}_2)$ is a saddle point with (leading order) eigenvalues,

$$\lambda_1(\mu) = -\mu |h_+| w_+ e^{-\mu\tilde{\ell}_+} = -\mu E_1 < 0, \quad \lambda_2(\mu) = (2 - \mu) |h_+| w_+ e^{-\mu\tilde{\ell}_+} = (2 - \mu) E_1 > 0 \quad (100)$$

(where we recall that we needed to assume in the existence analysis that $0 < \mu < 1$). A similar approach yields that the stability of a critical point $(\tilde{\psi}_1, \tilde{\psi}_2)$ of the second kind is determined by the eigenvalues of the Jacobian matrix

$$E_1 \begin{pmatrix} 1 - \frac{1}{\tilde{\psi}_1} \varepsilon^{-\frac{\mu}{1-\mu}} & -1 \\ -1 & 1 - \mu \end{pmatrix}, \quad (101)$$

with

$$E_1 = 16\varepsilon^{\frac{1}{1-\mu}} e^{\psi_* - \tilde{\varepsilon}_2}.$$

The eigenvalues are given to leading order by

$$\lambda_1 = -16\varepsilon^{\frac{1}{1-\mu}} e^{\psi_* - \tilde{\varepsilon}_2} = -\varepsilon S'(\psi_*), \quad \lambda_2 = 16(1 - \mu)\varepsilon^{\frac{1}{1-\mu}} e^{\psi_* - \tilde{\varepsilon}_2} = (1 - \mu)E_1 \quad (102)$$

(93), (96). Hence, $\lambda_2 > 0$ (again), so that $(\tilde{\psi}_1, \tilde{\psi}_2)$ is unstable. Moreover, the sign of λ_1 is determined by (the sign of) $S'(\psi_*)$. Hence, by (35), $(\tilde{\psi}_1, \tilde{\psi}_2)$ again is a saddle of the flow induced by (92) if the one-front pattern located at $\psi = \psi_*$ is stable; otherwise it is an unstable node. Essentially the same analysis yields the same instability results for the critical points of the third kind.

We refer to Section 5.3, Theorem 3, and App. D (applied to the case of $N = 2$ -front patterns) for a precise statement of this existence and stability result (and its proof).

We conclude that there are no stable stationary two-fronts in this case. As we have already seen in Fig. 1(b), this does not imply that the two-front dynamics of (2) with a localized topography is similar to that of the standard homogeneous Allen-Cahn Eq. (1). Fronts in (2) with localized topography may move (exponentially) slowly to $\pm\infty$, while the two fronts of a two-front pattern in (1) attract each other (see (92) with $S(\psi) \equiv 0$) and subsequently annihilate each other, see Fig. 1(a).

In the richest case with $h_- > 0$, $h_+ < 0$, and $S'(\psi_*) > 0$ (so that the stationary one-front pattern associated to ψ_* is stable), there are various scenarios – depending on the initial conditions in (92), or, equivalently, the initial positions of the two fronts:

- (i) The ψ_1 -front travels towards $-\infty$, the ψ_2 -front travels towards $+\infty$ (see Figs. 1(b), 11(b)).
- (ii) One of the fronts moves to a stable ψ_* one-front position, the other to $+\infty$ or to $-\infty$ (see Fig. 20(d)).
- (iii) The two fronts attract and annihilate each other (see Fig. 11(d)).
- (iv) The two fronts travel as a(n evolving) pair – or ‘train’ – to either $+\infty$ or $-\infty$.

(Note that the above given references to numerical simulations mostly concern simulations of (2) with localized topographies (4) with algebraically decaying tails and that the prescribed behavior often only occurs after several pairs of fronts of the initial N -front patterns (with $N > 2$) have merged.)

By considering the evolution of a solitary front $\psi(\tau)$ that starts out with $\psi(0) = \psi_0 \gg 1$, i.e., at the positive tail of $S(\psi)$, it is natural to conjecture that the fourth scenario cannot occur. Assuming that $h_+ < 0$, we see that the evolution of $\psi(\tau)$ is given (to leading order) by

$$\frac{d\psi}{d\tau} = -\varepsilon S(\psi) = \varepsilon |h_+| w_+ e^{-\mu\psi} \quad (103)$$

so that

$$\psi(\tau) = \frac{1}{\mu} \log(\varepsilon |h_+| w_+ \tau + e^{\mu\psi_0}). \quad (104)$$

As a consequence, the leading order distance between two non-interacting fronts, $\tilde{\psi}(\tau) < \psi(\tau)$ that start out sufficiently far from each other, evolves as

$$\psi(\tau) - \tilde{\psi}(\tau) = \frac{1}{\mu} \log\left(\frac{\varepsilon |h_+| w_+ \tau + e^{\mu\psi_0}}{\varepsilon |h_+| w_+ \tau + e^{\mu\tilde{\psi}_0}}\right) = \frac{e^{\mu\psi_0} - e^{\mu\tilde{\psi}_0}}{\varepsilon \mu |h_+| w_+ \tau} + \text{h.o.t. for } \tau \gg \frac{e^{\mu\psi_0}}{\varepsilon}. \quad (105)$$

Thus, on a (very) long timescale, the $\tilde{\psi}$ -front is expected to catch up with its forerunner ψ . Moreover, when the distance between ψ and $\tilde{\psi}$ becomes sufficiently ‘small’, the additional attractive front-interaction effect of the homogeneous Allen-Cahn equation will kick in, so that the distance between ψ and $\tilde{\psi}$ decreases even faster.

However, this reasoning implicitly assumes that the impact of the higher order terms $\varepsilon^2 R_2^{j,2}(\psi_1, \psi_2; \varepsilon)$ will always remain higher order, and thus negligible. Moreover, the simulations shown in Fig. 2(c) and in (upcoming) Fig. 22(c) indicate that – on long temporal and spatial scales – persistent traveling ‘trains’ consisting of slowly dispersing two-, three-, and five-front patterns may exist. However, these simulations were performed for (2) with topographical inhomogeneities (4) $H(x) = H_{\text{alg}}(x; p)$ with $p \in (0, 1)$, so that the term $H'(x)$ in (4) is not bounded (recall (5)). Thus, it is unclear whether or not there are circumstances for which the ‘trains’ mentioned under (iv) can exist. We refer to Section 7 for a discussion.

Finally, we note that if one of the conditions $h_- > 0$, $h_+ < 0$, $S'(\psi_*) > 0$ does not hold, some of the above scenarios are no longer possible or have to be adapted. For example, if $h_+ > 0$ (but still $h_- > 0$, $S'(\psi_*) > 0$), then fronts cannot travel to $+\infty$, so that scenario (i) will not occur and scenarios (ii) and (iv) only take place with a front(s) moving to $-\infty$. Moreover, if neither of these conditions holds, then the localized topography has no impact on the final outcome of the front interactions, but it may speed up the coarsening process, see Section 7 and Fig. 20.

5.2. Three- and four-front dynamics

As a bridge towards the general case of interacting N -fronts of Section 5.3, we now consider several aspects of the interactions of three-fronts and four-fronts, where we again only consider exponentially decaying topographies $H(x)$ for simplicity. Our main goal is to indicate by way of the $N = 3, 4$ -example systems how all stationary N -front patterns can be seen as extensions of four kinds of basic patterns, the three introduced in the previous section and a fourth one (that can be seen as a combination of the second and third). As in the previous section, it can be shown that the higher order terms of (84) can (a priori) not be neglected for $\mu > 1$. Therefore, in this section, we will only consider the case $0 < \mu < 1$.

By (84), the (leading order) interaction ODE for three-front patterns (that approach $u_\pm^\varepsilon(x) = \pm 1 + \mathcal{O}(\varepsilon)$ (9) as $x \rightarrow \pm\infty$) in (2) with driving term (4) is given by

$$\begin{cases} \frac{d\psi_1}{d\tau} = -\varepsilon S(\psi_1) + 16e^{-(\psi_2 - \psi_1)} \\ \frac{d\psi_2}{d\tau} = -\varepsilon S(\psi_2) + 16(e^{-(\psi_3 - \psi_2)} - e^{-(\psi_2 - \psi_1)}) \\ \frac{d\psi_3}{d\tau} = -\varepsilon S(\psi_3) - 16e^{-(\psi_3 - \psi_2)}. \end{cases} \quad (106)$$

Similarly, the system of ODEs governing the front locations in four-front patterns (homoclinic to $u_\pm^\varepsilon(x) = -1 + \mathcal{O}(\varepsilon)$) is given by

$$\begin{cases} \frac{d\psi_1}{d\tau} = -\varepsilon S(\psi_1) + 16e^{-(\psi_2 - \psi_1)} \\ \frac{d\psi_2}{d\tau} = -\varepsilon S(\psi_2) + 16(e^{-(\psi_3 - \psi_2)} - e^{-(\psi_2 - \psi_1)}) \\ \frac{d\psi_3}{d\tau} = -\varepsilon S(\psi_3) + 16(e^{-(\psi_4 - \psi_3)} - e^{-(\psi_3 - \psi_2)}) \\ \frac{d\psi_4}{d\tau} = -\varepsilon S(\psi_4) - 16e^{-(\psi_4 - \psi_3)}. \end{cases} \quad (107)$$

An immediate consequence of the form of these ODEs is that critical points $(\tilde{\psi}_1, \tilde{\psi}_2, \tilde{\psi}_3)$ of (106), respectively $(\tilde{\psi}_1, \tilde{\psi}_2, \tilde{\psi}_3, \tilde{\psi}_4)$ of (107), must have $\tilde{\psi}_1 \ll \tilde{\psi}_2 \ll \tilde{\psi}_3$, resp. $\tilde{\psi}_1 \ll \tilde{\psi}_2 \ll \tilde{\psi}_3 \ll \tilde{\psi}_4$, and that it is thus natural to adapt the ‘anchors’ p_j of the fronts (67) in a suitable fashion, according to the character of the critical point under consideration (see (113) below).

Since the number of fronts is also even in the case of four-front patterns, it is somewhat similar to the case of two-front solutions. Therefore, we start by studying the four-dimensional system (107). To determine the critical point $(\tilde{\psi}_1, \tilde{\psi}_2, \tilde{\psi}_3, \tilde{\psi}_4)$ that is a natural extension of the critical point of the first kind of (92), we first look for critical points with two-fronts located at the left and two at the right tail of the heterogeneity, i.e. with $\tilde{\psi}_1 \ll \tilde{\psi}_2 \ll -1 < 1 \ll \tilde{\psi}_3 \ll \tilde{\psi}_4$. For this, we follow (94), introduce $v_{1,2}^\pm$ – with $0 < v_2^\pm < v_1^\pm$ – and $\tilde{\varepsilon}_{1,2}^\pm$, and set

$$\tilde{\psi}_1 = -v_1^- |\log \varepsilon| + \tilde{\varepsilon}_1^- \ll \tilde{\psi}_2 = -v_2^- |\log \varepsilon| + \tilde{\varepsilon}_2^- \ll -1 < 1 \ll \tilde{\psi}_3 \quad (108)$$

$$= v_2^+ |\log \varepsilon| + \tilde{\varepsilon}_2^+ \ll \tilde{\psi}_4 = v_1^+ |\log \varepsilon| + \tilde{\varepsilon}_1^+,$$

where we note that indeed $\tilde{\psi}_2 - \tilde{\psi}_1 \gg 1$, $\tilde{\psi}_3 - \tilde{\psi}_2 \gg 1$, $\tilde{\psi}_4 - \tilde{\psi}_3 \gg 1$. It thus follows that (to leading order),

$$\begin{aligned} |S(\tilde{\psi}_1)| &= \varepsilon^{\mu v_1^-} |h_-| w_- e^{\mu \tilde{\varepsilon}_1^-} \ll \varepsilon^{\mu v_2^-} |h_-| w_- e^{\mu \tilde{\varepsilon}_2^-} = |S(\tilde{\psi}_2)| \\ |S(\tilde{\psi}_4)| &= \varepsilon^{\mu v_1^+} |h_+| w_+ e^{-\mu \tilde{\varepsilon}_1^+} \ll \varepsilon^{\mu v_2^+} |h_+| w_+ e^{-\mu \tilde{\varepsilon}_2^+} = |S(\tilde{\psi}_3)| \end{aligned} \quad (109)$$

(see (95)). Since by (107), $|16e^{-(\tilde{\psi}_2 - \tilde{\psi}_1)}| = |\varepsilon S(\tilde{\psi}_1)| \ll |\varepsilon S(\tilde{\psi}_2)|$ and $|16e^{-(\tilde{\psi}_4 - \tilde{\psi}_3)}| = |\varepsilon S(\tilde{\psi}_4)| \ll |\varepsilon S(\tilde{\psi}_3)|$, and $S(\tilde{\psi}_1) + S(\tilde{\psi}_2) + S(\tilde{\psi}_3) + S(\tilde{\psi}_4) = 0$, we deduce that to leading order

$$S(\tilde{\psi}_2) = -S(\tilde{\psi}_3) = \frac{16}{\varepsilon} e^{-(\tilde{\psi}_3 - \tilde{\psi}_2)}, \quad (110)$$

which is identical to (93). Hence, we conclude that $v_2^- = v_2^+ = v_2 = 1/(2 - \mu)$ and that $\tilde{\varepsilon}_2^-$ and $\tilde{\varepsilon}_2^+$ can be determined uniquely if (and only if) $h_+ < 0 < h_-$. It should be noted, however, that the two-front (sub)pattern thus formed by $\tilde{\psi}_2$ and $\tilde{\psi}_3$ is a reflected version of that in Section 5.1: at $\tilde{\psi}_2$, the pattern ‘jumps’ from $+1$ to -1 and back to $+1$ at $\tilde{\psi}_2$, while all two-front patterns in Section 5.1 jump from -1 to $+1$ back to -1 (by construction). This reflection is a completely symmetric operation and has no influence on existence nor stability. Under the assumption that indeed $h_+ < 0 < h_-$, we next substitute the outcome of (110) in the $\frac{d\psi_1}{d\tau} = 0$ and $\frac{d\psi_4}{d\tau} = 0$ equations, to obtain

$$\begin{aligned} 0 &= -\varepsilon S(\tilde{\psi}_1) + 16e^{-\tilde{\psi}_2} e^{\tilde{\psi}_1} = -\varepsilon^{1+\mu v_1^-} h_- w_- e^{\mu \tilde{\varepsilon}_1^-} + 16\varepsilon^{v_1^- - \frac{1}{2-\mu}} e^{-(\tilde{\varepsilon}_2^- - \tilde{\varepsilon}_1^-)} \\ 0 &= -\varepsilon S(\tilde{\psi}_4) - 16e^{\tilde{\psi}_3} e^{-\tilde{\psi}_4} = -\varepsilon^{1+\mu v_1^+} h_+ w_+ e^{-\mu \tilde{\varepsilon}_1^+} - 16\varepsilon^{v_1^+ - \frac{1}{2-\mu}} e^{(\tilde{\varepsilon}_2^+ - \tilde{\varepsilon}_1^+)}, \end{aligned} \quad (111)$$

find

$$v_1^- = v_1^+ := v_1 = \frac{3 - \mu}{(1 - \mu)(2 - \mu)}, \quad (112)$$

and conclude that $\tilde{\varepsilon}_{1,2}^-$ can again be determined uniquely (since $h_+ < 0 < h_-$). Once again, we note that it is natural – and even necessary – to adapt our choices for the anchors p_j (67) to the patterns under consideration:

$$p_1 = -v_1 < p_2 = -v_2 < p_3 = v_2 < p_4 = v_1, \quad (113)$$

which also implies that the lengths of two bounded intervals I_2 and I_3 become $\frac{1}{2}(v_1 + v_2) = 1/(1 - \mu)$ (70), thus longer than the standard length 1 (see (48)).

Especially since the ψ_1 - and ψ_4 -fronts are now deeper into the tails of $S(\psi)$ compared to those of the stationary two-front (of the first kind), we again need to check whether we can still neglect the higher order contributions to (107). Following the approach of Section 5.1, we denote the correction terms to the $\frac{d\psi_j}{d\tau}$ equation by $\varepsilon^2 R_2^{j,4}(\psi_1, \psi_2, \psi_3, \psi_4; \varepsilon)$ ($j = 1, 2, 3, 4$). We need to ensure that the stationary four-fronts can be constructed in circumstances under which the $\mathcal{O}(\varepsilon^2)$ correction terms $\varepsilon^2 R_2^{j,4}(\psi_1, \psi_2, \psi_3, \psi_4; \varepsilon)$ – i.e., that satisfy $\varepsilon^2 |R_2^{j,4}(\psi_1, \psi_2, \psi_1, \psi_2; \varepsilon)| < \varepsilon^2 C$ for some $C > 0$ – are indeed of higher order. Thus, we need to check whether $v_1 - v_2 = 2/((1 - \mu)(2 - \mu)) < 2$: if this is the case, then we know that there is no possible leading order contribution by neglected correction terms. It follows that the above leading order analysis is only valid for $\mu \in (0, \frac{1}{2}(3 - \sqrt{5})) \approx (0, 0.3819)$. In other words, like for the critical points of the second and third kind in (92), the original μ -interval $(0, 1)$ must be shrunk into a smaller interval to establish the existence of this kind of stationary four-front patterns. As for the critical points of the second kind discussed in the previous section, it is natural to expect that this domain may be extended, but a full higher order analysis would need to be performed to (possibly) come to such a conclusion (which, again, we refrain from doing in this study).

Like the existence, also the stability of the stationary four-front pattern is determined by the two-front that underlies it. By the relative magnitudes (109), we can write down the leading order part of the 4×4 matrix associated to the stability of $(\tilde{\psi}_1, \tilde{\psi}_2, \tilde{\psi}_3, \tilde{\psi}_4)$,

$$\begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & -\varepsilon S'(\tilde{\psi}_2) + E_2 & -E_2 & 0 \\ 0 & -E_2 & -\varepsilon S'(\tilde{\psi}_3) + E_2 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \quad (114)$$

(see (98), (99)). Hence, the two (relatively) large eigenvalues associated with the stability of $(\tilde{\psi}_1, \tilde{\psi}_2, \tilde{\psi}_3, \tilde{\psi}_4)$ are given by

$$\lambda_1 = -\mu \varepsilon^{\frac{2}{2-\mu}} |h_+| w_+ e^{-\mu \tilde{\varepsilon}_2^+} < 0, \quad \lambda_2 = (2 - \mu) \varepsilon^{\frac{2}{2-\mu}} |h_+| w_+ e^{-\mu \tilde{\varepsilon}_2^+} > 0 \quad (115)$$

(see (100)). The associated stationary four-front pattern in (2) is unstable. (We refrain from determining the other two smaller eigenvalues associated to the ψ_1 -front and the ψ_4 -front.)

Thus, we conclude that the kind of stationary four-front patterns can be regarded as straightforward extensions of the stationary two-front pattern of the first kind considered in Section 5.2: both the existence and the stability of these four-front patterns is determined by its two-front core.

Instead of considering critical points of (107) that correspond to the critical points of (92) of the second and third kind, we now first consider a new class of critical points $(\tilde{\psi}_1, \tilde{\psi}_2, \tilde{\psi}_3)$ of (106) – the critical points of the fourth kind – that can be seen as combinations of critical points of the second and third kind,

- critical points of the fourth kind have $\tilde{\psi}_1 \ll -1 < 1 \ll \tilde{\psi}_3$ and $\tilde{\psi}_2 = \mathcal{O}(1)$.

Thus, we combine (94) and (96) and set

$$\tilde{\psi}_1 = -v_- |\log \varepsilon| + \tilde{\varepsilon}_-, \quad \tilde{\psi}_2 = \psi_* + \varepsilon^{\nu_0} \tilde{\psi}_0, \quad \tilde{\psi}_3 = v_+ |\log \varepsilon| + \tilde{\varepsilon}_+. \quad (116)$$

Since also here $S(\tilde{\psi}_1) + S(\tilde{\psi}_2) + S(\tilde{\psi}_3) = 0$, we conclude by balancing the asymptotic magnitudes (in ε) that $v_- = v_+ = \nu$ and $\nu_0 = \mu \nu$. It then follows from (for instance) the $\frac{d\psi_3}{d\tau} = 0$ equation that

$$\nu_- = \nu_+ = \nu = \frac{1}{1 - \mu}, \quad \nu_0 = \frac{\mu}{1 - \mu},$$

where we note that this leading order analysis is only valid for $0 < \mu < \frac{1}{2}$ (as for the critical points of (106) of second and third kind). If (and only if) again $h_+ < 0 < h_-$ and assuming that $S'(\psi_*) \neq 0$, then the resulting equations for $\tilde{\varepsilon}_\pm$ and $\tilde{\psi}_0$ can be solved uniquely.

By defining

$$M_\pm = \mp h_\pm w_\pm e^{\mp \mu \tilde{\varepsilon}_\pm} > 0,$$

(by (87) and since $h_+ < 0 < h_-$), we can (after some algebra) write the 3×3 matrix associated with the stability of this critical point $(\tilde{\psi}_1, \tilde{\psi}_2, \tilde{\psi}_3)$ as,

$$\varepsilon^{\frac{1}{1-\mu}} \begin{pmatrix} -\mu M_- + M_- & -M_- & 0 \\ -M_- & -\varepsilon^{-\frac{\mu}{1-\mu}} S'(\psi_*) + M_- + M_+ & -M_+ \\ 0 & -M_+ & -\mu M_+ + M_+ \end{pmatrix}$$

This matrix has one (relatively) large eigenvalue and two smaller ones,

$$\lambda_0 = -S'(\psi_*) \varepsilon, \quad \lambda_\pm = (1 - \mu) M_\pm \varepsilon^{\frac{1}{1-\mu}} > 0$$

(to leading order and under the assumptions $h_+ < 0 < h_-$ and $S'(\psi_*) \neq 0$). Thus, also this stationary multi-front pattern is unstable (with two or three unstable eigenvalues according to the stability of the stationary one-front associated with ψ_*).

We may now proceed and determine all possible stationary three-fronts, using the stationary two-fronts of the first, second, and third kind introduced in Section 5.1 as foundation. We start with the extended fronts of the first kind – as we already did above for the four-front systems – and consider critical points $(\tilde{\psi}_1, \tilde{\psi}_2, \tilde{\psi}_3)$ of (106) that have all three fronts in the tails of $S(\psi)$, i.e., that either have $\tilde{\psi}_1 \ll -1 < 1 \ll \tilde{\psi}_2 \ll \tilde{\psi}_3$ or $\tilde{\psi}_1 \ll \tilde{\psi}_2 \ll -1 < 1 \ll \tilde{\psi}_3$. As for the four-fronts studied above, it follows from $S(\tilde{\psi}_1) + S(\tilde{\psi}_2) + S(\tilde{\psi}_3) = 0$ that (to leading order) $S(\tilde{\psi}_1) = -S(\tilde{\psi}_2)$ with $|S(\tilde{\psi}_3)| \ll |S(\tilde{\psi}_{1,2})|$ in the former case and $S(\tilde{\psi}_2) = -S(\tilde{\psi}_3)$ with $|S(\tilde{\psi}_1)| \ll |S(\tilde{\psi}_{2,3})|$ in the latter. Thus, in the first case, the positions of the $\tilde{\psi}_1$ - and $\tilde{\psi}_2$ -fronts coincide to leading order with those of the two-front of the first kind, with $\tilde{\psi}_3$ determined by

$$\varepsilon S(\tilde{\psi}_3) = 16e^{\tilde{\psi}_2} e^{-\tilde{\psi}_3},$$

similar to (111). Under the condition $h_+ < 0 < h_-$ we thus find a uniquely determined three-front pattern, that can be seen as a three-front extension of the two-front pattern of the first kind. The second case goes along

similar lines, the three-front pattern is built around a two-front pattern of the first kind – although this again is a reflected version of the two-front patterns considered in Section 5.1, i.e., it connects +1 to −1 back to +1. The fact that the ‘added’ third front goes deeper into the tail of $S(\psi)$ implies that the $\mathcal{O}(\varepsilon^2)$ correction term are only of higher order for μ in a certain subinterval if $(0, 1)$ – see Theorem 3 and especially its proof in App. D.

As before, also the stability of these three-fronts is completely determined by that of the underlying two-front: the 3×3 -matrices associated with the stability of the present three-front patterns have two (relatively) large eigenvalues that coincide (to leading order) with those of the underlying two-front pattern of the first kind (see (114)), (115)): both stationary three-front patterns are unstable.

The final kind of critical points $(\bar{\psi}_1, \bar{\psi}_2, \bar{\psi}_3)$ of (106) are points that have a two-front pattern of the second or third kind as underlying pattern (or a reflected version of these), i.e., that either have $\bar{\psi}_1 \approx \psi_*$ as in (96) and $\bar{\psi}_2 \ll \bar{\psi}_3$ or $\bar{\psi}_1 \ll \bar{\psi}_2 \ll \bar{\psi}_3 \approx \psi_*$. Again we refer to Section 5.3, Theorem 3 and App. D for more systematic constructions, statements and proofs. It also again follows that these patterns must be unstable, since the two large eigenvalues of the 3×3 stability matrices are determined by the underlying two-front patterns (of the second or third kind) to leading order.

Thus, apart from the new three-front patterns of the fourth kind, all other three-front patterns can be seen as (relatively straightforward) extensions of the two-front patterns. We wrap up the analysis of stationary three-front patterns by concluding that if $S(\psi)$ has one, respectively three, zeros and $h_+ < 0 < h_-$ – as for the unimodal hill (5) with $\mu \in (0, \mu_{PF})$, resp. $\mu \in (\mu_{PF}, 1)$ (Fig. 7) – system (106) has $\mathcal{N}(3) = 5$, resp. $\mathcal{N}(3) = 11$, critical points. More specifically, there are two critical points that are extended two-fronts of the first kind and do not have a front near a zero of $S(\psi)$ and three critical points that have a front near a given zero ψ_* of $S(\psi)$ – the three-fronts that are extended two-fronts of the second and third kind and the new three-front of the fourth kind (so that $5 = 2 + 3 \times 1$ and $11 = 2 + 3 \times 3$ – see (D.1)). All critical points/all possible stationary three-front patterns are unstable.

The stationary four-front patterns can all be constructed along the same lines, i.e., as ‘(relatively straightforward) extensions’ of the two-front patterns of the first, second, and third kind and of the three-front pattern of the fourth kind. Without going into the details – see Section 5.3 and App. D for the general set-up – we note that for $h_+ < 0 < h_-$ there are three critical points of the four-dimensional system (107) that do not have a front near a zero of $S(\psi)$ and thus can be seen as extensions of the two-front patterns of the first kind (above we worked out the case $\bar{\psi}_1 \ll \bar{\psi}_2 \ll -1 < 1 \ll \bar{\psi}_3 \ll \bar{\psi}_4$, the cases $\bar{\psi}_1 \ll -1 < 1 \ll \bar{\psi}_2 \ll \bar{\psi}_3 \ll \bar{\psi}_4$ and $\bar{\psi}_1 \ll \bar{\psi}_2 \ll \bar{\psi}_3 \ll -1 < 1 \ll \bar{\psi}_4$ follow along similar lines). Any two-front of the second kind has a unique extension as stationary four-front (with $\bar{\psi}_1 = \mathcal{O}(1) \ll \bar{\psi}_2 \ll \bar{\psi}_3 \ll \bar{\psi}_4$), and the same holds (as usual) for the two-fronts of the third kind. Finally, each three-front of the fourth kind can be extended to a four-front in two ways (either by placing the fourth ‘far away’ front to the left or to the right of the original three-front pattern). Thus, if $S(\psi)$ has 1, respectively 3, zeros then (107) has $\mathcal{N}(4) = 7$, resp. $\mathcal{N}(4) = 15$, critical points (see (D.1)). The stability of the associated stationary four-front patterns in (2) is again fully determined by the underlying ‘basic’ two- or three-front patterns. Since these are all unstable, it follows again that all critical points of system (107) – and thus all stationary four-front patterns in (2) – are unstable.

As in the two-front case, if only one of the conditions $h_+ < 0$ or $h_- > 0$ holds, then the number of critical points $\mathcal{N}(4)$ is equal to the number of zeros of $S(\psi)$ – both for the three-front patterns and for the four-front patterns (and the three- and four-front patterns are all extensions of the two-front patterns of the second kind – for $h_+ < 0$ – or the third kind – for $h_- > 0$). There are no critical points if neither of these conditions holds.

For both the interacting three-fronts and the interacting four-fronts, the different possible scenarios can be worked out as was done for the case of two-front solutions in Section 5.1. The scenarios depend on the initial conditions and also on the parity of

the number of fronts considered. We come back to this discussion in more detail in the general setting of the upcoming section. We refer to Fig. 11 for two four-front simulations of (2) with only slightly different initial conditions that show different behavior (where we used a localized topography $H_{\text{alg}}(x; p)$ with algebraically decaying tails, instead of a $H_{\text{exp}}(x; \mu)$ with exponentially decaying tails – recall (5)).

5.3. N -Front dynamics

Our first general result on the existence and stability of stationary N -front dynamics, which is a direct generalization of the insights of the previous sections, can be stated as follows.

Theorem 3. Consider the weakly heterogeneous Allen-Cahn Eq. (2) with inhomogeneity $F(U, U_x, x)$ given by sufficiently smooth topography $H(x)$ (4). Assume that $H(x)$ is localized and that there is a $\mu > 0$ so that for $x \rightarrow \pm\infty$ $H'(x)$ decays exponentially like $h_{\pm}(\mu)e^{\mp\mu\sqrt{2}x}$ (85). Assume furthermore that the Melnikov function $S(\psi)$ associated to $H(x)$ (36), (83) has $K \geq 0$ non-degenerate zeros $\psi_{*,k}$ (i.e., $S'(\psi_{*,k}) \neq 0$) and that $0 < \varepsilon \ll 1$ is sufficiently small. Let $\mathcal{N}(N)$ be the number of stationary N -front patterns of (2) that approach $u_{\pm}^{\varepsilon}(x) = -1 + \mathcal{O}(\varepsilon)$ as $x \rightarrow -\infty$ and $u_{\pm}^{\varepsilon}(-1)^{N+1}(x) = (-1)^{N+1} + \mathcal{O}(\varepsilon)$ as $x \rightarrow +\infty$ (9).

Then $\mathcal{N}(1) = K$, and for $N \geq 2$ and K odd,

- if $h_+(\mu) > 0$ and $h_-(\mu) < 0$, then $\mathcal{N}(N) = 0$;
- if $h_+(\mu) < 0 < h_-(\mu)$, then there is a $\mu_*(N) \in (0, 1)$ such that for $\mu \in (0, \mu_*(N))$,

$\mathcal{N}(N) = (K+1)N - 1$ (with $\mu_*(N) = 1 - 2^{-\frac{1}{N-1}} \in (0, 1)$);

- if either $h_+(\mu) < 0$ or $h_-(\mu) > 0$, then $\mathcal{N}(N) = K$ for $\mu \in (0, \mu_*(N))$.

For $N \geq 2$ and K even,

- $\mathcal{N}(N) = K$ for $\mu \in (0, \mu_*(N))$.

The one-front patterns are associated to the zeros $\psi_{*,k}$ and are spectrally stable if $S'(\psi_{*,k}) > 0$. All N -front patterns are unstable for $N \geq 2$.

An essentially identical result holds for stationary N -front patterns that connect $u_{\pm}^{\varepsilon}(x) = 1 + \mathcal{O}(\varepsilon)$ (as $x \rightarrow -\infty$) to $u_{\pm}^{\varepsilon}(-1)^N(x) = (-1)^N + \mathcal{O}(\varepsilon)$ (as $x \rightarrow +\infty$). Note also that the condition that $S(\psi)$ only has non-degenerate zeros implies that the richest case $h_+ < 0 < h_-$ can only occur for K odd: the signs of h_+ and h_- must be equal if $S(\psi)$ has an even number of zeros.

A proof of this theorem is given in App. D. Naturally, this proof cannot be directly based on an analysis of the N -front interaction ODE (84) – as was the case for the asymptotic analysis of the $N = 2, 3, 4$ -front patterns in the preceding subsections – without a rigorous validation of this reduction (see the brief discussion in the introduction of Section 4). However, the existence and stability of the stationary points of (84), i.e. of stationary N -front patterns, can be established by extending the approach of Secs. 2 and 3 for stationary one- and two-front patterns (see Theorems 1 and 2) without first establishing the validity of ODE (84): there is a direct correspondence between the construction and spectral properties of the stationary multi-front patterns as in Secs. 2 and 3 and the critical points of the multi-front interaction ODEs – see App. D.

A central ingredient of the existence part of the proof of Theorem 3 lies in establishing that each of these stationary N -front patterns indeed will exist if μ is sufficiently close to 0. As in the preceding subsections, this is for a large part a study of the potential impact of the $\mathcal{O}(\varepsilon^2)$ leading order correction terms that have not been (and will not be) determined: the occurrence of the bounds $\mu_*(N)$ on μ is caused by the fact that the magnitudes of the leading order terms in the asymptotic analysis decrease towards the upperbounds $\varepsilon^2 C$ (for some $C > 0$) that can be established for each of these correction terms. Only by going deeper into the analysis of the correction terms we can draw conclusions on the existence of (all) stationary N -fronts for $\mu \geq \mu_*(N)$ – which we have chosen not to do. (As follows from the proof of Theorem 3 in App. D, the upperbound $\mu_*(N)$ on μ typically varies with the character of the N -front pattern, $\mu_*(N)$ is determined explicitly as minimum over each of these character-dependent upperbounds.) However, since $\mu_*(N) > 0$ for all N , we know that stationary N -front patterns exist for exponentially decaying topographies $H(x)$ – as long as this exponential decay is sufficiently weak. This observation directly motivates the next corollary, which concerns N -front patterns for topographies with algebraically decaying tails.

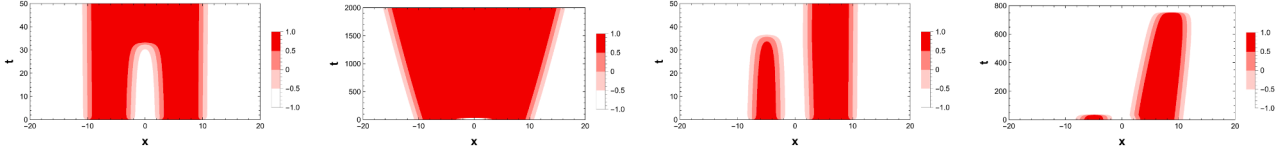


Fig. 11. Simulations of (2) for $x \in (-20, 20)$ (and homogeneous Neumann boundary conditions) with topographical inhomogeneity (4) given by $-H_{\text{alg}}(x; 0.5)$ (5) and $\varepsilon = 0.02$. (a)/(b) A four-front initial condition: $U(x, 0) = -1 + \tanh(5(x + 10)) - \tanh(5(x + 2.5)) + \tanh(5(x - 2.5)) - \tanh(5(x - 10))$; the two middle fronts $\psi_2(\tau) < 0 < \psi_3(\tau)$ attract each other and merge, while the outer fronts travel away, i.e., the initial four-front pattern $(\psi_1(\tau), \psi_2(\tau), \psi_3(\tau), \psi_4(\tau))$ evolves into a two-front pattern $(\psi_1(\tau), \psi_4(\tau))$ with $\psi_1(\tau) \rightarrow -\infty$ and $\psi_4(\tau) \rightarrow +\infty$ as $\tau \rightarrow \infty$. Note: $t \in (0, 50)$ in (a), $t \in (0, 2000)$ in (b). (c)/(d) A four-front initial condition: $U(x, 0) = -1 + \tanh(5(x + 7.5)) - \tanh(5(x + 2.5)) + \tanh(5(x - 2.5)) - \tanh(5(x - 10))$ that is equal to that of (a)/(b) except that $\psi_1(0)$ is slightly shifted to the right (from -10 to -7.5): now the two left hand side fronts $\psi_1(\tau)$ and $\psi_2(\tau)$ merge and both $\psi_{3,4}(\tau)$ of the resulting two-front pattern $(\psi_3(\tau), \psi_4(\tau))$ initially travel to the right until $\psi_3(\tau)$ catches up with $\psi_4(\tau)$ and also this pair annihilates each other. Note: $t \in (0, 50)$ in (c), $t \in (0, 800)$ in (d).

Corollary 1. Consider the situation of Theorem 3 but now for topographies $H(x)$ for which $H'(x)$ decays algebraically as $x \rightarrow \pm\infty$. Then, all statements of Theorem 3 hold true – with $\tilde{h}_{\pm}(p)$ (90) in the roles of $h_{\pm}(\mu)$ (85) – without any further conditions on the rate of decay.

As in the proof of Theorem 3, the proof of Corollary 1 is given in App. D. Here, we note that although the statement of Corollary 1 is less involved than that of Theorem 3, the asymptotic nature of the main properties of the N -front patterns, i.e., the positions of the fronts, is technically more involved in the case of algebraic decay. Therefore, the proof of Corollary 1 builds directly on that of Theorem 3.

Even for the homogeneous Allen-Cahn model (1), long term simulations of multi-front patterns on large domains are challenging, since the interaction between fronts is only exponentially weak so that the fronts typically get pinned (even in high precision simulations), i.e., the fronts stop traveling while the analysis shows that they must keep on moving (albeit exponentially slow). In the case of the inhomogeneous Allen-Cahn Eq. (2) with localized topography $H(x)$ (see (4)) this effect persists if it is assumed that $H(x)$ decays exponentially. However, the topographical driving forces are much stronger when $H(x)$ only decays algebraically slow (in fact, our theory applies if only $H'(x)$ decays, which implies that $H(x)$ may even grow as $x \rightarrow \pm\infty$, see (90) and Fig. 10). Thus, in addition to the ‘clean’ result of Corollary 1, there is a second reason to focus on algebraically decaying localized topographies: it is more tractable – easier – to numerically explore the nature of multi-front dynamics if it is assumed that $H(x)$ decays algebraically.

In Fig. 12 we present simulations of five-front dynamics in (2) that run along lines similar to those of scenarios (i) – (iii) in Section 5.1 for general two-front dynamics. In other words, like the two- and three-front dynamics of Fig. 1(b),(e) and the four-front dynamics of Fig. 11, the five-front dynamics shown in Fig. 12 corroborate the leading order nature of interaction ODEs (84) (for $N = 2, 3, 4$ and 5). This indicates that it will be possible to use (84) to give an approximate description of the expected long term dynamics of a given initial N -front configuration, as we did in Section 5.1 for $N = 2$. Moreover, we may do so for at least as long as the distance between successive fronts does not become too small or too large.

In the former case, the front interactions can no longer be considered weak (the assumption on which the derivation (and validation) of the system (84) is based). In fact, it is clear from (84) that before the distances between successive fronts become this small, the front evolution is already dominated by that of the homogeneous Allen-Cahn equation. This case is described and thoroughly analyzed in the classical articles [17–19] (and we note that the pairwise front-annihilation process that is especially relevant in the present setting was studied in [18]). Thus, we may claim that the dynamics of multi-front patterns in inhomogeneous PDE (2) for which the front interactions can no longer be considered weak can be analytically ‘controlled’ by existing methods. However, this is a priori not clear for the other limit situation in which the distances between successive fronts become so large that it can no longer be assumed that the higher order correction terms $\varepsilon^2 R_2^{j,N}(\psi_1, \psi_2, \dots, \psi_N; \varepsilon)$ to (84), which have not been determined, can be neglected, i.e. that the

asymptotic magnitudes of the $\varepsilon^2 R_2^{j,N}(\psi_1, \psi_2, \dots, \psi_N; \varepsilon)$ terms indeed are small compared to the magnitudes of the terms $\varepsilon S(\psi_j)$ ($j = 1, \dots, N$). According to (105), which is the assumption that $\varepsilon S(\psi_j)$ remains the leading order term in each $\frac{d\phi_j}{d\tau}$ equation, the distances between successive fronts (that are sufficiently far apart) must always decrease. The simulations shown in Fig. 2(c) seem to violate this prediction, which may suggest that $\varepsilon S(\psi_j)$ indeed cannot determine the leading order dynamics of a front (that is sufficiently distant from its neighbors). However, there is no solid foundation for such a conclusion, because the simulations of Fig. 2(c) are done for a topography $H(x)$ with algebraic behavior in its tails, while (105) has been derived for topographies that decay exponentially. We refer to Section 7 for further discussion.

6. The spatially periodically forced Allen-Cahn equation

In this section, we consider the case of multi-front dynamics (2) with a spatially periodic inhomogeneity, i.e., we assume that there is an $X > 0$ such that $F(U, V, x + X) \equiv F(U, V, x)$. First, we consider the dynamics of two-front patterns, again focusing on the special case of $F(U, V, x)$ given by (3) with $f_{1,2,3}(x)$ as in (43). Next, we further develop a general geometrical point of view; we especially apply it to provide a deeper understanding of the appearance of stationary two-fronts in periodically driven systems. As in the localized setting of Section 5, we conclude this section by deriving general insights in the dynamics of N -front patterns in periodically driven systems (focusing on those that differ essentially from those in Section 5).

6.1. Interacting two-fronts in the example system

In this subsection, we consider the strongly simplified case of front-interactions (again) in the setting of explicit example (3) with (43), and choose $N = 2$. We use this explicit form to be able to showcase the typical behavior in the most accessible and concrete way; the results are expected to be qualitatively similar for different periodically forced $F(U, V, x)$ (or $f_{1,2,3}(x)$), provided that the coefficients have two roots per period. (If they have more, then the dynamics is slightly more complex, but follows in a similar vein.) Thus, the Melnikov functions $\mathcal{R}_{\text{up/down}}(\phi)$ are given by (44) with (45), where we recall that the choice $\alpha_1 = k\alpha_2$ and $\alpha_3 = 0$ in (43) represents the topographic case (4) of (3).

Substitution in (78) gives the following ODE that describes the displacement of fronts over time. By rescaling time ($t' = 16t/\|u_{\text{up}}\|_2^2$, so that $t' = 8\sqrt{2}\tau$ (83)), we find by (78) that the dynamics of the two-front pattern $(\phi_1(t'), \phi_2(t'))$ in (2) – with the first front going up – is governed to leading order by

$$\begin{cases} \frac{d\phi_1}{dt'} = -\varepsilon(A+B)\sin(k\phi_1) & + e^{-\sqrt{2}(\phi_2-\phi_1)}; \\ \frac{d\phi_2}{dt'} = -\varepsilon(A-B)\sin(k\phi_2) & - e^{-\sqrt{2}(\phi_2-\phi_1)}, \end{cases} \quad (117)$$

with A and B determined by (45). Note that the critical points of (117) coincide with the configurations determined by (65). For the analysis of this system, it is convenient to introduce the new coordinates

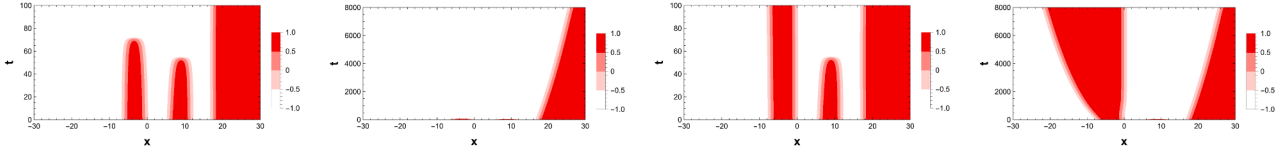


Fig. 12. Simulations of (2) for $x \in (-50, 50)$ (and homogeneous Neumann boundary conditions) with topographical inhomogeneity (4) given by $H_{\text{alg}}(x; 1.25)$ (5) and $\varepsilon = 0.2$. (a)/(b) A five-front initial condition: $U(x, 0) = \tanh(x + 6) - \tanh(x + 0.5) + \tanh(x - 6) - \tanh(x - 11.5) + \tanh(x - 17.5)$: first the ψ_3 - and ψ_4 -fronts merge, followed by the merging of ψ_1 and ψ_2 ; eventually, only the ψ_5 -front persists, it travels with decaying speed as a solitary front to $+\infty$. Note: $t \in (0, 100)$ in (a), $t \in (0, 8000)$ in (b). (c)/(d) Initially, the only difference with (a)/(b) is the position of ψ_1 – that is placed more to the left compared to (a)/(b) – $\psi_1(0) = -7$ instead of -6 in (a)/(b); after the ψ_3 - and ψ_4 -fronts have merged, ψ_2 settles at the stable stationary one-front location at $\psi = 0$, while ψ_1 travels (slowly) to $-\infty$ and ψ_5 to $+\infty$, both with decaying speeds. Note: $t \in (0, 100)$ in (c), $t \in (0, 8000)$ in (d). Thus, changing the initial condition of ϕ_1 changed the end configuration from one single front traveling to $+\infty$, to a three-front pattern with the middle one fixed at $\psi = 0$ and the other two traveling to $\pm\infty$.

$$(D(t'), S(t')) = (\phi_2(t') - \phi_1(t'), \phi_1(t') + \phi_2(t')),$$

$$\begin{cases} \frac{dD}{dt'} = -2\varepsilon A \sin\left(\frac{1}{2}kD\right) \cos\left(\frac{1}{2}kS\right) + 2\varepsilon B \cos\left(\frac{1}{2}kD\right) \sin\left(\frac{1}{2}kS\right) - 2e^{-\sqrt{2}D}, \\ \frac{dS}{dt'} = -2\varepsilon A \cos\left(\frac{1}{2}kD\right) \sin\left(\frac{1}{2}kS\right) + 2\varepsilon B \sin\left(\frac{1}{2}kD\right) \cos\left(\frac{1}{2}kS\right), \end{cases} \quad (118)$$

where we have used some standard trigonometric identities. Observe that for (asymptotically large) D such that $0 < \sqrt{2}D/|\log \varepsilon| \rightarrow \rho$ as $\varepsilon \rightarrow 0$ with $\rho > 1$, the fixed points $(\bar{D}, \bar{S})$ of (118) are determined to leading order by

$$\sin\left(\frac{1}{2}k\bar{D}\right) = \sin\left(\frac{1}{2}k\bar{S}\right) = 0 \quad \text{or} \quad \cos\left(\frac{1}{2}k\bar{D}\right) = \cos\left(\frac{1}{2}k\bar{S}\right) = 0. \quad (119)$$

This agrees with $(\phi_1, \phi_2) = (\phi_{\text{up}}, \phi_{\text{down}}) = (n_{\text{up}}(\varepsilon)\pi/k, n_{\text{up}}(\varepsilon)\pi/k)$, as found in Section 3.3.2.

To investigate the (dis)appearance of critical points as ρ passes through 1, we first consider the special case $B = 0$, so that $S(t')$ is symmetric under $S \rightarrow -S$ in (118), which is the case in the topographic setting (4). Note that the upcoming analysis is valid for all $f_{1,2,3}(x)$ with $B = 0$, i.e., with $\alpha_3 = 0$ (45), not specifically for a topographic choice of $f_{1,2,3}(x)$. In fact, the underlying simplification comes from the symmetry $F(-U, -V; x) = -F(U, V; x)$ in (2), which is trivially satisfied by (4). A similar analysis can be performed for $F(-U, -V; x) = F(U, V; x)$ that implies $A = 0$, so that (118) is symmetric under $S \rightarrow -S + \pi$. (Also, there is a third special case, $A = B$, which we do not discuss.)

In the symmetric/topographic case with $B = 0$, the S nullcline ($\frac{dS}{dt'} = 0$) of (118) is given by the straight lines for which $D = (2m_{S,D} + 1)\pi/k$ or $S = 2m_{S,S}\pi/k$ ($m_{S,D/S} \in \mathbb{Z}$). The D nullcline ($\frac{dD}{dt'} = 0$) satisfies

$$\varepsilon A \sin\left(\frac{1}{2}kD\right) \cos\left(\frac{1}{2}kS\right) = -e^{-\sqrt{2}D}.$$

Since the left hand side can only take values between $+\varepsilon A$ and $-\varepsilon A$, necessarily only points with $D > D_c := (|\log \varepsilon| - \log |A|)/\sqrt{2}$ can lie on this nullcline, which confirms the critical $\rho = 1$ growth rate of $D_c = D_c(\varepsilon)$. For $D(\varepsilon)$ with $\rho > 1$, these nullclines tend to the straight lines $D = 2m_{D,D}\pi/k$, $S = (2m_{D,S} + 1)\pi/k$ ($m_{D,D/S} \in \mathbb{Z}$) – (again) reconfirming Section 3.3.2 and (119). Decreasing D so that ρ becomes 1, while keeping $D > D_c$, causes these lines to break apart in closed curves. These closed ‘circles’ contract as D decreases further and subsequently disappear, see Fig. 13(a)–(g) for a sketch.

To understand the precise mechanisms driving the appearance of the critical points of (118) and/or the associated bifurcations, we focus on the case $\rho = 1$. Therefore, we introduce $(d(\tau'), s(\tau'))$, τ' , and constants $N = N_{\varepsilon,k} \in \mathbb{Z}$ and $R = R_{\varepsilon,k} \in [0, 2\pi)$ by

$$d(\tau') = D(t') - \frac{|\log \varepsilon|}{\sqrt{2}}, \quad s(\tau') = S(t'), \quad \tau' = 2\varepsilon t', \quad 2N\pi + R = \frac{k|\log \varepsilon|}{2\sqrt{2}}. \quad (120)$$

Here, we note that (d, s) correspond to $(\ell_{\text{down}} + \ell_{\text{up}}, \ell_{\text{down}} - \ell_{\text{up}})$ as introduced in (48) (and used throughout Section 3). In the symmetric topo-

graphic ($B = 0$) case, (118) becomes

$$\begin{cases} d' &= -A \sin\left(R + \frac{1}{2}kd\right) \cos\left(\frac{1}{2}ks\right) - e^{-d\sqrt{2}} \\ s' &= -A \cos\left(R + \frac{1}{2}kd\right) \sin\left(\frac{1}{2}ks\right). \end{cases} \quad (121)$$

This ODE is periodic under $s \rightarrow s + 4\pi/k$. We may consider it on a cylinder $(d, s) \in \mathbb{R} \times S^1$, with $s \in [0, 4\pi/k)$. We can distinguish between four families of fixed points, $(\bar{d}_{\pm}^a, \bar{s}_{\pm}^a)$ and $(\bar{d}_{\pm}^b, \bar{s}_{\pm}^b)$, distinguishable by setting either the cosine or the sine in the s -equation equal to zero:

$$\begin{aligned} \text{(a)} \quad \bar{s}_{\pm}^a &= 0, \bar{s}_{\pm}^a = \frac{2\pi}{k}, & A \sin\left(R + \frac{1}{2}k\bar{d}_{\pm}^a\right) &= \mp e^{-\bar{d}_{\pm}^a\sqrt{2}}, \\ \text{(b)} \quad \bar{d}_{\pm}^b &= \frac{(4m_b \pm 1)\pi - 2R}{k}, m_b \in \mathbb{Z} & A \cos\left(\frac{1}{2}k\bar{s}_{\pm}^b\right) &= \mp e^{-\frac{(4m_b \pm 1)\pi - 2R}{k}\sqrt{2}}. \end{aligned} \quad (122)$$

(to leading order in ε), where we immediately note that each family $(\bar{d}_{\pm}^{a/b}, \bar{s}_{\pm}^{a/b})$ contains countably many elements. The associated eigenvalues are given by

$$\begin{aligned} \text{(a)} \quad \lambda_{\pm,1}^a &= \mp \frac{1}{2}kA \cos\left(R + \frac{1}{2}k\bar{d}_{\pm}^a\right) + \sqrt{2}e^{-\sqrt{2}\bar{d}_{\pm}^a}, \quad \lambda_{\pm,2}^a = \mp \frac{1}{2}kA \cos\left(R + \frac{1}{2}k\bar{d}_{\pm}^a\right), \\ \text{(b)} \quad \lambda_{\pm,1/2}^b &= \frac{1}{2} \left(\sqrt{2}e^{-\sqrt{2}\bar{d}_{\pm}^b} \pm \sqrt{2e^{-2\sqrt{2}\bar{d}_{\pm}^b} + k^2 A^2 \sin^2\left(\frac{1}{2}k\bar{s}_{\pm}^b\right)} \right) \end{aligned} \quad (123)$$

(which for $\bar{d}_{\pm}^{a/b} \gg 1$ agrees with (66) (with $B = 0$) – after undoing all transformations and scalings). Hence, it immediately follows that all (b)-type critical points $(\bar{d}_{\pm}^b, \bar{s}_{\pm}^b)$ are saddles for all values of the parameters that are considered.

To study the nature of the (a)-type critical points $(\bar{d}_{\pm}^a, \bar{s}_{\pm}^a)$, we consider A as the bifurcation parameter. It follows from (122)(a) that, as A increases, new critical points are created at the bifurcational values $A = A_{\pm}^{\text{SN}}$ at which the curve $A \sin\left(R + \frac{1}{2}k\bar{d}\right)$ becomes tangent to one of the curves $\mp e^{-\sqrt{2}\bar{d}}$ (as a function of $\bar{d}$). Note that these new critical points have $\bar{d}$ -coordinates smaller than the $\bar{d}$ -coordinates of all (countably many) critical points that already exist before A passes through $A = A_{\pm}^{\text{SN}}$, i.e., the value of the minimal $\bar{d}$ -coordinate taken over all of the existing critical points decreases as A increases. For $\bar{d}_{+}^a$, respectively $\bar{d}_{-}^a$, the saddle node bifurcation occurs with a positive, resp. negative, derivative, which implies that $\cos\left(R + \frac{1}{2}k\bar{d}_{+}^a\right) > 0$, resp. $\cos\left(R + \frac{1}{2}k\bar{d}_{-}^a\right) < 0$. Hence, by (123)(a), $\lambda_{\pm,1}^a < 0$ holds at the tangency (and by construction $\lambda_{\pm,2}^a = 0$). It follows that as A increases through each of the values $A = A_{\pm}^{\text{SN}}$, two new critical points are created: $(\bar{d}_{\pm}^{a,1}, \bar{s}_{\pm}^{a,1})$ always appears as a saddle and $(\bar{d}_{\pm}^{a,2}, \bar{s}_{\pm}^{a,2})$ as a stable node (with $\bar{d}_{\pm}^{a,2} < \bar{d}_{\pm}^{a,1}$).

As A increases further, the stable node $(\bar{d}_{\pm}^{a,2}, \bar{s}_{\pm}^{a,2})$ does not undergo any further bifurcations (and $\bar{d}_{\pm}^{a,2}$ increases towards the next zero of $\sin\left(R + \frac{1}{2}k\bar{d}\right)$ as $A \rightarrow \infty$ (122)). However, $(\bar{d}_{\pm}^{a,1}, \bar{s}_{\pm}^{a,1})$ bifurcates: as A increases beyond $A_{\pm}^{\text{SN}}$, $\bar{d}_{\pm}^{a,1}$ decreases and reaches at $A = A_{\pm}^{\text{PF}} (> A_{\pm}^{\text{SN}})$ the point where the curve $\mp e^{-\sqrt{2}\bar{d}}$ intersects $A \sin\left(R + \frac{1}{2}k\bar{d}\right)$ in its nearest minimum (+) or maximum (−). As a consequence, $\lambda_{\pm,2}^a$ becomes zero at $A_{\pm}^{\text{PF}}$ (while $\lambda_{\pm,1}^a = \sqrt{2}\exp(-\sqrt{2}\bar{d}_{\pm}^a) > 0$) (123). Moreover, at this point, $(\bar{d}_{\pm}^{a,1}, \bar{s}_{\pm}^{a,1})$ coincide with the two emergent (symmetric) saddle points of

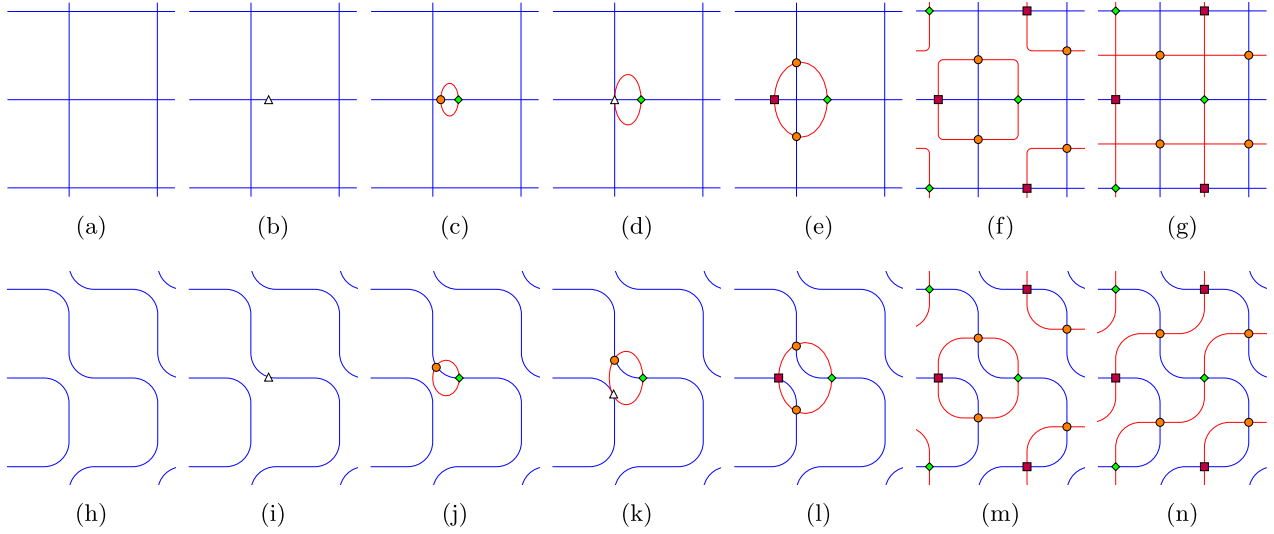


Fig. 13. Sketches of (possible) parts of the phase portrait of (118)/(121) in the symmetric case (with the topographic case $B = 0$ as example) – (a)-(g) – and with broken symmetry ($B \neq 0$) – (h)-(n). Blue lines denote the nullclines associated to the S/s -equation, red lines the nullclines associated to D/d . Circles, squares, diamonds and triangles indicate fixed points; green diamonds are stable nodes, orange circles are saddles, red squares are unstable nodes, and white triangles have one zero eigenvalue. (a)/(h) No fixed points (*i.e.* D/d is sufficiently low). (b)/(i) A saddle-node bifurcation. (c)/(j) A saddle and stable node created by the saddle node bifurcation. (d)/(k) A pitchfork bifurcation in the symmetric case/a second saddle node bifurcation in the asymmetric case. (e)/(l) Both cases have two saddles, a stable node, and an unstable node. (f)/(m) As D/d increases, several similar structures approach each other. (g)/(n) For $D/d \rightarrow \infty$, the structures merge and the nullclines form a regular grid.

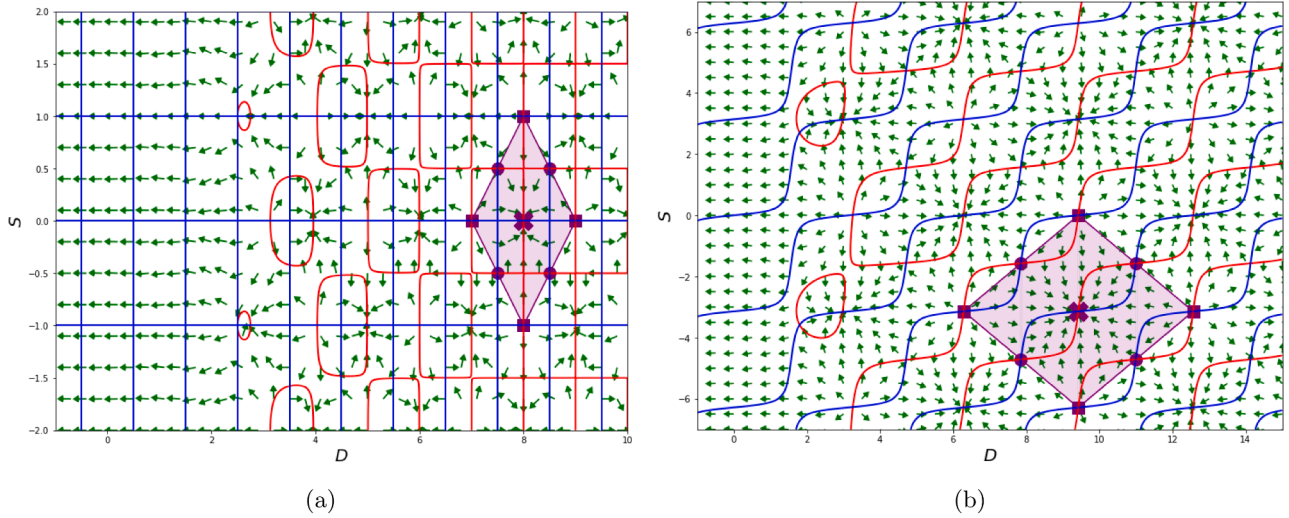


Fig. 14. (a) Phase portrait of (118) in a symmetric case: $\epsilon A = 0.029$, $\epsilon B = 0$, and $k = 2\pi$. (b) An asymmetric case: $\epsilon A = 0.09$, $\epsilon B = 0.01$ and $k = 2$. The red, respectively blue, lines denote the nullclines associated to the D , resp. S , equation. Green arrows denote the direction of the flow. The purple shaded area is the basin of attraction for the fixed point within that region (indicated with a purple cross); the purple lines forming its boundary are the stable manifolds of the surrounding fixed points of saddle type (indicated with purple circles), which limit to the unstable nodes (indicated with purple squares). (Numerical computations are made using Python's pyplot's plotting routines quiver and contour, and unstable manifolds are computed using backward time-integration with scipy's odeint).

(b)-type, $(\bar{d}_{\pm}^{b,1/2}, \bar{s}_{\pm}^{b,1/2})$ (122) that split off from $(\bar{d}_{\pm}^{a,1}, \bar{s}_{\pm}^{a,1})$ as A increases further beyond $A_{\pm}^{\text{PF}}$. The saddle $(\bar{d}_{\pm}^{a,1}, \bar{s}_{\pm}^{a,1})$ changes into an unstable node by this pitchfork bifurcation, its nature no longer changes as A increases further (in fact, $\bar{d}_{\pm}^{a,1}$ decreases towards the preceding zero of $\sin(R + \frac{1}{2}k\bar{d})$ as $A \rightarrow \infty$ (122)). We refer to Fig. 13(a)-(g) for visualizations of this process and to Fig. 14(a) for a phase portrait of the flow governed by (118) (or equivalently (121)) for A just below an $A_{\pm}^{\text{PF}}$.

Now, if system (118) does not possess a symmetry, *i.e.*, if $B \neq 0$ (or $A \neq 0$), then the above scenario breaks down. However, this is a standard broken symmetry scenario: the saddle node bifurcations persist and the pitchfork bifurcations break apart into a saddle-node bifurcation and a branch of persistent critical points. A precise distinction between (a)-

and (b)-type critical points can no longer be made, but the perturbed, asymmetric scenario can be described using these notions. For increasing A , first a saddle and a stable node corresponding to $(\bar{d}_{\pm}^{a,1}, \bar{s}_{\pm}^{a,1})$ ($\bar{d}_{\pm}^{a,2}, \bar{s}_{\pm}^{a,2}$) appear (with $\bar{d}_{\pm}^{a,1} < \bar{d}_{\pm}^{a,2}$). Neither of these undergo any additional bifurcations. However, as A increases further, two unstable critical points appear in a second saddle node bifurcation, a saddle and an unstable node. This saddle node bifurcation takes over the role of $A_{\pm}^{\text{PF}}$. The unstable node corresponds to $(\bar{d}_{\pm}^{a,1}, \bar{s}_{\pm}^{a,1})$ in the symmetrical case: to maintain the relation with the symmetric case, the persisting saddle that appeared at the first saddle node bifurcation has to be relabeled as a (b)-type saddle point as A passes through the second saddle node bifurcation, see Fig. 13(h)-(n) for a schematic sketch and Fig. 14(b) for a phase portrait.

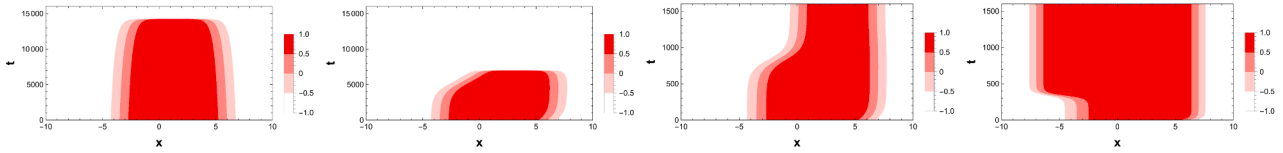


Fig. 15. Simulations of (2) on the x -interval $(-40, 40)$ (with homogeneous Neumann boundary conditions) plotted on $(-10, 10)$ with spatially periodic topographic inhomogeneity given by (4) with $H(x) = \sin kx$ with $k = 0.8977$ starting from the two-front initial condition $U(x, 0) = -1 + \tanh(x + 3.5) - \tanh(x - 6)$. (a) $\varepsilon = 0$, (b) $\varepsilon = 0.01$, (c) $\varepsilon = 0.1$ and (d) $\varepsilon = 0.4$.

Our insights so far on the existence, nature, and bifurcations of the critical points are a priori only the first steps towards understanding the dynamics of (two) interacting fronts governed by Eqs. (117)/(118)/(121). However, it is not hard to see from both the symmetrical and asymmetrical phase portraits of Fig. 14 that for fronts that are sufficiently far apart, i.e., for D (or d) sufficiently large, each stable ((a)-type) node $(\bar{d}_{\pm}^a, \bar{a}_{\pm}^b)$ is enclosed by a network formed by the unstable manifolds of 4 ((b)-type) saddle points $(\bar{d}_{\pm}^b, \bar{s}_{\pm}^b)$ – with 4 ((a)-type) unstable nodes $(\bar{d}_{\pm}^a, \bar{s}_{\pm}^a)$ at its vertices (i.e., the boundary of the basin of attraction of each stable ((a)-type) node $(\bar{d}_{\pm}^a, \bar{a}_{\pm}^b)$ is formed by the stable manifolds of the surrounding saddle-type fixed points; see Fig. 14). Hence, for any initial condition that is not chosen exactly on this network, the two-front pattern will converge to the nearby (enclosed) attractor. In other words, two-front patterns that are sufficiently far apart will always converge to a nearby stationary two-front pattern.

Here, ‘sufficiently far apart’ is as usual measured with respect to the standard distance scale $|\log \varepsilon|$ – see the simulations of Fig. 15 in which the initial position of a two-front pattern is kept fixed, while the magnitude of ε increases from 0, through 0.01, to 0.1 and 0.4, so that the distance between the fronts increases relative to the decreasing magnitude $|\log \varepsilon|$. For $\varepsilon = 0.001$ (Fig. 15(b)), the system behaves as the homogeneous Allen-Cahn case (with $\varepsilon = 0$, Fig. 15(a)): the fronts are attracted to each other, merge and disappear (nevertheless, the influence of the spatial inhomogeneity can clearly not be completely neglected for $\varepsilon = 0.01$). As ε increases further, the two-front pattern indeed settles in a nearby stationary pattern (Figs. 15(c), (d)). However, the value of ε also influences the position of the initial condition with respect to the above network of unstable manifolds and thus has impact on the stationary stable two-front state the PDE dynamics is attracted to – as can be seen by comparing Fig. 15(c) with $\varepsilon = 0.1$ to Fig. 15(d) with $\varepsilon = 0.4$.

In Section 6.3, a general statement about the existence of nearby stationary attractors for N -front patterns will be formulated for the case of a topographic spatially periodic driving term (4), under the condition that the fronts are sufficiently separated, see Claim 1. The proof is completely based on the observation that if (in the special case (117)) $\sqrt{2}(\phi_2 - \phi_1)/|\log \varepsilon| > 1$ (as $\varepsilon \rightarrow 0$), the front equations decouple (to leading order) and both $\phi_{1/2}$ evolve towards a zero $\phi_{*,1/2}$ of $\sin(k\phi)$ with $\cos(k\phi_{*,1/2}) < 0$ (so that $(\phi_{*,1}, \phi_{*,2})$ represents the stable, attracting, stationary two-front pattern to leading order). On the other hand, for fronts that are relatively close to each other, i.e., with $\sqrt{2}(\phi_2 - \phi_1)/|\log \varepsilon| < 1$, the terms associated to the driving heterogeneity become of higher order, and the two-front interaction equation is to leading order given by that for the classical homogeneous Allen-Cahn equation case [17–19], which explains why the $\varepsilon = 0.001$ dynamics of Fig. 15(b) are similar to that of Fig. 15(a) (in which $\varepsilon = 0$).

6.2. The existence problem as a weakly driven nonlinear oscillator: The geometric approach

In this section, we again restrict ourselves to the issue of the existence of stationary multi-front patterns, i.e., we do not consider the dynamics of multi-front patterns in (2) as we did at the end of the previous section (and will again in Section 6.3 below). Here, we take a geometric point of view by going deeper into the correspondence between the existence

question for localized stationary multi-front patterns (7) and the (classical) theory of weakly driven (planar) nonlinear oscillators (for example, see [28]). This point of view provides an alternative way to study the appearance of multi-front patterns – i.e., homoclinic and heteroclinic orbits of (7) – as functions of changing parameters. In fact, especially in the case that $F(u, v, x)$ is periodic in x , it provides a natural way to embed this analysis in that of the classical subject of ‘lobe dynamics’ of perturbed nonlinear oscillators.

Recapitulating the observations of Section 2 (especially Section 2.1), we note that for $\varepsilon = 0$, (7) possesses three fixed points, the two saddles $\mathcal{M}_{\pm}^0 := (u_{\pm}, 0) = (\pm 1, 0)$, with stable/unstable manifolds $W^{s/u}(\mathcal{M}_{\pm}^0)$, and the center $\mathcal{M}_0^0 := (u_0, 0) = (0, 0)$ (Fig. 5). There are two (unperturbed) heteroclinic orbits $W^u(\mathcal{M}_0^0) \cap W^s(\mathcal{M}_{\pm}^0)$ and $W^u(\mathcal{M}_{\pm}^0) \cap W^s(\mathcal{M}_0^0)$ between the saddles $\mathcal{M}_{\pm}^0$ that are explicitly given by the one-front solutions $u_{\text{up}}(x; \phi)$ and $u_{\text{down}}(x; \phi)$ (15).

For $0 < \varepsilon \ll 1$, (7) is non-autonomous, so that it is natural to consider the system in the extended (u, p, x) -phase space. Here, the saddles $\mathcal{M}_{\pm}^0$ persist (for $0 < \varepsilon$ sufficiently small) as unique, bounded solutions $u_{\pm}^{\varepsilon}(x)$ (given explicitly in (9)) which both have $2D$ stable/unstable manifolds $W^{s/u}(\mathcal{M}_{\pm}^{\varepsilon}) \subset \mathbb{R}^3$. These manifolds can be explicitly approximated by constructing the solutions $(u_{\pm, s/u}^{\varepsilon}, \frac{d}{dx} u_{\pm, s/u}^{\varepsilon}, x) \in W^{s/u}(\mathcal{M}_{\pm}^{\varepsilon})$. Focusing on $W^u(\mathcal{M}_{\pm}^{\varepsilon})$ we set

$$u_{\pm, u}^{\varepsilon}(x; \phi) = u_{\text{up}}(x; \phi) + \varepsilon u_{u,1}(x; \phi) + \varepsilon^2 u_{u,2}(x; \phi) + \text{h.o.t.} \quad (124)$$

and find, once again (see (22)),

$$\mathcal{L}_0 u_{u,1} = -F(u_{\text{up}}(x; \phi), u_{\text{up},x}(x; \phi), x).$$

The form of solutions $u_{u,1}(x; \phi)$ that remain bounded as $x \rightarrow -\infty$ (i.e. solutions u_u that tend to $u_{\pm}^{\varepsilon}$ as $x \rightarrow -\infty$) is given in (24), (25), (A.2) (see also App. A) and we have thus obtained the leading order approximation,

$$\bar{W}^u(\mathcal{M}_{\pm}^{\varepsilon}) := \left\{ \left(u_{\text{up}}(x; \phi) + \varepsilon u_{u,1}(x; \phi), \frac{d}{dx} u_{\text{up}}(x; \phi) + \varepsilon \frac{d}{dx} u_{u,1}(x; \phi), x \right) : x \in (-\infty, x_{\max}) \right\}. \quad (125)$$

Here, $x_{\max}$ is the maximum value for which the approximation is valid. More precisely, $x_{\max}$ is not uniquely determined, choosing $x_{\max}$ also determines the accuracy of the approximation. It is shown in App. A that

$$\lim_{x \rightarrow \phi \rightarrow \infty} u_{u,1}(x; \phi) e^{-\sqrt{2}|x-\phi|} = \lim_{x \rightarrow \phi \rightarrow \infty} \frac{1}{8} \int_{-\infty}^{\infty} F(u_{\text{up}}(z; \phi), u_{\text{up},x}(z; \phi), z) \Psi_b(z; \phi) dz e^{\pm \sqrt{2}|x-\phi|} = \frac{1}{8} \mathcal{R}(\phi) \quad (126)$$

(23). Thus, the choice $x_{\max} - \phi = |\log \varepsilon|/(2\sqrt{2})$ would yield an approximation that is accurate to $\mathcal{O}(\sqrt{\varepsilon})$, while (126) does not provide useful information if $x_{\max} - \phi = \rho |\log \varepsilon|/\sqrt{2}$ with $\rho > 1$ (unless $\mathcal{R}(\phi) = 0$, for which we have recovered the one-front connection between the limiting solutions $u_{\pm}^{\varepsilon}(x)$ of Section 2). Following the procedure of [44], it can be shown that considering the next order terms in the regular expansion (124) does not lead to a potential way out: $u_{u,2}(x; \phi)$ grows as $\exp(2\sqrt{2}|x-\phi|)$ as $x - \phi \rightarrow \infty$ so that $\varepsilon^2 u_{u,2}(x; \phi) = \mathcal{O}(\varepsilon^{2-2\rho})$ for $x_{\max} - \phi = \rho |\log \varepsilon|/\sqrt{2}$. In fact, it follows (in general) that all higher order terms yield $\mathcal{O}(1)$ contributions as $x_{\max} - \phi = |\log \varepsilon|/\sqrt{2}$. However, by restricting expansion (124) to $x_{\max} - \phi = |\log \varepsilon|/(2\sqrt{2}) + \mathcal{O}(1)$, an approximation can be obtained for the first ‘jump’ of $W^u(\mathcal{M}_{\pm}^{\varepsilon})$ from $u_{\pm}^{\varepsilon}(x)$ into the $\mathcal{O}(\sqrt{\varepsilon})$ region through which it passes $u_{\pm}^{\varepsilon}(x)$ (and more accurate approximations within this region can be obtained from expansion (124)). Beyond this region, further approximations of $W^u(\mathcal{M}_{\pm}^{\varepsilon})$ can be obtained by a similar approach based

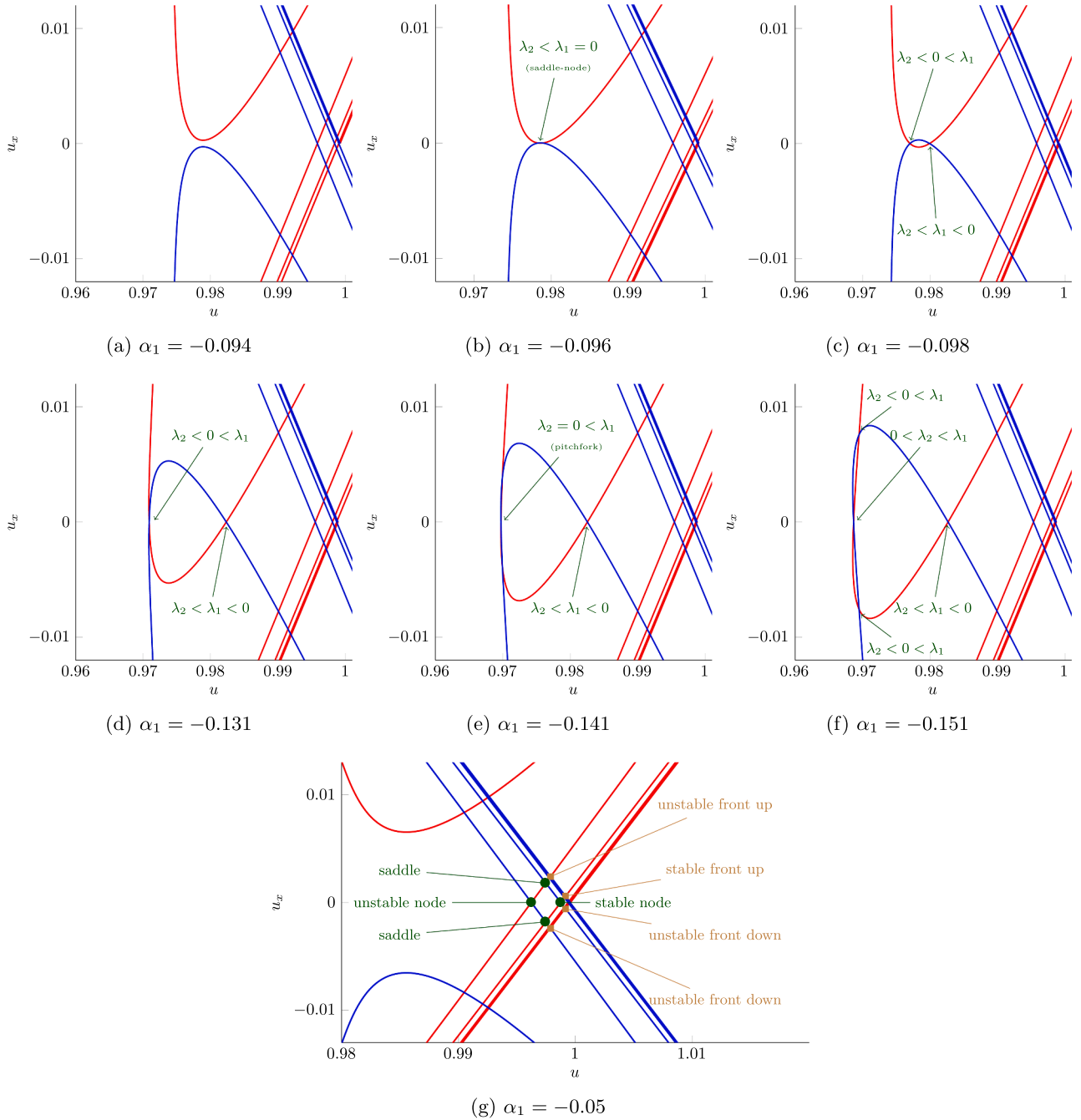


Fig. 16. Analytically obtained approximations of $W_0^{s/u}(\mathcal{M}_\pm^\varepsilon)$ evaluated at $\varepsilon = 0.1$ with $F(U, V, x) = \alpha_1 \cos(\pi x)U$, for various values for α_1 . The red lines denote the unstable manifold $W_0^u(\mathcal{M}_\pm^\varepsilon)$ and the blue lines the stable manifolds $W_0^s(\mathcal{M}_\pm^\varepsilon)$. (a) No intersection between manifolds, (b) saddle-node bifurcation, (c) two fixed points arise, (d) two fixed points, (e) pitchfork bifurcation, (f) four fixed points, (g) classification of two-front solutions in relation to 1-front solutions. Results on stability are entirely based on the analysis in Section 6.1.

on the ‘jump back orbit’ $u_{\text{down}}(x; \phi)$, etc., see again the procedure of [44].

Here, we do not go further into the details of approximating manifolds $W^{s/u}(\mathcal{M}_\pm^\varepsilon)$ beyond their first ‘jump’ from $\mathcal{M}_\pm^\varepsilon$ to $\mathcal{M}_\pm^\varepsilon$. (Though we refer the reader to [45] for a useful approach for doing this and for finding second-order, third-order, and higher-order intersections of these invariant manifolds.) Instead, we apply the above procedure to approximate $W^{s/u}(\mathcal{M}_\pm^\varepsilon)$ up to the $\mathcal{O}(\sqrt{\varepsilon})$ neighborhood of $\mathcal{M}_\pm^\varepsilon$, which allows us to investigate the first intersections $W^u(\mathcal{M}_-^\varepsilon) \cap W^s(\mathcal{M}_+^\varepsilon)$, i.e., the existence and (dis)appearance of homoclinic orbits of two-front type to $u^\varepsilon(x) = -1 + \mathcal{O}(\varepsilon)$ (9). Studying the impact of parameter variations on $W^u(\mathcal{M}_\pm^\varepsilon) \cap W^s(\mathcal{M}_\pm^\varepsilon)$ thus provides an alternative way to study the bifurcations of (stationary) two-

front patterns, compared to that of Section 6.1. As usual for these kinds of perturbed, integrable systems, especially for periodically driven systems, we will consider their restrictions to specific values of x , or equivalently, their intersections with a plane $\{x = x_0\}$, which we will denote by $W_{x_0}^{s/u}(\mathcal{M}_\pm^\varepsilon)$ and approximate by $\widetilde{W}_{x_0}^{s/u}(\mathcal{M}_\pm^\varepsilon)$. That is,

$$\begin{aligned} \widetilde{W}_{x_0}^u(\mathcal{M}_\pm^\varepsilon) &:= \widetilde{W}^u(\mathcal{M}_\pm^\varepsilon) \cap \{x = x_0\} \\ &= \left\{ \left(u_{\text{up}}(x_0; \phi) + \varepsilon u_{u,1}(x_0; \phi), \frac{d}{dx} u_{\text{up}}(x_0; \phi) + \varepsilon \frac{d}{dx} u_{u,1}(x_0; \phi), x \right) \right\}, \end{aligned} \quad (127)$$

and similarly for the approximations of the other stable and unstable manifolds. For functions $F(U, V, x)$ that are periodic in x , i.e., $F(U, V, x + X) \equiv F(U, V, x)$ for some $X > 0$, we consider the usual Poincaré section at, e.g., $\{x = 0\}$ and use $W_0^{s/u}(\mathcal{M}_\pm^\varepsilon)$ to study the geometry of the

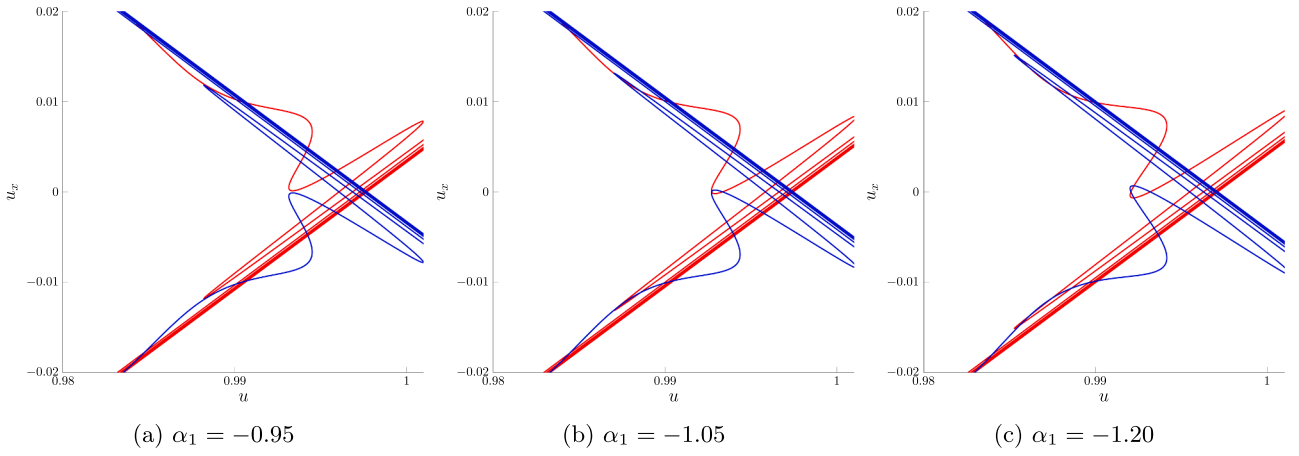


Fig. 17. Approximations of $W_0^{s/u}(\mathcal{M}_\varepsilon^\pm)$ with $\varepsilon = 0.1$ and $F(U, V, x) = \alpha_1 \cos(2\pi x)U$ for various values for α_1 . The red lines denote the unstable manifold $W_0^u(\mathcal{M}_\varepsilon^\pm)$, and the blue lines the stable manifolds $W_0^s(\mathcal{M}_\varepsilon^\pm)$. We focus on the small region in the middle where a small segment of the red manifold dips right to left down toward a small (symmetrically-disposed) segment of the blue manifold. (a) these local segments do not intersect, (b) these local segments have two intersections that are created in a saddle-node bifurcation, (c) these local segments have four intersections after a pitchfork bifurcation.

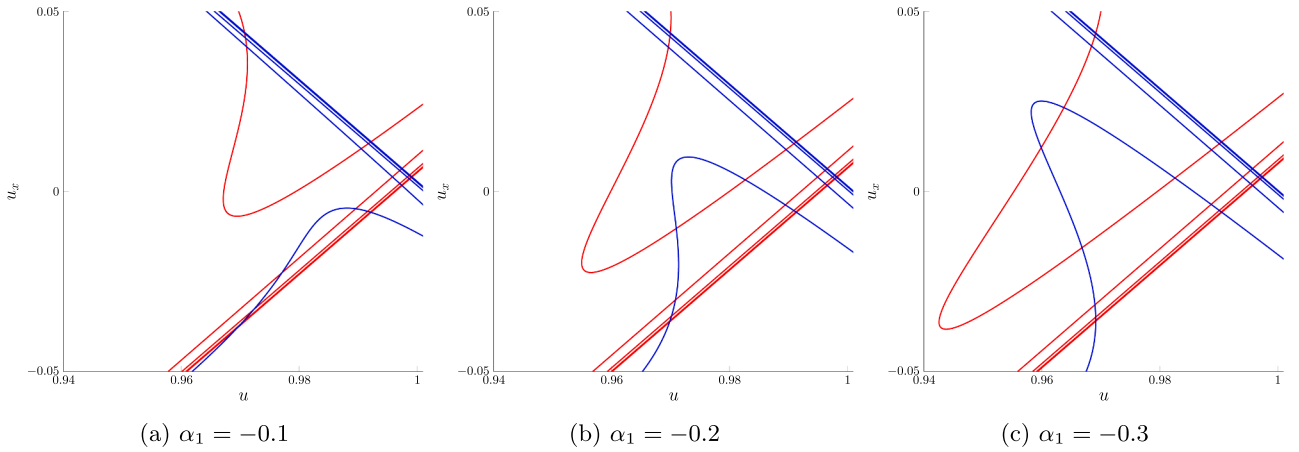


Fig. 18. Approximations of $W_0^{s/u}(\mathcal{M}_\varepsilon^\pm)$ with $\varepsilon = 0.1$ and $F(U, V, x) = \alpha_1 \cos(\pi x)U + 0.15 \sin(\pi x)$ for various values for α_1 . The red lines denote the unstable manifold $W_0^u(\mathcal{M}_\varepsilon^\pm)$ and the blue lines the stable manifolds $W_0^s(\mathcal{M}_\varepsilon^\pm)$. (a) No intersection between manifolds, (b) two new fixed points appear in a first saddle-node bifurcation, (c) four new fixed points exist after a second saddle-node bifurcation.

system (7). In Fig. 16, we present the results for a function F that possesses the symmetry $F(U, V, x) = -F(-U, -V, x)$ of the topographic case (although the example does not represent a topography $H(x)$). The case of functions $F(U, V, x)$ that possess the symmetry $F(U, V, x) = F(-U, -V, x)$ is similar. Our findings are in agreement with those of Section 6.1. For a changing bifurcation parameter, first a saddle-node bifurcation occurs at which $W_0^u(\mathcal{M}_\varepsilon^\pm)$ and $W_0^s(\mathcal{M}_\varepsilon^\pm)$ are tangent. This tangency indicates the beginning of the process that leads to a full intersection of the $W_0^u(\mathcal{M}_\varepsilon^\pm)$ - and $W_0^s(\mathcal{M}_\varepsilon^\pm)$ -lobes. Directly after the saddle-node bifurcation, the ‘nose’ of the $W_0^u(\mathcal{M}_\varepsilon^\pm)$ -lobe is inside that of $W_0^s(\mathcal{M}_\varepsilon^\pm)$ (and vice versa), it again punctures through $W_0^u(\mathcal{M}_\varepsilon^\pm)$ by the pitchfork bifurcation (due to the symmetry) – see again Fig. 16 and also Fig. 17, which confirms that this crossing through-scenario appears simultaneously at several – in fact countably many – locations in the (u, p) -plane.

This geometric point of view is especially helpful in understanding the distinction between systems (7) with or without symmetry. In Fig. 18, we present the generic results for a function $F(U, V, x)$ that does not possess one of the aforementioned symmetries. In line with the results in Section 6.1, it shows that generically having a second saddle node bifurcation is more natural than the pitchfork bifurcation of the symmetric case.

We conclude this (sub)section by the observation that the present geometrical approach based on the intersections $W_0^u(\mathcal{M}_\varepsilon^\pm) \cap W_0^s(\mathcal{M}_\varepsilon^\pm)$ does not yield any information on the stability of the associated multi-front patterns as solutions of PDE (2). It is natural to expect that there may be a relation between the geometrical nature of a (planar) intersection $W_0^u(\mathcal{M}_\varepsilon^\pm) \cap W_0^s(\mathcal{M}_\varepsilon^\pm)$ and the stability character of the associated N -front pattern in the infinite-dimensional setting. This is a subject of future research.

6.3. N -front dynamics: The existence of nearby attractors

In this section, we consider general aspects of the dynamics of multi-front patterns in periodically driven Allen-Cahn Eqs. (2). For simplicity we again focus on the topographic case, i.e. unlike the general case of the (directly) preceding subsections, we consider a driving term $F(U, V, x)$ given by (4) with topography $H(x)$ that is periodic with period $X > 0$. Thus, we know that $\mathcal{R}_{\text{up}}(\phi) = \mathcal{R}_{\text{down}}(\phi)$ ($= \mathcal{R}(\phi)$) (38) and that $\mathcal{R}(\phi)$ is also periodic with period X (39). On the other hand, we consider general periodic topographies, so that $\mathcal{R}(\phi)$ may have more than two zeros during each full period, unlike the examples considered so far.

As before, we first focus on the existence of stationary N -front patterns that will play a central role as attractors of the flow (see Claim

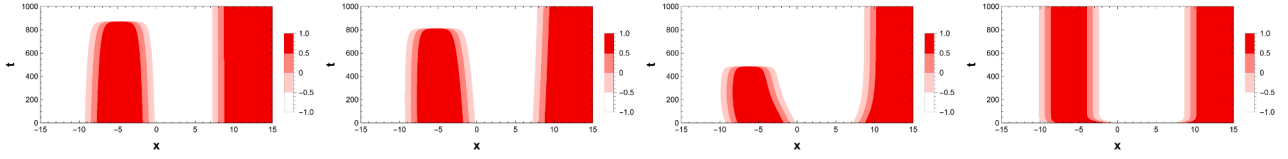


Fig. 19. Simulations of (2) on the x -interval $(-15, 15)$ (with homogeneous Neumann boundary conditions) with spatially periodic inhomogeneity given by (3) with $f_1(x) = \cos x$, $f_2(x) \equiv f_3(x) \equiv 0$ starting from the three-front initial condition $U(x, 0) = \tanh(x + 8.5) - \tanh(x + 1) + \tanh(x - 8)$ and $\varepsilon = 0$ (a), 0.001 (b), 0.01 (c), and 0.1 (d).

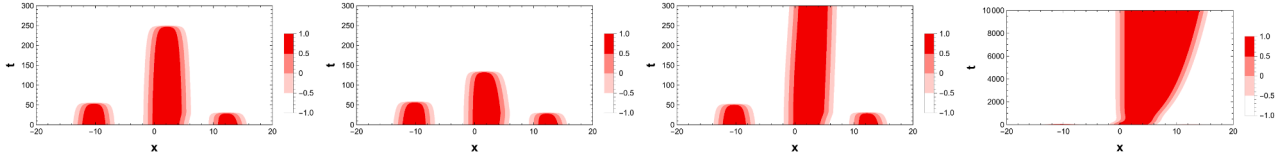


Fig. 20. Simulations of (2) with a localized topographical inhomogeneity (4) ($\varepsilon = 0.1$) on the interval $(-30, 30)$ with homogeneous Neumann boundary conditions (plotted on $(-20, 20)$) starting from the six-front initial condition $U(x, 0) = -1 + \tanh(x + 13.0) - \tanh(x - 7.5) + \tanh(x + 1.1) - \tanh(x - 5.0) + \tanh(x - 10.0) - \tanh(x - 15.9)$. (a) The flat homogeneous Allen-Cahn case: $H(x) \equiv 0$. (b) A weak valley: $H(x) = -H_{\text{alg}}(x; 2)$. (c)/(d) A weak hill: $H(x) = H_{\text{alg}}(x; 2)$.

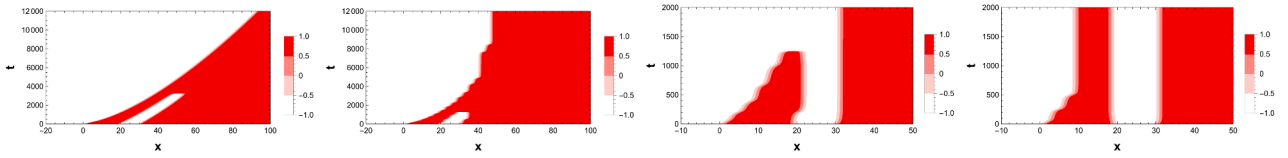


Fig. 21. Simulations of (2) with ‘mixed’ topographical inhomogeneity (4) given by $H(x; \delta) = -H_{\text{alg}}(x; 0.5) + \delta \sin 3x$ (5), $\varepsilon = 0.1$ and three-front initial condition $U(x, 0) = \tanh(x - 1) - \tanh(x - 19) + \tanh(x - 30)$; simulations are done for $x \in (-200, 200)$ with homogeneous Neumann boundary conditions. (a) $\delta = 0$, (b) $\delta = 0.025$, (c) $\delta = 0.04$, and (d) $\delta = 0.06$.

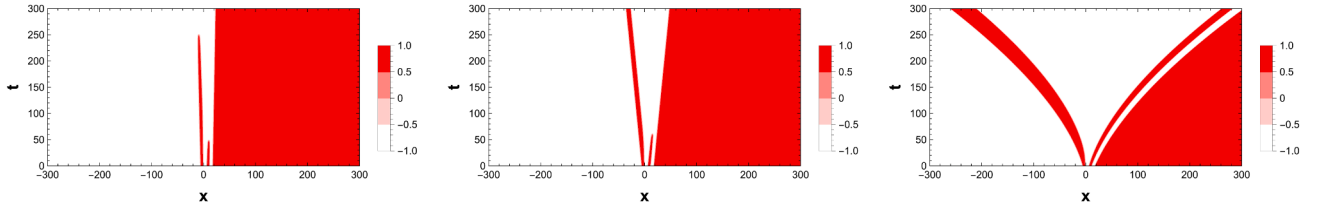


Fig. 22. Simulations of (2) with topographical inhomogeneity (4) given by $-H_{\text{alg}}(x; p)$ (5), $\varepsilon = 0.1$ and five-front initial condition $U(x, 0) = \tanh(x + 6) - \tanh(x + 0.5) + \tanh(x - 6) - \tanh(x - 11.5) + \tanh(x - 17.5)$; $p = .4$ in (a), $p = .0$ in (b) and $p = -0.4$ in (c); unlike in Fig. 12(c)/(d), the solitary one-front centered around $x = 0$ is unstable. All simulations are done for $x \in (-500, 500)$ (with homogeneous Neumann boundary conditions) while only the middle $x \in (-300, 300)$ part of the pattern is plotted.

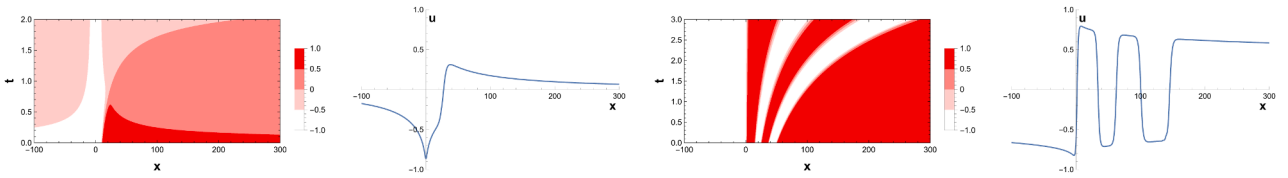


Fig. 23. Simulations of (2) with topographical inhomogeneity (4) $H(x) = -H_{\text{alg}}(x; p)$ (5) for $p < -1$: $H''(x)$ has also become unbounded (with $\varepsilon = 0.1$, $x \in (-100, 300)$, homogeneous Neumann boundary conditions). (a)/(b) One-front dynamics for $p = -1.5$ (with initial condition $U(x, 0) = \tanh(x - 10)$); $t = 1.0$ in (b). (c)/(d) Five-front dynamics for $p = -1.2$ (with initial condition $U(x, 0) = \tanh(x - 1) - \tanh(x - 15) + \tanh(x - 25) - \tanh(x - 37) + \tanh(x - 50)$); $t = 2.0$ in (d). Note that there seems to be a sharp change in $U_x(x, t)$ near $x = 0$ in both simulations (i.e., in (b) and (d)), however zooming in shows that the solution remains smooth (in the t -intervals considered).

1 below). The situation is different from that of localized topographies considered in Theorem 3. There are countably many stationary N -fronts from which (roughly) one out of 2^N is stable.

Theorem 4. Consider the weakly heterogeneous Allen-Cahn Eq. (2) with $F(U, U_x, x)$ given by the topography $H(x)$ (4). Assume that $H(x)$ is sufficiently smooth and is periodic with (minimal) period $X > 0$, so that the Melnikov function $S(\psi)$ associated with $H(x)$, which is periodic with period $Y = \sqrt{2}X$ (recall (39), (83)), has an even number $K \geq 0$ of non-degenerate zeros $0 \leq \psi_{s,1} < \psi_{s,2} < \dots < \psi_{s,K} < Y$ on the interval $[0, Y)$ (i.e., $S(\psi_{s,k}) = 0$ with $S'(\psi_{s,k}) \neq 0$ and necessarily $\text{sign } S'(\psi_{s,k}) \neq \text{sign } S'(\psi_{s,k+1})$). For $N \geq 2$, let $(\tilde{\psi}_1, \tilde{\psi}_2, \dots, \tilde{\psi}_N)$ denote a stationary N -front pattern that connects $u_-^\varepsilon(x) = -1 + \mathcal{O}(\varepsilon)$ as $x \rightarrow -\infty$ to $u_+^\varepsilon(x) = 1 + \mathcal{O}(\varepsilon)$ as $x \rightarrow \infty$.

as $x \rightarrow +\infty$ (9), where $\tilde{\psi}_j$ is the location of the j -th front (with $\tilde{\psi}_j < \tilde{\psi}_{j+1}$, $j = 1, \dots, N$). Let $\Delta\tilde{\psi}_j = \tilde{\psi}_{j+1} - \tilde{\psi}_j$ be the distance between two successive fronts. Let $0 < \varepsilon \ll 1$.

- If $(\tilde{\psi}_1, \tilde{\psi}_2, \dots, \tilde{\psi}_N)$ is a stationary N -front pattern with $|\Delta\tilde{\psi}_j|/\log \varepsilon = \rho_j$ as $\varepsilon \rightarrow 0$, then $\rho_j \geq 1$ for all $j = 1, \dots, N$.
- Consider a sequence $(\tilde{\psi}_1, \tilde{\psi}_2, \dots, \tilde{\psi}_N) = (\psi_{s,j_1} + n_1(\varepsilon)Y, \psi_{s,j_2} + n_2(\varepsilon)Y, \dots, \psi_{s,j_N} + n_N(\varepsilon)Y)$ of N consecutive zeros of $S(\psi)$, i.e., $\tilde{\psi}_j < \tilde{\psi}_{j+1}$ for $j = 1, 2, \dots, N$ (with $n_j(\varepsilon) \in \mathbb{Z}$, $i_j = 1, 2, \dots, K$). If $\Delta\tilde{\psi}_j = \tilde{\psi}_{j+1} - \tilde{\psi}_j$ increases faster than $|\log \varepsilon|$ as $\varepsilon \rightarrow 0$, i.e., if $(n_{j+1}(\varepsilon) - n_j(\varepsilon))/|\log \varepsilon| > Y$ as $\varepsilon \rightarrow 0$ ($j = 1, \dots, N-1$), then there exists a stationary N -front pattern $(\tilde{\psi}_1, \tilde{\psi}_2, \dots, \tilde{\psi}_N)$ asymptotically close to $(\tilde{\psi}_1, \tilde{\psi}_2, \dots, \tilde{\psi}_N)$, i.e., $\tilde{\psi}_j = \tilde{\psi}_j + r_j(\varepsilon)$ with $r_j(\varepsilon) \rightarrow 0$ as $\varepsilon \rightarrow 0$ ($j = 1, \dots, N$).
- Associated to any ordered sequence $(n_1(\varepsilon), n_2(\varepsilon), \dots, n_N(\varepsilon)) \in \mathbb{Z}^N$ (i.e., $n_j(\varepsilon) < n_{j+1}(\varepsilon)$) with

$(n_{j+1}(\epsilon) - n_j(\epsilon))/|\log \epsilon| > Y$ as $\epsilon \rightarrow 0$ ($j = 1, \dots, N-1$) there are K^N distinct stationary N -front solutions denoted by $(\tilde{\psi}_{1,j_1}, \dots, \tilde{\psi}_{N,j_N}) = (\psi_{s,j_1} + n_1(\epsilon)Y + r_{1,j_1}(\epsilon), \dots, \psi_{s,j_N} + n_N(\epsilon)Y + r_{N,j_N}(\epsilon))$, with $r_{i,j_i}(\epsilon) \rightarrow 0$ as $\epsilon \rightarrow 0$ ($i = 1, 2, \dots, K$, $j = 1, \dots, N$). The spectral stability of $(\tilde{\psi}_{1,j_1}, \dots, \tilde{\psi}_{N,j_N})$ is determined by N (asymptotically small) eigenvalues $\lambda_j(\epsilon; i_1, \dots, i_N)$ that are given to leading order by

$$\lambda_j(\epsilon; i_1, \dots, i_N) = -\epsilon S'(\psi_{s,j_j}), \quad j = 1, 2, \dots, K, \quad j = 1, \dots, N.$$

Thus, $(\frac{1}{2}K)^N$ of all K^N stationary N -front patterns associated to the sequence $(n_1(\epsilon), n_2(\epsilon), \dots, n_N(\epsilon))$ are spectrally stable, all $(1 - \frac{1}{2N})K^N$ others are unstable.

We refer to App. E for a sketch of the proof of this theorem. An essentially identical result holds for stationary N -front patterns that connect $u_+^*(x) = 1 + \mathcal{O}(\epsilon)$ at $-\infty$ to $u_{-1}^*(x) = (-1)^N + \mathcal{O}(\epsilon)$ at $+\infty$.

Transitions between the cases of zero stationary N -fronts and of countably many stationary N -front patterns occur as one of the ρ_j passes through one, where we recall that the ρ_j are defined in Theorem 4 by $\Delta\tilde{\psi}_j/|\log \epsilon| = \rho_j$ as $\epsilon \rightarrow 0$. In Secs. 6.1 and 6.2, we studied the appearance of critical points for the specific $N = 2$ case with a topography $H(x)$ determined by $H'(x) = f_2(x) = \alpha_2 \sin(kx)$. In the general N -front setting – with an $S(\psi)$ that may have more than two zeros, unraveling the transitions becomes technically more involved, but the essential concepts are the same, especially regarding the geometry of the manifolds and their intersections, as shown in Section 6.2. Therefore, we refrain from further details.

Remark 14. It cannot be excluded, a priori, that some critical points may exist for $\rho_j = 1$ in the case that $S(\psi)$ does not have zeros (i.e., for $K = 0$ in Theorem 4); however, $S(\psi)$ cannot change sign, thus there can be no stationary N -front configurations $(\tilde{\psi}_1, \tilde{\psi}_2, \dots, \tilde{\psi}_N)$ with both $S(\tilde{\psi}_1) = 16e^{-(\tilde{\psi}_2 - \tilde{\psi}_1)/\epsilon} > 0$ and $S(\tilde{\psi}_N) = -16e^{-(\tilde{\psi}_N - \tilde{\psi}_{N-1})/\epsilon} < 0$.

The abundance of stable stationary N -front patterns established by Theorem 4 motivates the upcoming claim as a natural next step. It shows that any N -front configuration with successive fronts that are sufficiently far apart must evolve toward a nearby attracting stationary N -front. This is corroborated by the simulations shown in Fig. 19 of an evolving three-front pattern (similar to the two-front example of Fig. 15): for ϵ sufficiently large, i.e., if the separation distance between successive fronts is sufficiently large with respect to the $|\log \epsilon|$ distance scale, the three-front pattern evolves towards a nearby attracting stationary patterns as described by the upcoming theorem (see Fig. 19(d)), for smaller ϵ (Fig. 19(b),(c)), the three-front dynamics are similar those of the homogeneous Allen-Cahn equation (Fig. 19(a)) in which two-fronts merge and only a stable attracting one-front pattern remains.

Claim 1. Let the hypotheses of Theorem 4 hold and assume $K \geq 0$. Let $(\psi_{0,1}(0), \psi_{0,2}(0), \dots, \psi_{0,N}(0))$ (with $\psi_{0,j} < \psi_{0,j+1}$) indicate the initial position of an evolving N -front pattern in (2) that connects $u_-^*(x) = -1 + \mathcal{O}(\epsilon)$ at $x \rightarrow -\infty$ to $u_{-1}^{N+1}(x) = (-1)^{N+1} + \mathcal{O}(\epsilon)$ at $x \rightarrow +\infty$. If $N \geq 2$ and $\Delta\psi_{0,j} = \psi_{0,j+1} - \psi_{0,j}$ increases faster than $|\log \epsilon|$ as $\epsilon \rightarrow 0$ (for all $j = 1, \dots, N-1$), then the N -front pattern given by $(\psi_{0,1}(\tau), \psi_{0,2}(\tau), \dots, \psi_{0,N}(\tau))$ is attracted to a nearby stable stationary N -front pattern $(\tilde{\psi}_1, \tilde{\psi}_2, \dots, \tilde{\psi}_N)$. More specifically, under the flow of (2),

$$\lim_{\tau \rightarrow \infty} (\psi_{0,1}(\tau), \psi_{0,2}(\tau), \dots, \psi_{0,N}(\tau)) = (\tilde{\psi}_1, \tilde{\psi}_2, \dots, \tilde{\psi}_N) \quad \text{with } |\psi_{0,j} - \tilde{\psi}_j| < Y, \quad j = 1, 2, \dots, N.$$

If $N = 1$, then $\lim_{\tau \rightarrow \infty} \psi_{0,1}(\tau) = \tilde{\psi}_1$ with $|\psi_{0,j} - \tilde{\psi}_j| < Y$ – without any additional conditions.

It should be noted that, in the formulation of this claim, we excluded (non-generic) situations in which an N -front pattern is exactly in the stable manifold of an unstable critical point $(\tilde{\psi}_1, \tilde{\psi}_2, \dots, \tilde{\psi}_N)$ of saddle type.

To formulate and prove this claim as a theorem, one would need first to establish the validity of the N -front interaction ODE (84). Such a result will include the existence of an N -dimension manifold J_N (in an appropriately chosen function space H) to which the dynamics of N -front patterns driven by (2) are attracted (exponentially). Based on this, a precise description of the N -front pattern $(\psi_{0,1}(\tau), \psi_{0,2}(\tau), \dots, \psi_{0,N}(\tau))$ can be given; more specifically: pattern $(\psi_{0,1}(\tau), \psi_{0,2}(\tau), \dots, \psi_{0,N}(\tau))$ is determined by choosing its initial condition $(\psi_{0,1}(0), \psi_{0,2}(0), \dots, \psi_{0,N}(0))$ sufficiently close (in the norm associated to H) to J_N . The validation of Claim 1 then follows from the validity of N -front interaction ODE (84) for the dynamics on the (exponentially) attracting manifold J_N .

As was already argued in the beginning of Section 4, the existence of manifold J_N and the validity of the interaction ODE (84) can be established by methods developed in the literature, see [26,27]. We do not

see any fundamental obstructions to apply these methods to the present set-up, however, we refrain from going into the analytical details here.

7. Discussion

In this work, we have analyzed several aspects of multi-front dynamics of the Allen-Cahn Eq. (2) driven by general, small-amplitude spatial heterogeneities. Our research is not comprehensive. For simplicity, we emphasized heterogeneities (4) that are either purely periodic or spatially localized (with uniformly decaying tails that decay with the same rates as $x \rightarrow \pm\infty$). Nevertheless, the methods we have employed may be used to investigate more general (and more technical) heterogeneities. As examples, in this section, we briefly discuss the effect of (localized) heterogeneous driving terms on the process of coarsening and the ‘mixed’ case of a localized inhomogeneity limiting on a spatially periodic background. In addition, we discuss the apparent validity of our analysis in the regime where it is not a priori expected to hold, i.e., for localized topographies with ‘tails’ that no longer decay. Finally, we briefly return to the ecosystem setting that motivated part of our research.

The impact on coarsening. The Cahn-Hilliard and Allen-Cahn equations are the two prototypical, and thus most studied, models for the process of coarsening – see [46,47] and the references therein. As follows from the classical literature [17–19], any multi-front initial condition of the homogeneous Allen-Cahn Eq. (1) evolves towards either one of the stable steady states $U(x, t) \equiv u_{\pm} = \pm 1$ (Section 1.4) or towards a stationary one-front solution $u_{\text{up,down}}(x; \phi)$ (15).

In this article, we have shown that the coarsening process can be altered or even stopped by spatial heterogeneities. We found that spatially periodic heterogeneities stop the coarsening process completely in multi-front patterns in which there is a sufficiently large separation distance between the fronts, recall Claim 1. (We add that, while Claim 1 only considers topographic heterogeneities, a similar claim can be formulated for general spatially periodic inhomogeneities $\epsilon F(U, V, x)$ in (2).) Also, the simulations of Figs. 2(a) and 19 show that there is an interplay between coarsening and pinning for (spatially) periodically driven multi-front patterns with fronts that are so close to each other that the general claim of Claim 1 cannot hold.

We also studied spatial heterogeneities which consist of a spatially localized term, and we showed that here the picture is more complex. In Fig. 20, simulations of (2) with two different localized topographies, $H(x) = \pm H_{\text{alg}}(x; 2)$, are presented, and these show that two different outcomes. The topography $H(x) = -H_{\text{alg}}(x; 2)$ speeds up, and thus strengthens, the coarsening process, as can also be deduced from (84). Fronts that are far away from the origin are driven towards the origin by the inhomogeneity, so that the full extent of the multi-front pattern shrinks by a persistent ‘squeezing process’. More precisely, the extent of an N -front pattern is determined by the distance $\psi_N - \psi_1$ between its two outermost fronts. In fact, in this case, (81) generalizes to

$$\frac{d}{dt}(\psi_N - \psi_1) = -\epsilon [S(\psi_N) - S(\psi_1)] - 16[e^{(\psi_2 - \psi_1)} + e^{(\psi_N - \psi_{N-1})}],$$

recall also (83) and (84). Thus, it follows from Lemma 2 for $\psi_1 \ll -1 < 1 \ll \psi_N$ and $H'(x)$ localized with algebraically decaying tails that (to leading order),

$$\frac{d}{dt}(\psi_N - \psi_1) = -\frac{2^{\frac{1}{2}(p+3)}}{3} \epsilon [\tilde{h}_+(p)\psi_N^{-p} - \tilde{h}_-(p)|\psi_1|^{-p}] - 16[e^{(\psi_2 - \psi_1)} + e^{(\psi_N - \psi_{N-1})}]. \quad (128)$$

Since $\tilde{h}_{\pm}(p) = \pm(1-p)$ for $H_{\text{alg}}(x; p)$, it follows that $H(x) = -H_{\text{alg}}(x; p)$ with $p > 1$ indeed strengthens the homogeneous squeezing process of (81). Similarly, the topography $H(x) = +H_{\text{alg}}(x; p)$ with $p > 1$ counteracts the homogeneous process. In fact, due to the exponential decay of the homogeneous terms, the inhomogeneous effects of (128) dominate when the fronts are sufficiently far apart: the extent of a multi-front pattern will grow for $\tilde{h}_+(p) < 0 < \tilde{h}_-(p)$ – see the simulation in Fig. 20(c),(d) that shows an evolving two-front pattern consisting of a stationary front pinned at $x = 0$ and a second front traveling towards $x = +\infty$.

Additional examples are presented in Figs. 1(b), (e), and 2(b). Each of these examples shows that a spatially localized inhomogeneity may yield an endstate that is significantly different from the endstate of the homogeneous system. Here, for example, we compare Fig. 1(a) to 1(b) and Fig. 1(d) to 1(e); also, in Fig. 2(b), the homogeneous one-front endstate of a five-front initial condition is replaced by a situation in which eventually $U \rightarrow -1$ (to leading order in ε) on the entire domain, while in Fig. 20(d) $U \rightarrow +1$ (to leading order in ε) on the positive x -axis, instead of $U \rightarrow -1$ on $\mathbb{R}$ as in Fig. 20(a). Apart from these effects, spatially localized inhomogeneities may also significantly delay the coarsening process (see again Fig. 2(b)).

These phenomena can be understood by the approach developed here in combination with the classical insights of [17–19], see also the final paragraph of Section 5.3. This is not the case for the simulations shown in Fig. 2(c) and below in Fig. 22(c), which suggest that spatial inhomogeneities may not only stop the coarsening process, they may also drive a spreading process that (continuously) increases the extent of the multi-front pattern without reducing the number of fronts. See the discussion below.

General topographies. Other topographies, beyond the spatially localized and periodic topographies cases considered in Secs. 5 and 6, involving general inhomogeneous terms $F(U, V, x)$ (2) can be studied by our methods. As a simple example of this, we present in Fig. 21 simulations of (2) with a mixed topographical inhomogeneity consisting of the sum of a localized topography and δ times a spatially periodic topography. The associated driving Melnikov function $S(\psi)$ (84) can be written as $S(\psi) = S_{\text{loc}}(\psi) + \delta S_{\text{per}}(\psi)$. For δ sufficiently small, the pinning impact of $S_{\text{per}}(\psi)$ only kicks in for sufficiently large ψ , i.e., when $\log \psi = p |\log \delta|$, to leading order in δ (recall Lemma 2 and Eq. (83)). Thus, while the ψ_2 -fronts and ψ_3 -fronts merge and ψ_1 travels as a solitary front to $+\infty$ (with decaying speed) in the homogeneous three-front simulation of Fig. 21(a) (see Section 5.3 and Figs. 12, 22(a)), a small spatially periodic term slows down the $\psi_{1,2,3}$ -fronts a bit and the remaining ψ_1 -front eventually gets pinned. Hence, the configuration now approaches a (pinned) stationary ψ_1 one-front pattern (Fig. 21(b)). As δ increases further, ψ_3 gets pinned before it can be reached by ψ_2 . However, since ψ_2 is also slowed down by the periodic effects, it is overtaken by ψ_1 . That is, the solution also approaches a pinned stationary one-front pattern, but now originating from ψ_3 . Finally, for still larger values of δ , the ψ_1 -front is pinned before it reaches ψ_2 front (which is also pinned), and the entire configuration approaches a stationary three-front.

The validity of the N -front interaction equations. As noted at the beginning of Section 4 and discussed in somewhat more detail at the end of Section 4.1, both the existence of an attracting, approximate, N -dimensional manifold $\mathcal{M}_N^0$ for the (semi-)flow generated by (2) and the validity of Eq. (78)/(84) as leading order descriptions of the flow on $\mathcal{M}_N^0$ can be established by a direct application of existing methods [26,27]. However, manifold $\mathcal{M}_N^0$ has boundaries. The dynamics near the boundaries associated with fronts getting too close to each other (or to fronts beyond these boundaries) can again be controlled by existing classical methods, see especially [18]. Then, for the opposite situations in which the fronts are widely separated, one also needs to verify that the neglected $\mathcal{O}(\varepsilon^2)$ terms $\varepsilon^2 R_2^{j,N}(\psi_1, \dots, \psi_N; \varepsilon)$ ($j = 1, \dots, N$) in (84) remain of higher order. In general, one expects that the magnitudes of the $R_2^{j,N}(\psi_1, \dots, \psi_N; \varepsilon)$ terms will decrease as $|\psi_j| \gg 1$. Nevertheless, at this depth of the asymptotic analysis, it could be that the interactions of the fronts involve more than just nearest-neighbor interactions and depend on other fronts further away, or even on all other fronts, as is the case for localized structures in semi-strong interaction [43,48].

Relaxation of the hypothesis that F is bounded provides another compelling reason to analyze the validity of the ODE system (84) beyond the classical weak interaction cases considered in [26,27]. Application of these existing methods requires that the inhomogeneous term $\varepsilon F(U, V, x)$

in (2) is bounded. Hence, for topographical inhomogeneities (4), we have required in this article that $H'(x)$ is bounded (for all $x \in \mathbb{R}$). However, as we already noted in (10) at the beginning of our analysis, the asymptotic results we derived may extend beyond the case in which $H'(x)$ is bounded. It may suffice to require only that $H''(x)$ is bounded.

To study the impact of this relaxation of the hypothesis on the (coarsening) dynamics of N -front configurations, we redo the one-front analysis of Section 5.1 and consider the evolution of a solitary front $\psi(\tau)$ that starts out with $\psi(0) = \psi_0 \gg 1$, but now at the (positive) tail of a Melnikov function $S(\psi)$ associated to a topography with algebraic tails, i.e., we use Lemma 2 instead of Lemma 1. We find that, under the assumptions that $\tilde{h}_+(p) < 0$ and that the higher order corrections to (84) remain of higher order as $S(\psi)$ becomes small, the algebraic counterparts of (103) and (104) are given by

$$\frac{d\psi}{d\tau} = \varepsilon \tilde{H}_+ \psi^{-p}, \quad \tilde{H}_+(p) = \frac{2^{\frac{1}{2}(p+3)}}{3} |\tilde{h}_+(p)| > 0, \quad \psi(\tau) = \left(\varepsilon \tilde{H}_+ \tau + \psi_0^{p+1} \right)^{\frac{1}{p+1}}, \quad p \neq -1. \quad (129)$$

Hence, the leading order distance between adjacent fronts $\tilde{\psi}(\tau) < \psi(\tau)$, which start out sufficiently far from each other, evolves as

$$\psi(\tau) - \tilde{\psi}(\tau) = \frac{1}{(p+1)^{\frac{2p+1}{p+1}}} \frac{\psi_0^{p+1} - \tilde{\psi}_0^{p+1}}{(\varepsilon \tilde{H}_+ \tau)^{\frac{1}{p+1}}} + \text{h.o.t.} \quad \text{for } \tau \gg \frac{\psi_0^{p+1}}{\varepsilon}. \quad (130)$$

This is the algebraic counterpart of (105). We now examine the dynamics for $p > 0$, $p = 0$, and $p < 0$, respectively, beginning with the former.

For $p > 0$, i.e., for the case in which the validity of (84) can be settled following [26,27], the situation is similar to that of exponentially decaying topographies. The distance $\psi(\tau) - \tilde{\psi}(\tau)$ decreases (algebraically in time), $\tilde{\psi}(\tau)$ catches up (eventually) with $\psi(\tau)$, and the two fronts merge and disappear. This scenario is confirmed by the simulations shown in Figs. 2(b) and 22(a), in which the initial data consist of five-front data.

The case $p = 0$ is special, since (130) predicts that $\psi(\tau) - \tilde{\psi}(\tau)$ will approach a constant value. Although this is only a leading order result, it too is corroborated by simulations. As shown for example in Fig. 22(b), the ψ_1 -front and ψ_2 -front seem to ‘lock’ to each other and travel with constant speed to the left maintaining a constant distance, while the ψ_3 -front travels as a solitary front to the right with the same constant speed. (We add that this all happens after the ψ_3 -front and ψ_4 -front have merged, as in Fig. 22(a)).

A significant change occurs as p takes on negative values. Namely, for $-1 < p < 0$, (130) predicts that the distance $\psi(\tau) - \tilde{\psi}(\tau)$ between two successive fronts will increase. Although the inhomogeneous term $F(U, V, x)$ in (2) is no longer bounded (since $H'(x)$ is not bounded), this prediction is again confirmed by the numerics. See Figs. 2(c) and 22(c). In both simulations, all of the distances between neighboring fronts increase. As a consequence, no coarsening takes place (if the initial distances between neighboring fronts are not too small). The end-states of the simulations are formed by dispersing trains of fronts, either moving to $-\infty$ on the left (see the (ψ_1, ψ_2) -train in Fig. 22(c)) or to $+\infty$ on the right (see the (ψ_3, ψ_4, ψ_5) -train in Fig. 22(c) and the $(\psi_1, \psi_2, \psi_3, \psi_4, \psi_5)$ -train in Fig. 2(c)).

Thus, the simulations over this entire range of p values strongly suggest that the front interaction system (84) remains valid beyond the natural condition that inhomogeneity $F(U, V, x)$ in (2) is bounded (for all $x \in \mathbb{R}$). However, both the above analysis and further simulations indicate that there is a second critical value of p for which this may no longer be the case. As p decreases below -1 , $H''_{\text{alg}}(x; p)$ also is no longer bounded – and thus also the first step in our asymptotic approach needs to be reconsidered (see (10), the background state $u_b^*(x)$ is no longer bounded on $\mathbb{R}$ if $H''(x)$ is not). The simulations of Fig. 23 show that the front dynamics of (2) indeed have gone through an essential change in character; namely, the multi-front patterns can no longer be seen as perturbations of combinations of the homogeneous one-front patterns $u_{\text{up}}(x; \phi)$ and $u_{\text{down}}(x; \phi)$ (15), which is the backbone of our asymptotic approach. Nevertheless, the patterns exhibited by (2) still have a clear multi-front nature, and one might expect that also the case $p < -1$, perhaps combined with the requirement that $|p+1| \ll 1$, could be studied analytically. Such a study

would be especially interesting, since the (now quite formal) approximations in (129) suggest that the position $\psi(\tau)$ of a front may have a finite-time blow-up character, i.e., its speed may become unbounded as τ approaches a critical value τ_c . Although simulating (2) under these conditions becomes quite subtle, the results of these simulations do not contradict this possibility. We refer to Fig. 23 where a simulation is shown in which the fronts travel fast and continue to speed up.

Thus, a careful study of the dynamics of N -front patterns in (2) with inhomogeneity $F(U, V, x)$ – either in the form of (4) or not – that may become unbounded (in x) would be both interesting and challenging.

Implications for ecosystem dynamics. The validity of the pulse interaction equations for algebraically ‘decaying’ inhomogeneities with $p < 0$, i.e., for topographies with inclinations that grow faster than linear, is a mathematical issue without obvious ecological relevance. However, our investigations into coarsening and pinning of multi-front dynamics within interaction Eqs. (78) and/or its simplification (84) can be expected to be relevant in the context of ecosystems. The most simple ecosystem models are singularly perturbed two-component reaction-diffusion models (see [8,49] and the references therein). By the methods of [8,43,50], equivalents of (78)/(84) can be derived (and validated) that govern the multi-pulse or multi-front interactions of vegetation patches in ecosystem models (although one should not underestimate the necessary technicalities). Moreover, these systems will not be identical to (78)/(84). However, as (78)/(84), these systems will have an evident division into terms governed by the ‘external’ topography and ‘internal effects’ representing the pulse- or front-interactions that also take place in the homogeneous setting [8,43]. In fact, the latter effects will not be exponentially small – as in the case in (78)/(84) – due to the semistrong nature of these interactions in singularly perturbed systems [48].

Thus, the semistrong two-component equivalents of (78)/(84) will provide insight into the dynamics of coarsening and pinning of vegetation patches in response to the nature of the topography of the domain. Understanding the mechanisms that distinguish between coarsening and pinning is relevant for ecology, since desertification of dryland ecosystems can be seen as a coarsening process of vegetation patches in which the vegetation patches are replaced step by step by bare soil states [51] and since pinned vegetation patches persist. Further, realistic topographies are not necessarily of small amplitude and will vary on various spatial scales; nevertheless, the present work is a step towards building a fundamental understanding of these mechanisms.

CRedit authorship contribution statement

Robbin Bastiaansen: Writing – review & editing, Writing – original draft, Visualization, Software, Formal analysis; **Arjen Doelman:** Writing – review & editing, Writing – original draft, Visualization, Software, Project administration, Formal analysis, Conceptualization; **Tasso J. Kaper:** Writing – review & editing, Writing – original draft, Formal analysis, Conceptualization.

Data availability

Data will be made available on request.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Properties of the system at $\mathcal{O}(\epsilon)$

In the study of the stationary forced ($\epsilon \neq 0$) system (see especially Secs. 2.2, 3.1, and 6.2), it is necessary to understand the properties of solutions to the inhomogeneous problem

$$\mathcal{L}_0 u_1 = -F(u_h, u'_h, x), \quad (\text{A.1})$$

with $\mathcal{L}_0$ as defined in (17). We focus here on the case $u_h(x; \phi) = u_{\text{up}}(x; \phi) = +\tanh\left(\frac{\sqrt{2}}{2}[x - \phi]\right)$ and observe that the situation for $u_h = u_{\text{down}}$ is equivalent upon replacement of $F(u_{\text{up}}, u'_{\text{up}}, x)$ by $F(-u_{\text{up}}, -u'_{\text{up}}, x)$. (We refer to Lemma 2.5 of [44] for comparable more general statements and results.) Without loss of generality, we choose u_1 such that $u_1(\phi; \phi) = 0$. Foremost, the homogeneous equation $\mathcal{L}_0 \Psi = 0$ has a bounded solution Ψ_b and an unbounded solution Ψ_u , which are given by

$$\begin{aligned} \Psi_b(x; \phi) &= \partial_x u_{\text{up}}(x; \phi) = \frac{\sqrt{2}}{2} \operatorname{sech}^2\left(\frac{\sqrt{2}}{2}[x - \phi]\right) \\ \Psi_u(x; \phi) &= v(x; \phi) \Psi_b(x; \phi), \end{aligned} \quad (\text{A.2})$$

where v satisfies

$$\partial_x v(x; \phi) = \Psi_b(x; \phi)^{-2} = 2 \cosh^4\left(\frac{\sqrt{2}}{2}[x - \phi]\right).$$

By direct integration, v can thus be defined as

$$v(x; \phi) = \int_{\phi}^x \Psi_b(z; \phi)^{-2} dz = \frac{3}{4}[x - \phi] + \frac{\sqrt{2}}{2} \sinh\left(\sqrt{2}[x - \phi]\right) + \frac{\sqrt{2}}{16} \sinh\left(2\sqrt{2}[x - \phi]\right). \quad (\text{A.3})$$

The leading order terms for the limit behavior of the functions v , Ψ_b , and Ψ_u can then be found, confirming the assertion that Ψ_b is bounded and Ψ_u is unbounded in both limits $x \rightarrow \infty$ and $x \rightarrow -\infty$:

$$\begin{aligned} v(x; \phi) &\rightarrow \pm \frac{\sqrt{2}}{32} e^{\pm 2\sqrt{2}[x - \phi]} + \text{h.o.t.} & \text{as } x - \phi \rightarrow \pm\infty; \\ \Psi_b(x; \phi) &\rightarrow 2\sqrt{2} e^{\pm\sqrt{2}[x - \phi]} + \text{h.o.t.} & \text{as } x - \phi \rightarrow \pm\infty; \\ \Psi_u(x; \phi) &\rightarrow \pm \frac{1}{8} e^{\pm\sqrt{2}[x - \phi]} + \text{h.o.t.} & \text{as } x - \phi \rightarrow \pm\infty. \end{aligned} \quad (\text{A.4})$$

The inhomogeneous problem (A.1) can be solved using variation of parameters, which yields the solutions

$$u_1(x; \phi) = A(x; \phi) \Psi_b(x; \phi) + B(x; \phi) \Psi_u(x; \phi),$$

with

$$\begin{aligned} A(x; \phi) &= \int_{\phi}^x F(u_h(z; \phi), u_{h,x}(z; \phi), z) \Psi_u(z; \phi) dz, \\ B(x; \phi) &= B_0 - \int_{\phi}^x F(u_h(z; \phi), u_{h,x}(z; \phi), z) \Psi_b(z; \phi) dz. \end{aligned}$$

Here, B_0 is a constant that is determined by the boundary conditions (note A does not carry a constant A_0 as ϕ has been chosen such that $A(\phi; \phi) = 0$ and thus such that $u_1(\phi; \phi) = 0$). Using the limiting behavior of Ψ_b and Ψ_u (A.4), we find that the limiting behavior of A and B is as follows,

$$\begin{aligned} |A(x; \phi)| &\leq \left\| F(u_h(\cdot; \phi), u_{h,x}(\cdot; \phi), \cdot) \right\|_{\infty} \frac{\sqrt{2}}{16} e^{\pm\sqrt{2}[x - \phi]} + \text{h.o.t.} & \text{as } x - \phi \rightarrow \pm\infty; \\ |B(x; \phi) - B_{\pm}(\phi)| &\leq \left\| F(u_h(\cdot; \phi), u_{h,x}(\cdot; \phi), \cdot) \right\|_{\infty} 2e^{\pm\sqrt{2}[x - \phi]} + \text{h.o.t.} & \text{as } x - \phi \rightarrow \pm\infty; \end{aligned}$$

where

$$B_{\pm}(\phi) = B_0 - \int_{\phi}^{\pm\infty} F(u_h(z; \phi), u_{h,x}(z; \phi), z) \Psi_b(z; \phi) dz. \quad (\text{A.5})$$

Therefore, the limiting behavior of u_1 is given by

$$u_1(x; \phi) = \pm \frac{1}{8} B_{\pm} e^{\pm\sqrt{2}[x - \phi]} + \mathcal{O}(1) \text{ as } x - \phi \rightarrow \pm\infty. \quad (\text{A.6})$$

Thus, boundedness of u_1 for $x \rightarrow -\infty$ (respectively $x \rightarrow +\infty$) is guaranteed when $B_{-}(\phi) = 0$ (respectively $B_{+}(\phi) = 0$). That is, B_0 needs to be chosen as

$$B_0 = - \int_{-\infty}^{\phi} F(u_h(z; \phi), u_{h,x}(z; \phi), z) \Psi_b(z; \phi) dz, \quad \text{resp.} \quad (A.7)$$

$$B_0 = \int_{\phi}^{\infty} F(u_h(z; \phi), u_{h,x}(z; \phi), z) \Psi_b(z; \phi) dz$$

to ensure boundedness as $x \rightarrow -\infty$, resp. $x \rightarrow \infty$. Thus, if we consider an $u_1(x, \phi)$ that remains bounded as $x - \phi \rightarrow -\infty$, it follows that its limit behavior for $x - \phi \rightarrow \infty$ is indeed given by (126) (and thus that boundedness in both limits only holds if the solvability condition $\mathcal{R}(\phi) = 0$ (23) holds).

Appendix B. Proof of Theorem 1

The existence part of this theorem corresponds to finding a heteroclinic orbit in a weakly driven, planar nonlinear oscillator – see Remark 2. Thus, the existence of one-front patterns $U_{1-f}(x)$ follows by a straightforward application of the Melnikov method (see for instance [28]). This is done in Section 2.2 – see also Section 6.2 for a more extensive treatment in the standard Melnikov case in which the driving term $F(u, u_x, x)$ is spatially periodic in x .

In order to establish the spectral stability of $U_{1-f}(x)$, we need to distinguish between two cases: either $F(U, U_x, x)$ is spatially localized, i.e., $\lim_{x \rightarrow \pm\infty} F(U, U_x, x) = C_{\pm}(U, U_x)$ exists, or $F(U, U_x, x)$ is spatially periodic, i.e., there exists an $X > 0$ such that $F(U, U_x, x + X) = F(U, U_x, x)$ for all $x \in \mathbb{R}$. (Note that the concept/definition of spectrum is not necessarily clear for general $F(U, U_x, x)$.)

We first consider the former case, i.e., $F(U, U_x, x)$ is spatially localized. It follows from the analysis in Section 1.4 – see especially (9) – that the background states $u_{\pm}^e(x)$ are also localized, i.e., that the limits $\lim_{x \rightarrow \pm\infty} u_{\pm}^e(x) =: u_{\pm, \infty}^e$ exist, or equivalently, that the limits $\lim_{x \rightarrow \pm\infty} U_{1-f}(x) = u_{\pm, \infty}^e$ exist. Thus, the spectrum of $\mathcal{L}_*$ (29) consists of essential and point spectrum [37]. Since $u_{\pm, \infty}^e = \pm 1 + \mathcal{O}(\epsilon)$, it follows – by the theory of exponential dichotomies [38,39] – that the essential spectrum $\Sigma_{\text{ess}, \epsilon}$ of $\mathcal{L}_*$ is $\mathcal{O}(\epsilon)$ close to the essential spectrum $\Sigma_{\text{ess}, 0}$ of the operator $\mathcal{L}_0$ associated to the stability of the unperturbed one-front pattern $u_{\text{up}}(x; \phi)$ (15). Hence, by (18), we conclude that the essential spectrum cannot cause instability.

To control the eigenvalues/point spectrum of (28), we follow the Evans function approach of [7], in which the existence and spectral stability of one-pulse patterns in a weakly heterogeneous, singularly perturbed, two-component reaction-diffusion equation of Gray-Scott/Klausmeier type is studied. This Evans function approach is based on the ‘homogeneous’ Evans function associated with the spectral stability of the unperturbed homogeneous pattern. This is quite involved for systems of reaction-diffusion equations, but is much more straightforward in the present setting of a scalar equation. Indeed, the Evans function constructed in [7] is based on the methods originally developed in [40,52] for homogeneous, singularly perturbed, two-component reaction-diffusion systems, while we can simply build from the Evans function $D_0(\lambda)$ (20) associated to the Sturm-Liouville problem (16) studied in Section 2.1. By definition, the Evans function $D_*(\lambda)$ associated to the spectral problem (28) is a determinant that measures whether the (linear) space $\mathcal{V}_*^-(\lambda)$ of solutions of (28) that decay (exponentially) for $x \rightarrow -\infty$ intersects the space $\mathcal{V}_*^+(\lambda)$ of solutions that decay for $x \rightarrow \infty$ (see (20)). Typically, $\mathcal{V}_*^+(\lambda)$ is obtained from $\mathcal{V}_*^-(\lambda)$ by setting $x \rightarrow -x$. This is also the case for $D_0(\lambda)$ (20), for which the spaces $\mathcal{V}_0^{\pm}(\lambda)$ are spanned by $\tilde{u}_{\pm}(x; \lambda)$, and indeed $\tilde{u}_+(x; \lambda) = \tilde{u}_-(-x; \lambda)$ by (19).

The Evans function $D_*(\lambda)$ associated to the perturbed spectral problem (28) is a perturbation of $D_0(\lambda)$ (20). The perturbed spaces $\mathcal{V}_*^{\pm}(\lambda)$ can be constructed from their unperturbed counterparts $\mathcal{V}_0^{\pm}(\lambda)$ by a straightforward asymptotic analysis based on the Sturm-Liouville problem (16). This immediately implies that $D_*(\lambda)$ can only have zeros that are asymptotically close to those of $D_0(\lambda)$ (a non-empty intersection $\mathcal{V}_*^-(\lambda) \cap \mathcal{V}_*^+(\lambda)$ must limit on a non-empty intersection $\mathcal{V}_0^-(\lambda) \cap \mathcal{V}_0^+(\lambda)$). Hence, the elements of the point spectrum of operator $\mathcal{L}_*$ must be asymptotically close to those of $\mathcal{L}_0$. Thus, by (20), the (spectral) stability of a one-front pattern $U_{1-f}(x)$ is

determined by the eigenvalue that merges with the trivial translation eigenvalue $\lambda = 0$ in the limit $\epsilon \rightarrow 0$, as expected.

We set $\lambda = \epsilon \tilde{\lambda}$. The elements of $\mathcal{V}_*^{\pm}(\lambda)$ for asymptotically small λ can be directly approximated by an asymptotic analysis along the lines of Section 2.2 and App. A. It follows that applying the solvability condition on (34) is, in geometrical terms, equivalent to imposing that the solutions of (28) that decay as $x \rightarrow -\infty$, i.e., the elements of $\mathcal{V}_*^-(\epsilon \tilde{\lambda})$, do not grow exponentially as $x \rightarrow \infty$. (Note that both the derivation and the interpretation of this argument is completely similar to that of the derivation and interpretation of the Melnikov function/solvability criterion $\mathcal{R}(\phi)$ in Section 2.2 for the existence of $U_{1-f}(x) = u_*(x)$.) Thus, application of the solvability condition for (34) is equivalent to imposing the condition $\mathcal{V}_*^-(\epsilon \tilde{\lambda}) \cap \mathcal{V}_*^+(\epsilon \tilde{\lambda}) \neq \emptyset$. Hence, $\lambda = \epsilon \tilde{\lambda}(\phi_*) + \mathcal{O}(\epsilon^2)$ as given in (35) is a zero of $D_*(\lambda)$ and, thus, an eigenvalue of (28). Moreover, it also follows that $\lambda = \epsilon \tilde{\lambda}(\phi_*) + \mathcal{O}(\epsilon^2)$ is the only eigenvalue of (28) asymptotically close to 0.

The spectral analysis for the spatially periodic case can be done along the same lines. Naturally, it is based on an Evans function defined for a spectral problem with spatially periodic terms. We refrain from going into the details here. See [25,53,54] and the references therein. In addition, we refer to the alternative (but also similar) approach developed in [25]. This paper considers a general class of semilinear evolution systems on the real line with spatially periodic coefficients that includes (2). In it, the existence and spectral stability of stationary multi-front or multi-pulse patterns is established – based on general and natural assumptions. Building on the existence of $U_{1-f}(x)$, the statement of Theorem 1 for spatially periodic $F(U, U_x, x)$ also follows by the approach developed in [25]. $\square$

Appendix C. Proof of Theorem 2

Here, we follow a procedure that is similar to that used to prove Theorem 1 in App. B.

Establishing the existence of a two-front pattern $U_{2-f}(x)$ corresponds to constructing a (one-pulse) homoclinic orbit in the weakly driven planar system (7). This is achieved by the Melnikov analysis of Section 3.1, where we note that the application of the continuity conditions in (47) and (49) is – in geometrical terms – equivalent to imposing that the stable and unstable manifolds $W^s(\mathcal{M}_*^{\pm})$ and $W^u(\mathcal{M}_*^{\pm})$ of the invariant manifold $\mathcal{M}_*^{\pm}$ (that in the limit $\epsilon \rightarrow 0$ corresponds to the line of saddle points $\{(-1, 0, x) | x \in \mathbb{R}\}$ in the extended (u, u_x, x) -phase space) intersect (see Secs. 1.4, 2.2, and 6.2). Thus, condition (54) – i.e., the outcome of the application of the continuity conditions – establishes the existence of a homoclinic orbit in the intersection $W^s(\mathcal{M}_*^{\pm}) \cap W^u(\mathcal{M}_*^{\pm})$.

Since the asymptotic Melnikov analysis only considers leading order terms, it cannot be applied to situations in which the magnitudes of these leading order expressions become comparable to those of the next order correction terms. Therefore, it is necessary to impose the extra assumption $|\mathcal{R}_{\text{up}}(\phi_{\text{up}})| = |\mathcal{R}_{\text{down}}(\phi_{\text{down}})| \gg \epsilon$, i.e., for any $C > 0$ there is an $\epsilon_C > 0$ such that $|\mathcal{R}_{\text{up}}(\phi_{\text{up}})|/\epsilon = |\mathcal{R}_{\text{down}}(\phi_{\text{down}})|/\epsilon > C$ for all $0 < \epsilon < \epsilon_C$, in the statement of the theorem. See Section 5 and especially Theorem 3 and its proof in App. D for more details.

To study the spectral stability of $U_{2-f}(x)$, we assume that $F(U, U_x, x)$ is either spatially localized or spatially periodic. In both cases, the (spectral) stability is settled by linearizing around the two-front pattern $u_{2-f}(x)$,

$$(\mathcal{L}_{2-f} - \lambda)\tilde{u} = 0. \quad (C.1)$$

Here, $\mathcal{L}_{2-f} : H^2(\mathbb{R}) \rightarrow L^2(\mathbb{R})$ is given by

$$\mathcal{L}_{2-f} = \partial_x^2 + (1 - 3(u_{2-f}(x))^2) + \epsilon [F_u(u_{2-f}(x), \partial_x u_{2-f}(x), x) + F_{u_x}(u_{2-f}(x), \partial_x u_{2-f}(x), x) \partial_x]. \quad (C.2)$$

See (28) and (29).

The analysis of the operator differs between the two cases. We first study the case in which $F(U, U_x, x)$ is spatially localized. In this case, we may immediately conclude that the essential spectrum of the operator $\mathcal{L}_{2-f}$ is asymptotically close to the unperturbed essential spectrum $\Sigma_{\text{ess}, 0}$

(18), for the same reasons as for one-front solutions, recall App. B. Hence, the essential spectrum cannot cause instabilities for two-front solutions.

The point spectrum of the perturbed operator $\mathcal{L}_{2-f}$ can be determined by an Evans function approach, and the construction of the Evans function $D_{2-f}(\lambda)$ associated to the two-front pattern can be based on the homogeneous one-front (Sturm-Liouville) ‘building block’ $D_0(\lambda)$ (20). We refer to [55] for details on the relations between the spaces $\mathcal{V}_{2-f}^\pm(\lambda)$ of solutions of the two-front spectral problem (C.1) that decay exponentially to 0 as $x \rightarrow \pm\infty$ – i.e., the spaces that define and determine $D_{2-f}(\lambda)$ – and the basic unperturbed spaces $\mathcal{V}_0^\pm(\lambda)$. It follows that elements of the spaces $\mathcal{V}_{2-f}^\pm(\lambda)$ must be asymptotically close to shifted versions of elements of $\mathcal{V}_0^\pm(\lambda)$, which implies that $D_{2-f}(\lambda)$ can only have zeros asymptotically close to those of $D_0(\lambda)$ [55], as was also the case for one-front solutions in the proof of Theorem 1.

Thus, also here, the spectral stability of a two-front pattern $U_{2-f}(x)$ is determined by the eigenvalues of the operator $\mathcal{L}_{2-f}$ that merge with the translation eigenvalue $\lambda = 0$ as $\varepsilon \rightarrow 0$. Although the algebra is a bit more involved than for the one-front case, the essence of the analysis is the same. The end products, namely the conditions (57), determine the values of $\lambda = \varepsilon \tilde{\lambda} + \mathcal{O}(\varepsilon^2)$ for which $\mathcal{V}_{2-f}^-(\lambda) \cap \mathcal{V}_{2-f}^+(\lambda) \neq \emptyset$. Thus, $D_{2-f}(\varepsilon \tilde{\lambda}_{1,2} + \mathcal{O}(\varepsilon^2)) = 0$ with $\tilde{\lambda}_{1,2}$ as in (58) (with (59)). That is, $\lambda_{1,2} = \tilde{\lambda}_{1,2} + \mathcal{O}(\varepsilon^2)$ are the two eigenvalues of the operator $\mathcal{L}_{2-f}$ that determine the spectral stability of a two-front pattern $U_{2-f}(x)$.

Lastly, the case in which $F(U, U_x, x)$ is periodic as a function of x can be treated either by following the above set-up with an Evans function for spatially periodic spectral problems or by following [25], as was also the case for one-fronts in the proof of Theorem 1 (recall App. B). However, rather than state the proof here specifically for two-front solutions, we refer to the proof for the case of general N -front solutions in Theorem 4 in App. E. There, the approach of [25] is applied for general multi-front (and/or multi-pulse) patterns in systems with spatially periodic heterogeneities, which includes the present case of $N = 2$. $\square$

Appendix D. Proofs of Theorem 3 and Corollary 1

Stationary N -front patterns correspond to homoclinic ($N = \text{even}$) or heteroclinic ($N = \text{odd}$) orbits in the weakly driven planar system (7). For even N (respectively, for odd N), establishing the existence of these patterns is equivalent to imposing that the stable and unstable manifolds $W^s(\mathcal{M}_-^\varepsilon)$ and $W^u(\mathcal{M}_-^\varepsilon)$ of the invariant manifolds $\mathcal{M}_-^\varepsilon$ (that correspond to the line of saddle points $\{(-1, 0, x) | x \in \mathbb{R}\}$ in the extended (u, u_x, x) -phase space as $\varepsilon \rightarrow 0$) intersect in a suitable manner (respectively, that $W^s(\mathcal{M}_+^\varepsilon)$ and $W^u(\mathcal{M}_+^\varepsilon)$ intersect).

In geometrical terms, determining stationary N -front patterns corresponds to studying the intersections $W^s(\mathcal{M}_-^\varepsilon) \cap W^u(\mathcal{M}_-^\varepsilon)$ (for N even) and $W^s(\mathcal{M}_+^\varepsilon) \cap W^u(\mathcal{M}_+^\varepsilon)$ (for N odd). By construction, these intersections directly correspond to the critical points of the N -front interaction ODE (84). In fact, in these geometrical terms, the ‘algebra’ by which the stationary 2, 3, 4-front patterns have been constructed in Theorem 2 (App. C) and in Secs. 5.1 and 5.2 is identical to the algebra induced by the Melnikov analysis that establishes the associated intersections $W^s(\mathcal{M}_-^\varepsilon) \cap W^u(\mathcal{M}_-^\varepsilon)$ and $W^s(\mathcal{M}_+^\varepsilon) \cap W^u(\mathcal{M}_+^\varepsilon)$. This is also the case for the higher order intersections corresponding to the stationary N -front patterns with $N \geq 5$: establishing the existence of stationary N -front patterns by this Melnikov approach is equivalent to studying the critical points of the N -front interaction ODE (84). In other words, the (leading order) equations for the critical points $(\tilde{\psi}_1, \tilde{\psi}_2, \dots, \tilde{\psi}_N)$ of (84) correspond directly (83) to the Melnikov conditions obtained on the locations $\phi_1, \phi_2, \dots, \phi_N$ of the fronts of a stationary multi-front pattern $u_{N-f}(x; \phi_1, \phi_2, \dots, \phi_N)$.

This equivalence immediately allows us to conclude that the total number, $\mathcal{N}(N)$, of stationary N -front patterns can be deduced from the observation in the previous subsections that all multi-front patterns can be seen as direct extensions of only four kinds of basic patterns – the first, second, and third kinds of two-front patterns studied in Section 5.1, and the fourth kind, which are the three-front patterns of Section 5.2.

First, in the richest case, i.e., assuming that $h_+ < 0 < h_-$, we note that – either by direct observations or by inductive arguments – it can be shown that there are $N - 1$ distinct extensions of the two-front pattern of the first kind to a (stationary) N -front pattern (where it may be necessary to flip the original two-front pattern from one that approaches -1 as $x \rightarrow \pm\infty$ to one that approaches $+1$ as $x \rightarrow \pm\infty$). These patterns are different from all other stationary N -front patterns in the sense that they do not have a front asymptotically close to a zero of $S(\psi)$. Next, each of the two-front patterns of the second and third kinds can be extended uniquely to an N -front pattern, as is also the case for the four-front patterns. In the upcoming existence analysis, we will especially focus on these patterns (both for the exponential decay case of Theorem 3 and for the algebraically decaying heterogeneities of Corollary 1). These give rise to the critical value $\mu_*(N)$ in the case of Theorem 3. Lastly, all three-front patterns of the fourth kind can be extended in $N - 2$ ways to N -front patterns. (This is similar to the two-front patterns of the first kind.)

Thus, summing all of the possibilities, we conclude that for $h_+ < 0 < h_-$ and $N \geq 2$ the number of distinct N -front patterns totals

$$\mathcal{N}(N) = (N - 1) + K[1 + 1 + (N - 2)] = (K + 1)N - 1. \quad (\text{D.1})$$

Moreover, $\mathcal{N}(N)$ reduces simply to K if $h_+ < 0$ or if $h_- > 0$, and it reduces to 0 if $h_+ > 0$ and $h_- < 0$.

We can also establish the spectral stability of a stationary N -front pattern $u_{N-f}(x; \phi_1, \phi_2, \dots, \phi_N)$ as a solution of the PDE (2) by considering the N eigenvalues of the associated critical points in (84) (recall the final paragraph of Section 4.1). This may be seen by taking the geometrical Evans function point of view, which was also used in the proofs of Theorems 1 and 2. Since the spatial homogeneity is localized, we do not have to distinguish between two cases as in (the proofs of) Theorems 1 and 2. Moreover, the upcoming Evans function approach is independent of the decay rates of the localized inhomogeneities. Hence, also here, we can simultaneously treat both cases: the case of exponential decay in Theorem 3 and that of algebraic decay in Corollary 1.

The spectral problem associated to $u_{N-f}(x)$ is

$$(\mathcal{L}_{N-f} - \lambda)\tilde{u} = 0.$$

Here,

$$\mathcal{L}_{N-f} = \partial_x^2 + (1 - 3(u_{N-f}(x))^2) + \varepsilon [F_u(u_{N-f}(x), \partial_x u_{N-f}(x), x) + F_{u_x}(u_{N-f}(x), \partial_x u_{N-f}(x), x) \partial_x]. \quad (\text{D.2})$$

(See (28), (29), (C.1), (C.2)). This problem is similar to the spectral problem for the one-front and two-front patterns. The essential spectrum of $\mathcal{L}_{N-f}$ is determined by the asymptotic background states, as was also the case in the proofs of Theorems 1 and 2. Thus, it is asymptotically close to $\Sigma_{\text{ess},0}$ (18) and cannot be the source of instability. Next, the discrete spectrum of the perturbed operator $\mathcal{L}_{N-f}$ is determined by an Evans function $D_{N-f}(\lambda)$ based on the homogeneous (Sturm-Liouville) Evans function $D_0(\lambda)$ (20). For all $N > 1$, we can apply the general approach of [55] to conclude that the eigenvalues of $\mathcal{L}_{N-f}$ must be asymptotically close to those of Sturm-Liouville operator $\mathcal{L}_0$ (17). This is exactly what we also found in the $N = 2$ case, recall the proof of Theorem 2. Thus, also for general N , the spectral stability of a stationary N -front pattern $u_{N-f}(x)$ is determined by the $\mathcal{O}(\varepsilon)$ eigenvalues of $D_{N-f}(\lambda)$ (that in the limit $\varepsilon \rightarrow 0$ merge with the trivial translation eigenvalue $\lambda = 0$ of the homogeneous (Sturm-Liouville) Evans function $D_0(\lambda)$).

These $\mathcal{O}(\varepsilon)$ eigenvalues coincide by construction with the eigenvalues of the critical points of the N -front interaction Eq. (84) (taking into account the scalings introduced in (83)). This can be seen by realizing that these critical points are elements of a finite-dimensional attracting manifold $\mathcal{J}_N$ in an (appropriately chosen) function space H in which the PDE dynamics of (2) take place [27]. The fact that manifold $\mathcal{J}_N$ is attracting near such a critical point is equivalent to the fact that the spectrum of $\mathcal{L}_{N-f}$ that does not approach $\lambda = 0$ as $\varepsilon \rightarrow 0$ is in the stable half-plane, bounded away from the imaginary axis (also in the limit $\varepsilon \rightarrow 0$). The (spectral) stability of the N -front pattern as critical point of (2) on

$\mathcal{J}_N \subset H$ is thus determined by the eigenvalues associated to the linearization around this point as critical point of (84). As we saw, this stability is also determined by the $\mathcal{O}(\epsilon)$ eigenvalues of $\mathcal{L}_{N-f}$. By the complete equivalence of both ways of setting up the (linearized) stability analysis, we conclude that the $\mathcal{O}(\epsilon)$ eigenvalues of $\mathcal{L}_{N-f}$ indeed coincide (up to scaling) with the eigenvalues of the associated critical points of (84) on $\mathcal{J}_N$ (see again [27]).

These observations can for instance be confirmed quantitatively by considering symmetric two-front patterns. Condition (62) on the existence of symmetric two-front patterns agrees with condition (93) with $\tilde{\psi}_1 = -\tilde{\psi}_2 < 0$ on the existence of a critical point of the first kind in the two-front interaction Eq. (92) (where ϕ and ψ are related by (83)). The spectral stability of these symmetric two-front patterns is determined by their $\mathcal{O}(\epsilon)$ eigenvalues (Theorem 2) given by (63), while the eigenvalues of the corresponding critical point of two-front interaction Eqs. (92) are given by (100), which agrees exactly with (63) (again, taking into account the scalings introduced in (83)).

Thus, we conclude that both the (leading order) existence and spectral stability of the stationary N -front patterns $u_{N-f}(x; \phi_1, \phi_2, \dots, \phi_N)$ can be established by studying the critical points of interaction ODE (84). In the rest of this appendix, we use this to first give a proof of Theorem 3 and then a proof of Corollary 1.

Proof of Theorem 3. To establish the existence part of Theorem 3, we first focus on N -front patterns that are extensions of the two-front patterns of the second kind. Specifically, we assume that $h_+ < 0$ and that the N -fronts satisfy

$$\tilde{\psi}_1 = \psi_* + \epsilon^{\nu_1} \tilde{\psi}_1, \quad \tilde{\psi}_2 = \nu_2 |\log \epsilon| + \tilde{\epsilon}_2, \quad \dots, \quad \tilde{\psi}_N = \nu_N |\log \epsilon| + \tilde{\epsilon}_N, \quad \text{with } \nu_2 < \dots < \nu_N,$$

so that $\tilde{\psi}_1 = \mathcal{O}(1) \ll \tilde{\psi}_2 \ll \dots \ll \tilde{\psi}_N$. In other words, we focus on N -front patterns that have one front near a zero of $S(\psi)$ and the other fronts all lie to one side of this zero.

The system that determines these N -fronts as critical points of (84) can be solved inductively, with the two-front of the second kind as starting point. We first consider $N = 3$, observe that $1 \ll \tilde{\psi}_2 - \tilde{\psi}_1 \ll \tilde{\psi}_3 - \tilde{\psi}_2$, and hence conclude that we may copy the leading order result from Section 5.1. Namely, from (97), we see that $\nu_1 = \mu/(1 - \mu)$ and $\nu_2 = 1/(1 - \mu)$. Moreover, $\tilde{\epsilon}_1$ and $\tilde{\epsilon}_2$ can now also be determined uniquely.

Next, ν_3 and $\tilde{\epsilon}_3$ are (to leading order) determined by

$$\epsilon^{1+\mu\nu_3} |h_+| w_+ e^{-\mu\tilde{\epsilon}_3} = 16\epsilon^{\nu_3-\nu_2} e^{\tilde{\epsilon}_2-\tilde{\epsilon}_3}.$$

Thus, it follows that $\nu_3 = \frac{1}{1-\mu}(1 + \nu_2) = \frac{2-\mu}{(1-\mu)^2}$ and that $\tilde{\epsilon}_3$ can be determined uniquely. This process can be repeated for the 4-, ..., N -front patterns. The pair $(\nu_{j+1}, \tilde{\epsilon}_{j+1})$ is determined inductively by

$$\epsilon^{1+\mu\nu_{j+1}} |h_+| w_+ e^{-\mu\tilde{\epsilon}_{j+1}} = 16\epsilon^{\nu_{j+1}-\nu_j} e^{\tilde{\epsilon}_j-\tilde{\epsilon}_{j+1}}, \quad j = 2, \dots, N-1. \quad (\text{D.3})$$

Hence, $\nu_{j+1} = \frac{1}{1-\mu}(1 + \nu_j)$ for $j = 2, \dots, N-1$, and $\tilde{\epsilon}_{j+1}$ can be explicitly and uniquely determined via $\tilde{\epsilon}_j$.

From this induction, we conclude that

$$\nu_N(\mu) = \frac{1}{\mu} \left[\frac{1}{(1-\mu)^{N-1}} - 1 \right] \quad (\text{D.4})$$

Hence, we observe that $\lim_{\mu \downarrow 0} \nu_N(\mu) = N-1$, so that ν_N grows linearly in N in this regime. In addition, and more importantly, we observe that $\nu_j(\mu) - \nu_{j-1}(\mu) = 1/(1-\mu)^{j-1}$. Therefore, for all $\mu \in (0, 1)$, the distance between successive fronts increases with j .

To be certain that this leading order analysis is well-established, we again need to impose that the construction that determines the j th front takes place at a magnitude in ϵ for which the correction/error terms – that are *prima facie* of $\mathcal{O}(\epsilon^2)$ and have not been taken into account – indeed remain of higher order (and cannot become of leading order). It follows from (D.3) that we need to impose that $1 + \mu\nu_{j+1} = \nu_{j+1} - \nu_j < 2$ for all $j = 2, \dots, N-1$. By (D.4), we conclude that this is the case if

$$0 < \mu < \mu_{s,1}(N) := 1 - 2^{-\frac{1}{N-1}} \quad (N \geq 2), \quad (\text{D.5})$$

where the subscript 1 is an index indicating that we have labeled this N -front solution as the first of the $\mathcal{N}(N)$ N -front solutions in total.

Thus, to establish the existence of the present type of N -front patterns, we need to impose that $0 < \mu < \mu_{s,1}(N) < 1$. Clearly, $\mu_{s,1}(N)$ is a decreasing function of N . Also, for $N \gg 1$, $\mu_{s,1}(N) = \log 2/(N-1) + \mathcal{O}(1/N^2)$. Hence, $\mu_{s,1}(N) \downarrow 0$ as $N \rightarrow \infty$, and the width of the allowed μ -region decays monotonically to 0 as $N \rightarrow \infty$. Nevertheless, we may conclude that the two-front patterns of the second kind can indeed be extended to these N -front patterns for any given $N > 2$ as long as μ is sufficiently close to 0, i.e. if the (exponential) decay of the topography $H(x)$ is sufficiently weak (and if $h_+ < 0$). Moreover, we may immediately conclude that these patterns are unstable, since the largest two of the N asymptotically small eigenvalues associated to their spectral stability are (to leading order) given by those of the two-front, i.e., by (102). This established the results for the N -front solutions that are extensions of the two-front solutions of the second kind.

The third kind of critical points may, under the assumption that $h_- > 0$, also be extended to N -front patterns. First, these may be extended to N -fronts that have one front near a zero of $S(\psi)$ and all other fronts to the left of this zero (i.e., $\tilde{\psi}_1 \ll \tilde{\psi}_2 \ll \dots \ll \tilde{\psi}_{N-1} \ll -1$ and $\tilde{\psi}_N = \mathcal{O}(1)$), as may be shown in the same manner used in Section 5.1. We find the same critical value of μ as in (D.5), i.e., $\mu_{s,2}(N) = \mu_{s,1}(N)$. Hence, we obtain the same (in)stability result. All other $\mathcal{N}(N) - 2$ N -front patterns can be constructed in a similar inductive fashion, with initial data that is given either by a two-front of the first, second, or third type or by a three-front of the fourth type. For each of these N -fronts, there will be a critical value $\mu_{s,k}(N) > 0$ of μ so that their existence can be established in the μ -interval $(0, \mu_{s,k}(N))$, i.e., for these values μ it can be shown that the $\mathcal{O}(\epsilon^2)$ corrections to (83) remain of that order (cannot become of leading order).

Also, the critical value $\mu_s(N)$ introduced in the statement of Theorem 3 is the minimal value of the various $\mu_{s,k}(N)$'s, for $k = 1, 2, \dots, \mathcal{N}(N)$. The fact that the above-considered N -front patterns of the second and third kind have $N-1$ fronts to one side of the first (or last) front at $\psi = \mathcal{O}(1)$ implies that its last (or first) front reaches deeper into the tail of $S(\psi)$ than that of any of the other N -front patterns. In other words, the above considered patterns have fronts at positions for which $S(\psi)$ is smaller than for any of the other N -front patterns. Thus, the necessary assumption that the next order corrections to (83) remain of higher order (and cannot leading order), i.e. the condition $\epsilon |S(\psi)| \ll \mathcal{O}(\epsilon^2)$, is stronger for these N -front patterns than it is for any of the other N -front patterns. In other words, $\mu_{s,k}(N) \geq \mu_{s,1}(N) = \mu_{s,2}(N) = \mu_s(N)$, as given by (D.5) – which also appears in the statement of the theorem.

Since the spectral (in)stability of all N -fronts patterns is determined by the instability of the underlying patterns of the first, second, third, or fourth kind, the statement of Theorem 3 follows. $\square$

Proof of Corollary 1. As in the above proof of Theorem 3, we again first consider the construction of the N -front patterns by extending the two-fronts of the second kind to generate N -fronts that have $N-1$ fronts to the right of the first front at $\psi = \mathcal{O}(1)$.

For $N = 2$, i.e., for the two-front of the second kind, we replace (96) by

$$\tilde{\psi}_1 = \psi_* + \tilde{\psi}_1 \Theta_1(\epsilon) \quad \tilde{\psi}_2 = \frac{\tilde{\nu}_2}{\Theta_2(\epsilon)}, \quad (\text{D.6})$$

with $0 < \Theta_{1,2}(\epsilon) \ll 1$ two (asymptotically small) order functions of a priori unknown magnitude in ϵ and $\tilde{\psi}_1 = \tilde{\psi}_1(\epsilon) = \mathcal{O}(1)$, $\tilde{\nu}_2 = \tilde{\nu}_2(\epsilon) = \mathcal{O}(1)$ two constants to be determined. Assuming that $H'(x)$ decays as $1/|x|^p$ (90) it follows by Lemma 2 (and (83)) that also $S(\psi)$ decays as $1/|\psi|^p$ (for $|\psi| \gg 1$). Thus, under the condition that $\tilde{h}_\pm(p) < 0$ (90), we find by the second equation of (92) that

$$\epsilon(\Theta_2(\epsilon))^p e^{\frac{\tilde{\nu}_2}{\Theta_2(\epsilon)}} = C_2 = C_2(\tilde{\nu}_2), \quad (\text{D.7})$$

for some $\mathcal{O}(1)$ constant $C_2 > 0$. Hence,

$$\frac{\tilde{\nu}_2}{\Theta_2(\epsilon)} = |\log \epsilon| + p |\log \Theta_2(\epsilon)| + \mathcal{O}(1), \quad (\text{D.8})$$

so that we can determine the order function $0 < \Theta_2(\epsilon) \ll 1$ and constant $\tilde{\nu}_2(\epsilon)$ by

$$\Theta_2(\epsilon) = \frac{1}{|\log \epsilon| + p \log |\log \epsilon|} = \frac{1}{|\log \epsilon|} - p \frac{\log |\log \epsilon|}{|\log \epsilon|^2}, \quad \tilde{\nu}_2(\epsilon) = 1 \quad (\text{D.9})$$

to leading order in ε . Similarly, we find from the first equation of (92) that

$$\varepsilon \Theta_1(\varepsilon) e^{\frac{\tilde{v}_2}{\Theta_2(\varepsilon)}} = C_1 = C_1(\tilde{\psi}_1) = \mathcal{O}(1).$$

By (D.7) and (D.9), this yields that

$$\Theta_1(\varepsilon) = \left(\frac{1}{|\log \varepsilon|} - p \frac{\log |\log \varepsilon|}{|\log \varepsilon|^2} \right)^p = \frac{1}{|\log \varepsilon|^p} - p^2 \frac{\log |\log \varepsilon|}{|\log \varepsilon|^{p+1}} \quad (\text{D.10})$$

to leading order in ε . Note that the findings of (D.9) and (D.10) are natural extensions of the case of a vanishing exponential decay in Theorem 3. There, $v_2(\mu) = 1/(1-\mu) \rightarrow 1$ as $\mu \downarrow 0$ (96), (97), which corresponds to $\tilde{v}_2(\varepsilon)/\Theta_2(\varepsilon) = |\log \varepsilon|$ (D.6) (i.e., to leading order, $\tilde{v}_2 = v_2(0) = 1$ and $1/\Theta_2(\varepsilon) = |\log \varepsilon|$). Moreover, $v_1(\mu) = \mu/(1-\mu) \rightarrow 0$ as $\mu \downarrow 0$, which would imply that $\tilde{\psi}_1$ is no longer asymptotically close to a zero ψ_* of $S(\psi)$ (96). In the algebraic case this is replaced by $\tilde{\psi}_1$ being $\mathcal{O}(1/|\log \varepsilon|^p)$ close to ψ_* .

For N -front patterns that extend this two-front solution of the second kind to patterns that have one front near a zero of $S(\psi)$ and all other fronts to the right of this zero, we find in a similar way (to leading order),

$$\frac{\tilde{v}_j}{\Theta_j(\varepsilon)} = (j-1)(|\log \varepsilon| + p \log |\log \varepsilon|), \quad j = 2, \dots, N$$

(see (D.8)). Thus, we can choose all order functions $\Theta_j(\varepsilon)$, $j = 3, \dots, N$, to be equal to $\Theta_2(\varepsilon)$ as given by (D.9). The position of the j th front is then determined by (the pre-factor) $\tilde{v}_j(\varepsilon) = j-1$ to leading order, which again agrees with the $\mu \downarrow 0$ limit in the above case of exponential decay: $v_j(0) = \tilde{v}_j = j-1$ and $1/\Theta_j(\varepsilon) = |\log \varepsilon|$, $j = 2, \dots, N$.

This implies that in the algebraic decay case, the undetermined terms at $\mathcal{O}(\varepsilon^2)$ of the front interaction system (84) remain of higher order for all stationary N -front patterns, since (to leading order)

$$\varepsilon S(\tilde{\psi}_N) = \varepsilon \frac{\tilde{C}_N}{(\Theta_N(\varepsilon))^p} = \varepsilon \tilde{C}_N |\log \varepsilon|^p \gg \varepsilon^2$$

for some $\mathcal{O}(1)$ constant $\tilde{C}_N = \tilde{C}_N(\varepsilon) > 0$. Here, we have again used that N -front patterns that have one front near a zero of $S(\psi)$ and all other fronts to the right (or left) of this zero extend deeper into the tail of $S(\psi)$ than any of the other N -front patterns.

As in the case of exponential decay, the (in)stability of the stationary N -front patterns builds on that of the underlying two-front patterns of the first, second, and third kind and on that of the three-front pattern of the fourth kind. However, one needs to be careful since the leading order results on the eigenvalues of the underlying basic patterns in the case of exponential decay cannot be automatically extended to the limit $\mu = 0$.

In the case of the two-front patterns of the second kind, the leading order $\mathcal{O}(\varepsilon^{-\mu/(1-\mu)})$ expression in the matrix (101) associated to the (spectral) stability of the two-front pattern of the second kind merges with the next $\mathcal{O}(1)$ terms in the limit $\mu \downarrow 0$. Thus, one may not take the $\mu \downarrow 0$ limit of the eigenvalues $\lambda_{1,2}(\mu)$ (100), since these are obtained by leading order arguments. Nevertheless, it is straightforward to obtain the counterpart of the matrix (101) in the case of algebraic decay,

$$E_1 \begin{pmatrix} 1 - \varepsilon \frac{S'(\psi_*)}{E_1} & -1 \\ -1 & 1 - \frac{p}{\tilde{\psi}_2} \end{pmatrix}, \quad (\text{D.11})$$

with

$$E_1 = 16e^{-(\tilde{\psi}_2 - \tilde{\psi}_1)} = -\varepsilon S(\tilde{\psi}_2) = \mathcal{O}(\varepsilon \Theta_2(\varepsilon)^p) = \mathcal{O}\left(\frac{\varepsilon}{|\log \varepsilon|^p}\right).$$

Thus, similar to the case of exponential decay, the $\varepsilon S'(\psi_*)/E_1$ -term in matrix (D.11) is dominant and we may conclude

$$\lambda_1 = -\varepsilon S'(\psi_*), \quad \lambda_2 = E_1 > 0,$$

to leading order (see (102)); that is, the two-front of the second kind represented by $(\tilde{\psi}_1, \tilde{\psi}_2)$ is unstable.

The analysis of the two-front patterns of the third kind is essentially identical to the above analysis. In the case of two-front patterns of the first kind, the matrix given in (99) also degenerates, in the sense that $\lambda_1(\mu) \uparrow 0$ as $\mu \downarrow 0$ (100). Although the unstable eigenvalue $\lambda_1(\mu) \downarrow 2 > 0$ – and

one is thus inclined to conclude that the instability persists into the case of algebraic decay – we need to consider also this case somewhat more carefully. We find that the algebraic counterpart of matrix (99) is similar to (99), namely

$$E_1 \begin{pmatrix} 1 - \frac{p}{|\tilde{\psi}_1|} & -1 \\ -1 & 1 - \frac{p}{\tilde{\psi}_2} \end{pmatrix} \quad (\text{D.12})$$

with $1/|\tilde{\psi}_{1,2}| \ll 1$ (in fact, it can be checked that $\tilde{\psi}_1 = -1/2|\log \varepsilon|$ and $\tilde{\psi}_2 = 1/2|\log \varepsilon|$ to leading order, which again agrees with the $\mu \downarrow 0$ limit of the case of exponential decay). It follows that

$$\lambda_1(p) = -\frac{p}{2} \left(\frac{1}{|\tilde{\psi}_1|} + \frac{1}{\tilde{\psi}_2} \right) < 0, \quad \lambda_2(p) = 2 - \frac{p}{2} \left(\frac{1}{|\tilde{\psi}_1|} + \frac{1}{\tilde{\psi}_2} \right) > 0,$$

(to leading order). Thus, also in the case of algebraic decay, the two-front pattern of the third kind is unstable.

A similar analysis shows that the three-front patterns of the fourth kind remain unstable in the case of algebraic decay. Finally, it follows along exactly the same lines as in the exponential case that the eigenvalues of the underlying two-front patterns of the first, second, and third kind and those of the three-front patterns of the fourth kind persist as leading order eigenvalues for all N -front patterns that are based on it. $\square$

Appendix E. Proof of Theorem 4

In this appendix, we prove the existence of stationary N -front patterns of (2) using Melnikov arguments that determine the existence of homoclinic and/or heteroclinic orbits in weakly, periodically driven planar system (7). Following the same arguments as those in the proof of Theorem 3 in App. D, we conclude that the Melnikov conditions correspond directly, i.e., are identical, to the conditions that determine the critical points of interaction ODE (84). Hence, as in App. D, we can study the existence of stationary N -front patterns by determining the stationary points of (84), without assuming the validity of (the dynamic version of) these equations.

Since $S(\tilde{\psi})$ is periodic, it is bounded – and thus $\mathcal{O}(1)$ with respect to ε – for all $\tilde{\psi} \in \mathbb{R}$. As a consequence, all front locations $\tilde{\psi}_j$ of a critical point $(\tilde{\psi}_1, \tilde{\psi}_2, \dots, \tilde{\psi}_N)$ of (84) must satisfy

$$\varepsilon^{\rho_j} = e^{\log \varepsilon \rho_j} = e^{-\Delta \tilde{\psi}_j} = e^{-(\tilde{\psi}_{j+1} - \tilde{\psi}_j)} = \mathcal{O}(\varepsilon).$$

Thus, for $\rho_j < 1$, there cannot be any stationary N -front patterns (as in the homogeneous Allen-Cahn equation), or, the other way around: stationary N -front patterns can only exist if

$$\lim_{\varepsilon \rightarrow 0} \frac{\tilde{\psi}_{j+1} - \tilde{\psi}_j}{|\log \varepsilon|} = \rho_j \geq 1.$$

On the other hand, if $\rho_j > 1$ for all $j = 1, 2, \dots$, i.e., if the distances between all successive fronts grow faster than $|\log \varepsilon|$ (as $\varepsilon \downarrow 0$), then the $\varepsilon S(\psi_j)$ terms in (84) represent the leading order effects. In other words, the existence of a critical point $(\tilde{\psi}_1, \tilde{\psi}_2, \dots, \tilde{\psi}_N)$ is determined to leading order purely by the heterogeneity,

$$S(\tilde{\psi}_j) = 0, \quad j = 1, 2, \dots, N.$$

The conclusions about the existence of stationary N -front patterns in Theorem 4 follow directly from these observations.

Next, the spectral stability of the stationary multi-front patterns can be determined by setting up an Evans function approach for the spatially periodic spectral problem, as referred to in the proofs of Theorems 1 and 2. Under the assumption that $\rho_j > 1$, i.e., the case considered for (the stability statements of) Theorem 4, such an asymptotic approach (in ε) immediately yields that the impact of the inhomogeneity is the leading order effect (as in the existence case), which implies that the asymptotically small spectrum of the N -front is to leading order determined by

$$\phi_j = -\varepsilon S'(\tilde{\psi}_j) \phi_j, \quad j = 1, 2, \dots, N,$$

which yields the stability statements of Theorem 4.

However, we refrain from working out the full details of this (quite technical) enterprise. Instead, we appeal to the recent work [25] in which the existence and stability of multi-front and/or multi-pulse patterns is studied in a general class of evolutionary systems with spatially periodic coefficients that includes the inhomogeneous Allen-Cahn model (2) with the spatially-periodic $F(U, U_x, x)$ here. The main theorem of [25] (see Theorem 3.1 there) is based on a number of assumptions on the existence and stability of a stationary one-front (or one-pulse) pattern. These assumptions are satisfied by our analysis in Section 2 and especially by Theorem 1.

We observe that the analysis of [25] is on the natural scales of the PDEs studied there. There is no explicit small parameter there, like the ε here. As a consequence, the requirement in Theorem 3.1 of [25] that the distance between successive fronts is sufficiently large is given in terms of the natural scale of the systems there. Here, sufficiently large is measured in terms of $|\log \varepsilon|$, for the asymptotic analysis. So, one natural way to apply the approach of [25] in the present setting is first to define a value ε_1 so that the existence and stability results of Theorem 1 hold for all ε that satisfy $0 < \varepsilon < \varepsilon_1$ (note that such an ε_1 must exist). Then, we fix a value of ε that satisfies this inequality. We label that fixed value ε_0 , and we apply Theorem 3.1 of [25] to the PDE with the fixed ε_0 . This yields a proof of a version of Theorem 4 in which the stability statements hold for $\tilde{\psi}_{j+1} - \tilde{\psi}_j$ ($j = 2, 3, \dots, N-1$) ‘sufficiently large’, but does not provide an explicit condition in terms of ρ_j . That is, Theorem 3.1 of [25] does not directly validate the explicit lower limit $\rho_j > 1$ ($j = 1, 2, \dots, N$) of the formulation of Theorem 4 in Section 6.3. Given the relative simplicity of the present setting – compared to that of [25] – this ‘gap’ can be closed by a further use of asymptotic analysis. We refrain from going into the details. $\square$

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